

Stochastic Finance

Financial Mathematics for Data Science — Lecture Notes

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About these notes

Scope

These are lecture notes for *Stochastic Finance — Financial Mathematics for Data Science* (third-semester course, Master in Physics of Data, University of Padova, taught by Prof. Claudio Fontana). The course is a first mathematical treatment of financial derivatives: how they are priced, how their risks are hedged, and which mathematical machinery — no-arbitrage arguments, stochastic processes, stochastic calculus, risk-neutral probabilities — makes the treatment rigorous. Numerical methods and two practitioner lectures connect the theory to how banks and insurers actually work.

Reading the notes

- **Part I (Chapters 1–3): markets and no-arbitrage.** Markets, clearing and margins, interest rates and bonds; derivatives and the model-free consequences of no arbitrage (forward prices, put–call parity, option bounds); the binomial model, where hedging, risk-neutral valuation, martingales and American options appear in their simplest form.
- **Part II (Chapters 4–8): continuous-time models.** Brownian motion and Itô calculus; the Black–Scholes–Merton model, its PDE and formula, the Greeks and hedging in practice; Monte Carlo methods and variance reduction; local and stochastic volatility; interest rate products and short-rate models.
- **Part III (Chapters 9–10): practitioner lectures.** Risk-free rate curves after the end of Libor and their use in insurance; automated market making for FX options.

Each chapter separates definitions, theorems, examples and remarks in styled boxes, includes the derivations done in class, and ends with exercises (solutions collapsible in the HTML version) and a *Sources* section. Numerical examples were checked by direct computation; where the course slides contain a misprint, the text says so.

Notation

- S_t : price of the underlying at time t ; B_t : bank account; $P_t(T)$: price at t of the zero-coupon bond maturing at T .
- r (constant) or r_t (stochastic): continuously compounded risk-free (short) rate; $R_t(\tau)$: zero rate at t for horizon τ .
- W_t : Brownian motion under $\mathbb{P}$; W_t^Q under the risk-neutral probability Q ; $\mathcal{F}_t$: information at time t .
- $\theta = (\theta^B, \theta^S)$: trading strategy in bank account and stock; $V_t(\theta)$: its value.
- $\Pi_t(G)$: arbitrage-free price at t of the payoff G ; $F(t, x)$: pricing function.
- Φ, φ : standard normal distribution function and density.
- $\Delta, \Gamma, \Theta, \mathcal{V}, \rho$: the Greeks; σ_{imp} : implied volatility.
- $L_t(S, T)$: simple forward rate; $f_t(T)$: instantaneous forward rate.

Sources

The notes reorganise Prof. Fontana's slides (`slides_intro`, `slides_1-slides_8`) and his handwritten lecture notes (8, 14 and 21 October, 4 and 19 November 2025), together with the guest lectures of V. Lunina (FX options market making, 12 November 2025) and V. Russo (risk-free rates, 9 December 2025). Explanations and examples are drawn from:

- J. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson, 2022 (Chapters 1–3);
- T. Björk, *Arbitrage Theory in Continuous Time*, 4th ed., Oxford University Press, 2020 (Chapters 3–5 and 8);
- M. Musiela and M. Rutkowski, *Martingale Methods in Financial Modelling*, 2nd ed., Springer, 2005;
- R. Seydel, *Tools for Computational Finance*, 6th ed., Springer (Chapter 6);
- A. Turner and D. Zeindler, *Stochastic Finance: An Introduction with Examples*, Cambridge University Press, 2023.

The course outline also lists stochastic models for credit risk; no material on it was distributed, so it is not covered here.

Part I.

Part I: Markets and no-arbitrage

1. An introduction to financial markets

This course is about valuing financial contracts and managing their risks with mathematical models. Before any model, we need the institutional picture: what is traded, where, how counterparty risk is controlled, and how money at different dates is compared. This chapter covers the four asset classes, exchange-traded and over-the-counter markets, clearing houses and the margin mechanism, and the arithmetic of interest rates — compounding, discounting, bond pricing and the bootstrap of the zero curve.

1.1. Why study financial markets mathematically

Modern finance raises a few recurring questions: how do markets work, what are the main sources of risk, which instruments are available and how can they be used, and — the focus of this course — how to value complex contracts and manage their risks. Probability is the natural language for risk, and a mathematical model gives rigorous, quantitative answers; but **no model is perfect**, and a recurring theme will be to understand what each model leaves out.

A few facts motivate the effort:

- **Markets are largely automated.** The trading floor of the NYSE is today “largely a TV studio” (Financial Times, 2021); most orders are generated by algorithms, sometimes at high frequency. On 23 April 2013 a fake tweet from the Associated Press account about explosions at the White House caused a *flash crash* of the US market within seconds.
- **Interbank risk is real.** Before 2007 the spread between Euribor and overnight-indexed (OIS) rates of the same maturity was negligible; it exploded during the 2007–2009 financial crisis and again during the 2010–2012 European debt crisis, revealing that lending between banks is not risk-free (Chapter 8).
- **Derivatives are huge.** The derivatives market was essentially non-existent before 1973; its notional size today is a multiple of world GDP.
- **Derivatives are everywhere.** Many common products embed derivatives: a floating-rate mortgage with an option to switch to a fixed rate (an interest-rate swaption), life insurance with a guaranteed return (a put option on a portfolio), equity-linked insurance (an equity option), convertible bonds (a call option on the issuer’s stock), callable bonds (an issuer’s call option), retail structured notes (custom option packages). Each such product is “base asset + derivatives”, its valuation requires option-pricing techniques, and accounting standards (IFRS) require the embedded derivatives to be recognised separately.

1.2. Asset classes

The securities traded in financial markets fall into four main categories.

1. **Stocks.** A share gives the right to part of the cash flows of a company, paid as dividends. Its price can be decomposed as the net present value of expected future dividends plus a resale value.
2. **Fixed-income securities** (bonds). They pay predetermined amounts at predetermined dates in a given currency; their price is essentially determined by interest rates.
3. **Currencies** (EUR, USD, GBP, JPY, CHF, ...). Holding a currency gives the right to earn the interest rate of bank deposits in that currency; currencies are related by exchange rates.

4. **Derivatives** (forwards, futures, options, swaps, ...). A derivative is a contract whose value is *derived* from another asset, the **underlying**; its price must be consistent with that of the underlying.

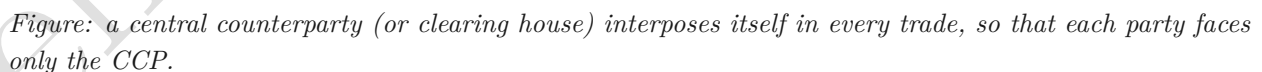
1.3. Exchange-traded markets

Investors submit orders of different types:

- ### 1.3.1. Clearing houses

- every transaction is centralised: each trade is split into two trades with the clearing house;
- traders deposit funds (**margins**) to guarantee that they can meet their future obligations;
- members also contribute to a common **guarantee fund**.

Trader A agrees with trader B to buy 1m USD from B in one year. In practice A contracts with the clearing house to buy 1m USD in one year, and B contracts with the clearing house to sell 1m USD in one year. If B defaults, A still receives the dollars from the clearing house. The clearing house's exposure is covered by the margins posted by A and B and, as a last resort, by the guarantee fund.



- **bilaterally**, through a *master agreement* detailing all mutual obligations, including when the transaction can be terminated; or

- through a **central counterparty** (CCP), which, like an exchange's clearing house, stands between the two parties so that neither bears the other's default risk.

Exchange-traded markets have more transactions; OTC markets have larger average trade size and larger total volume.

Before and after 2008. Before the 2007–2009 crisis the OTC market was largely unregulated, mostly bilateral, and grew enormously from 2000 to 2007; Lehman Brothers alone had over one million outstanding OTC transactions with more than 8000 counterparties when it failed. Today, in the US, standardised OTC derivatives must whenever possible be traded on *swap execution facilities* (exchange-like platforms with posted quotes), all trades must be reported to a central repository, and a CCP must be used for most derivatives between financial institutions (similar rules apply in the EU under EMIR). The goals are to reduce counterparty and systemic risk and to increase transparency.

1.4.1. The margin mechanism

How does a CCP control the default risk it takes on? Through daily **marking to market**: every day each position is revalued at the current settlement price, gains and losses are credited to or debited from the trader's **margin account**, and the account must stay above a **maintenance margin**; if it falls below, the trader receives a **margin call** to top it up to the **initial margin**.

Example 1.2 (A futures position marked to market)

A trader buys futures on 20 units of an asset at the futures price of 5000 € per unit for delivery in one year. The initial margin is 12 000 €, the maintenance margin 9000 €.

day	futures price	daily gain	cumulative gain	margin account	margin call
0	5000			12 000	
1	4970	−600	−600	11 400	
2	4900	−1400	−2000	10 000	
3	4950	+1000	−1000	11 000	
4	4830	−2400	−3400	8600	3400
5	4800	−600	−4000	11 400	
6	4850	+1000	−3000	12 400	
7	4780	−1400	−4400	11 000	
8 (1y)	4870	+1800	−2600	12 800	

On day 4 the account falls below 9000 € and the trader must deposit 3400 € to restore 12 000 €. At maturity the total result is $(4870 - 5000) \times 20 = -2600$ €, and it has been realised day by day: $12\,800 - 12\,000 - 3400 = -2600$ €. If the trader had failed to meet the margin call, the position would have been closed using the margin already posted — the CCP never loses more than one day's price move beyond the margin.

1.5. Interest rates and compounding

Depositing A € for one year, how much do we get? It depends on the **interest rate** R (in % per year) and on the **compounding frequency** m (how many times per year interest is credited):

- $R = 8\%$, annual compounding ($m = 1$): $A \mapsto A(1.08)$;

1. An introduction to financial markets

- semiannual ($m = 2$): $A \mapsto A(1 + 0.08/2)^2$;
- quarterly ($m = 4$): $A \mapsto A(1 + 0.08/4)^4$.

Definition 1.1 (Compounding). With rate R and compounding frequency m , an amount A grows in T years to

$$A\left(1 + \frac{R}{m}\right)^{mT}.$$

The map $m \mapsto A(1 + R/m)^{mT}$ is increasing: more frequent compounding pays interest on interest sooner. Two rates are **equivalent** if they produce the same wealth. Given R_1 with frequency m_1 , the equivalent rate with frequency m_2 solves $(1 + R_1/m_1)^{m_1} = (1 + R_2/m_2)^{m_2}$:

$$R_2 = m_2 \left[\left(1 + \frac{R_1}{m_1}\right)^{m_1/m_2} - 1 \right].$$

Definition 1.2 (Continuous compounding). Letting $m \rightarrow \infty$, $(1 + R/m)^m \rightarrow e^R$: with a continuously compounded rate R , an amount A grows to Ae^{RT} in T years. The continuously compounded rate equivalent to R_1 with frequency m_1 is

$$R_c = m_1 \log\left(1 + \frac{R_1}{m_1}\right), \quad R_1 = m_1(e^{R_c/m_1} - 1).$$

Daily compounding is already very close: with $R = 5\%$, $(1 + 0.05/365)^{365} = 1.051267$ against $e^{0.05} = 1.051271$.

Example 1.3 (Converting rates)

- 8% with quarterly compounding is equivalent to $(1 + 0.08/4)^4 - 1 = 8.243\%$ with annual compounding.
- 10% with semiannual compounding is equivalent to $2\log(1.05) = 9.758\%$ continuously compounded.
- 8% continuously compounded is equivalent to $4(e^{0.02} - 1) = 8.08\%$ with quarterly compounding.

From now on, unless stated otherwise, rates are **continuously compounded** (mathematically the most convenient) and $\log$ is the natural logarithm.

1.6. Discounting

The reverse question: what is a future cash flow worth today?

Definition 1.3 (Discounting). If A € will be received in T years and the continuously compounded rate is R , its present value is Ae^{-RT} €. Indeed, depositing Ae^{-RT} today yields exactly A in T years.

This reasoning assumes no risk: the rate does not change and the future cash flow is known for sure. Random cash flows require the probabilistic machinery of the following chapters.

1.7. Bonds and the zero curve

Definition 1.4 (Zero-coupon and coupon bonds). A **zero-coupon bond** (ZCB) with maturity T pays one unit of money at T and nothing before. A **coupon bond** with maturity T , coupon rate c and payment frequency m pays c/m every $1/m$ years and, at T , the principal 1 plus the last coupon.

For example, a 3-year bond paying a 10% coupon semiannually pays 0.05 every six months and 1.05 at $T = 3$. For a principal N all prices scale by N .

Interest rates depend on the horizon, which leads to the central object of fixed income.

Definition 1.5 (Zero rate). The T -year **zero rate** $R_0(T)$ is the rate applying to an investment from today to T with no intermediate payments: 1 € invested today becomes $e^{TR_0(T)}$ € at T , and 1 € paid at T is worth $e^{-TR_0(T)}$ € today. The map $T \mapsto R_0(T)$ is the **zero curve** (or yield curve).

Proposition 1.1 (Bond valuation principle). *The price of a bond is obtained by discounting each cash flow at the zero rate of its payment date:*

$$P_0(T) = e^{-TR_0(T)}, \quad P_0(T, c, m) = \frac{c}{m} \sum_{k=1}^{mT} e^{-\frac{k}{m} R_0(\frac{k}{m})} + e^{-TR_0(T)} = \frac{c}{m} \sum_{k=1}^{mT} P_0\left(\frac{k}{m}\right) + P_0(T).$$

The second form shows that a coupon bond is a portfolio of zero-coupon bonds — a first instance of *static replication*.

Example 1.4 (Pricing a coupon bond)

With zero rates $R_0(0.5) = 5.0\%$, $R_0(1) = 5.8\%$, $R_0(1.5) = 6.4\%$, $R_0(2) = 6.8\%$, a bond with principal 100, maturity 2 years and 6% coupon paid semiannually is worth

$$3e^{-0.5 \times 0.05} + 3e^{-1 \times 0.058} + 3e^{-1.5 \times 0.064} + 103e^{-2 \times 0.068} = 98.39.$$

1.7.1. Bootstrapping the zero curve

In practice zero rates are not observed directly; they are extracted from bond prices by the **bootstrap**: price the bonds in order of increasing maturity, each time solving for the one new unknown rate.

Example 1.5 (Bootstrap)

Market prices (principal 100, coupons paid semiannually):

maturity	coupon rate	price
0.25	0	99.6
0.5	0	99.0
1	0	97.8
1.5	4%	102.5
2	5%	105.0

The zero-coupon bonds give the short rates directly: $99.6 = 100e^{-0.25R_0(0.25)}$, so $R_0(0.25) = 1.603\%$; similarly $R_0(0.5) = 2.010\%$ and $R_0(1) = 2.225\%$. For 1.5 years, using the rates already found,

$$102.5 = 2e^{-0.5R_0(0.5)} + 2e^{-R_0(1)} + 102e^{-1.5R_0(1.5)} \implies R_0(1.5) = 2.284\%,$$

and for 2 years,

$$105 = 2.5e^{-0.5R_0(0.5)} + 2.5e^{-R_0(1)} + 2.5e^{-1.5R_0(1.5)} + 102.5e^{-2R_0(2)} \implies R_0(2) = 2.416\%.$$

(The course slides print 2.461%; solving the equation gives 2.416%.)

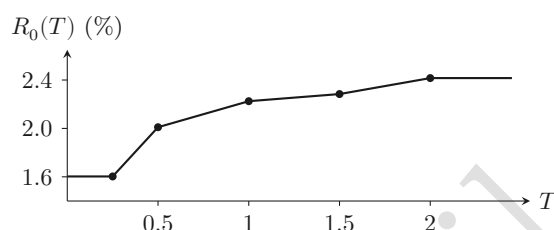


Figure: bootstrapped zero curve, linearly interpolated between the bond maturities and assumed flat outside them.

Outside the range of observed maturities the curve has to be extrapolated (here: flat); between them it must be interpolated. The choice of interpolation is a modelling decision with real consequences — Chapter 9 revisits it for modern risk-free-rate curves.

1.8. Forward rates and par yields

Zero rates for different maturities imply the rates that can be locked in today for **future** periods.

Proposition 1.2 (Forward rates from zero rates). *With continuously compounded zero rates $R_1 = R_0(T_1)$ and $R_2 = R_0(T_2)$, $T_1 < T_2$, the forward rate for the period $[T_1, T_2]$ is*

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} = R_2 + (R_2 - R_1) \frac{T_1}{T_2 - T_1}.$$

Proof. Investing for T_2 years yields $e^{R_2 T_2}$; investing for T_1 years and rolling over at the forward rate yields $e^{R_1 T_1} e^{R_F (T_2 - T_1)}$. Absence of arbitrage forces the two to coincide. $\square$

The second form shows that when the zero curve is increasing, forward rates lie **above** zero rates; letting $T_2 \rightarrow T_1$ gives the **instantaneous forward rate** $f(T) = R_0(T) + T R_0'(T)$, studied in Chapter 8.

Definition 1.6 (Par yield). The **par yield** for maturity T is the coupon rate c that makes a bond's price equal to its principal. With coupons paid m times a year and $A = \sum_{k=1}^{mT} P_0(k/m)$, the condition $\frac{c}{m} 100 A + 100 P_0(T) = 100$ gives

$$c = \frac{m(1 - P_0(T))}{A}.$$

This is the same formula as the swap rate of Chapter 8: a par bond and a swap are two views of the same cash flows. For an increasing curve, par yield $\leq$ zero rate $\leq$ forward rate (Björk, Exercise 22.5).

1.9. Duration and convexity

How sensitive is a bond's price to interest rates? Let a bond pay c_i at times t_i and have **yield** y (the single continuously compounded rate that discounts all cash flows to the market price): $B = \sum_i c_i e^{-yt_i}$.

Definition 1.7 (Duration and convexity). The (Macaulay) **duration** and the **convexity** of the bond are

$$D = \frac{1}{B} \sum_i t_i c_i e^{-yt_i} = -\frac{1}{B} \frac{dB}{dy}, \quad C = \frac{1}{B} \sum_i t_i^2 c_i e^{-yt_i} = \frac{1}{B} \frac{d^2 B}{dy^2}.$$

D is a weighted average of the payment times, with weights the present values of the cash flows: a zero-coupon bond has duration equal to its maturity. A second-order Taylor expansion gives

$$\frac{\Delta B}{B} \approx -D \Delta y + \frac{1}{2} C (\Delta y)^2.$$

Duration measures the first-order sensitivity to a **parallel shift** of the yield curve; convexity corrects for curvature (bond prices are convex in the yield, so the linear approximation always underestimates the price). With non-continuous compounding the derivative gives the **modified duration** $D/(1 + y/m)$. Portfolio durations are value-weighted averages, which makes duration the basic tool of **immunisation**: matching the duration of assets and liabilities (Chapter 9). Its limitations: it only captures parallel shifts, and for large moves the convexity term matters.

1.10. Exercises

Problems from Hull, *Options, Futures, and Other Derivatives* (numbering of the 11th edition).

Exercise 1.1

(Hull 2.1) You enter into a short futures contract to sell July silver at \$17.20 per ounce. The contract is for 5000 ounces; the initial margin is \$4000 and the maintenance margin \$3000. What change in the futures price leads to a margin call? What happens if you do not meet it?

Solution

A short position loses when the price rises. A margin call occurs when the loss exceeds $4000 - 3000 = 1000$, i.e. a price rise of $1000/5000 = 0.20$: a futures price above \$17.40. If the call is not met, the broker closes out the position.

Exercise 1.2

(Hull 2.11) At the end of one day a clearing house member is long 100 contracts with settlement price \$50 000 each; the original margin is \$2000 per contract. The next day the member clears 20 additional long contracts entered at \$51 000. The settlement price at the end of that day is \$50 200. How much must the member add to its margin account?

Solution

New contracts require initial margin $20 \times 2000 = 40\,000$. Variation margin: the old position gains $100 \times 200 = 20\,000$; the new contracts lose $20 \times 800 = 16\,000$; net gain 4000. The member must add $40\,000 - 4000 = 36\,000$.

Exercise 1.3

(Hull 4.1, 4.8) A bank quotes 7% per annum with quarterly compounding. Find the equivalent rate with continuous and with annual compounding. Which continuous rate is equivalent to 8% with monthly compounding?

Solution

Continuous: $4 \log(1.0175) = 6.939\%$; annual: $1.0175^4 - 1 = 7.186\%$. For 8% monthly: $12 \log(1 + 0.08/12) = 7.973\%$.

Exercise 1.4

(Hull 4.4) Continuously compounded zero rates for 3, 6, 9, 12, 15, 18 months are 3.0, 3.2, 3.4, 3.5, 3.6, 3.7%. Compute the forward rates for the second to sixth quarters.

Solution

With $R_F = (R_2 T_2 - R_1 T_1)/(T_2 - T_1)$: second quarter $(3.2 \times 0.5 - 3.0 \times 0.25)/0.25 = 3.4\%$; then 3.8%, 3.8%, 4.0%, 4.2%.

Exercise 1.5

(Hull 4.10) Zero rates for 6, 12, 18, 24, 30 months are 4.0, 4.2, 4.4, 4.6, 4.8% (continuous). Price a 30-month bond with face value 100 paying a 4% coupon semiannually.

Solution

$$2e^{-0.04 \times 0.5} + 2e^{-0.042} + 2e^{-0.044 \times 1.5} + 2e^{-0.046 \times 2} + 102e^{-0.048 \times 2.5} = 98.04.$$

Exercise 1.6

(Hull 4.12) The 6-, 12-, 18- and 24-month zero rates are 5, 6, 6.5 and 7% (continuous). What is the 2-year par yield (semiannual coupons)?

Solution

$A = e^{-0.025} + e^{-0.06} + e^{-0.0975} + e^{-0.14} = 3.6935$ and $P_0(2) = e^{-0.14} = 0.8694$, so $c = 2(1 - 0.8694)/3.6935 = 7.07\%$.

Exercise 1.7

(Hull 4.14) A 10-year 8% coupon bond sells for 90 and a 10-year 4% coupon bond for 80 (same payment dates). What is the 10-year zero rate?

Solution

Two 4% bonds minus one 8% bond: the coupons cancel and the portfolio pays $2 \times 100 - 100 = 100$ at year 10, costing $2 \times 80 - 90 = 70$. So $100e^{-10R} = 70$ and $R = \log(100/70)/10 = 3.567\%$.

Exercise 1.8

(Hull 4.18) A 5-year bond with yield 7% (continuous) pays an 8% annual coupon. Find its price, duration and the price change predicted by duration for a 0.2% decrease in yield; compare with the exact change and with the convexity-corrected estimate.

Solution

$B = 8 \sum_{t=1}^4 e^{-0.07t} + 108e^{-0.35} = 103.05$. Duration $D = \frac{1}{B} \sum_t t c_t e^{-0.07t} = 4.323$ years. Prediction: $\Delta B \approx -103.05 \times 4.323 \times (-0.002) = 0.891$. Exact: repricing at 6.8% gives +0.895. Convexity $C = 20.31$ adds $\frac{1}{2} \times 103.05 \times 20.31 \times 0.002^2 = 0.004$, giving 0.895.

Exercise 1.9

(Hull 4.19) Treasury bills maturing in 6 and 12 months cost 94.0 and 89.0 (face 100). A 1.5-year bond paying \$4 every six months costs 94.84; a 2-year bond paying \$5 every six months costs 97.12. Bootstrap the continuously compounded zero rates.

Solution

$R(0.5) = 2 \log(100/94) = 12.375\%$, $R(1) = \log(100/89) = 11.653\%$. Then $4e^{-0.5R(0.5)} + 4e^{-R(1)} + 104e^{-1.5R} = 94.84$ gives $R(1.5) = 11.502\%$, and the 2-year bond gives $R(2) = 11.299\%$: a downward-sloping curve.

Exercise 1.10

(Hull 4.6, 4.16) For an upward-sloping term structure, order (a) the 5-year zero rate, (b) the yield of a 5-year coupon bond, (c) the forward rate for $[4.75, 5]$. What changes if the curve slopes downward?

Solution

Upward: (c) > (a) > (b). The coupon bond's yield is an average of zero rates over all payment dates, weighted towards earlier (lower) ones; the forward rate is the marginal rate at the long end. For a downward-sloping curve the order is reversed: (b) > (a) > (c).

1.11. Sources

Prof. Fontana, *Stochastic Finance* slides, `slides_intro` and Chapter 1 (`slides_1`), and the handwritten lecture notes of 8 October 2025 (clearing houses, marking to market, compounding, bond pricing). Hull, *Options, Futures, and Other Derivatives*, Sections 1.1–1.2 (markets), 2.4–2.5 and 2.8 (clearing, margins, orders) and Chapter 4 (interest rates and bonds).

2. Financial derivatives

Derivatives are the central object of the course: contracts whose payoff depends on the value of an underlying asset. This chapter introduces the main types — forwards, futures and options, vanilla and exotic — and the three kinds of traders who use them. Its main tool is the **no-arbitrage principle**, which already yields a surprising number of exact results without any probabilistic model: the law of one price, forward and futures prices, put–call parity, model-free bounds on option prices, and the fact that an American call on a non-dividend stock should never be exercised early. Static option strategies close the chapter.

2.1. Hedgers, speculators and arbitrageurs

Three broad categories of market participants trade derivatives:

- **hedgers** reduce their exposure to adverse price movements (companies, banks, funds);
- **speculators** take positions to bet on future price movements (banks, hedge funds, investors);
- **arbitrageurs** look for riskless profits by combining transactions across counterparties or markets (banks, hedge funds).

Example 2.1 (*A hedger: forward vs. put*)

An Italian textile company will receive 100k USD from a US client in six months. Today 1 USD = 0.86 EUR, but the future rate is unknown: the company is exposed to **currency risk**, and its business is textiles, not betting on exchange rates.

- **Forward contract.** The company agrees today, at zero cost, to pay 100k USD in six months and receive 86k EUR. Its income is locked at 86k EUR whatever happens. The price of certainty: if the rate rises to 0.90, the company forgoes the 90k EUR it would have received.
- **Put option.** For 2k EUR today, the company buys the *right* (not the obligation) to sell 100k USD for 86k EUR in six months. If the rate falls below 0.86 it exercises and receives 86k EUR; otherwise it lets the option expire and converts at the market rate. Its net income is at least $86 - 2 = 84$ k EUR, with unlimited upside.

The forward eliminates risk in both directions for free; the option removes only the downside, at a price.

Example 2.2 (*A speculator: leverage*)

An investor believes the USD/EUR rate (now 0.86) will fall within six months.

1. *Borrow 100k USD, convert into 86k EUR.* At 0.81 the debt is repaid with 81k EUR: profit 5k EUR. At 0.88 it costs 88k EUR: loss 2k EUR. Gains and losses are small relative to the volume, and a large credit line is needed.
2. *Forward contract* to pay 100k USD and receive 86k EUR: the same profile, but with no cash flow today.
3. *Borrow 10k EUR and buy five puts* (each on 100k USD at 0.86). At 0.81 the puts are exercised: buy 500k USD for 405k EUR, sell them for 430k EUR, gain 25k EUR, net 15k EUR after the

premium. At 0.88 the puts expire worthless: loss 10k EUR.

Options **amplify** gains and losses (*leverage*); losses are limited to the premium, while gains are potentially unlimited.

Example 2.3 (An arbitrageur)

A US bank pays 2% on 6-month USD deposits, a European bank lends EUR at 1% over 6 months, and the forward above (100k USD for 86k EUR) is available in any quantity. Today:

- borrow 86k EUR, convert into 100k USD, deposit them in the US;
- enter into 1.02 forward contracts.

Net cost today: zero. In six months the deposit holds 102k USD; the forwards convert them into 87 720 EUR; repaying the loan costs 86 860 EUR. The **riskless profit** is 860 EUR, whatever the exchange rate does.

This is an **arbitrage opportunity**: zero initial cost, no risk of loss, a positive profit with positive probability. Scaling all transactions by 100 yields 86k EUR. In reality there are transaction costs and some default risk, and above all every market participant would exploit the opportunity, moving prices through supply and demand until it disappears. **If arbitrage opportunities ever exist, they vanish almost immediately.**

2.2. General features and the no-arbitrage principle

Every derivative has a **maturity** T (the termination date), an **underlying** (usually a traded asset: a stock, an index, a currency), a **payoff** (the cash flow it delivers, as a function of the underlying, usually at maturity) and a **price** (its market value before maturity). We denote by S_t the price of the underlying at time t ; for $T > t$, S_T is unknown at t , i.e. random. t denotes a generic valuation date.

Definition 2.1 (Arbitrage opportunity). An **arbitrage opportunity** is a trading strategy with zero initial cost, no risk of losses at any time, and a strictly positive profit with positive probability at some future time.

Definition 2.2 (No-arbitrage principle). The financial market admits no arbitrage opportunities, however assets and derivatives are combined into an investment strategy.

We assume the principle throughout. In this chapter it yields **model-free** results: valid whatever the probabilistic dynamics of the underlying.

Theorem 2.1 (Law of one price). Let X and Y be the payoffs at T of two assets or portfolios, with prices Π_t^X, Π_t^Y . If $\mathbb{P}(X = Y) = 1$ and there is no arbitrage, then $\mathbb{P}(\Pi_t^X = \Pi_t^Y) = 1$ for all $t \in [0, T]$.

Proof. Suppose that $\mathbb{P}(\Pi_{t_0}^X \neq \Pi_{t_0}^Y) > 0$ for some t_0 . Consider the strategy: do nothing until t_0 ; at t_0 , (a) if $\Pi_{t_0}^X > \Pi_{t_0}^Y$, buy one unit of Y , sell one unit of X and deposit $\Pi_{t_0}^X - \Pi_{t_0}^Y$ in the bank until T ; (b) if the prices are equal, do nothing; (c) if $\Pi_{t_0}^X < \Pi_{t_0}^Y$, do the opposite of (a). The cost at t_0 is always zero. At T the outcome is, in case (a),

$$Y - X + (\Pi_{t_0}^X - \Pi_{t_0}^Y) e^{(T-t_0)R_{t_0}(T-t_0)} = (\Pi_{t_0}^X - \Pi_{t_0}^Y) e^{(T-t_0)R_{t_0}(T-t_0)} > 0$$

almost surely (since $X = Y$), zero in case (b), and positive in case (c). The strategy is an arbitrage with positive probability of profit, a contradiction. $\square$

Here $R_t(\tau)$ is the zero rate prevailing at time t for horizon τ , and $P_t(T) = e^{-(T-t)R_t(T-t)}$ the price at t of the ZCB maturing at T .

2.3. Forward contracts

Definition 2.3 (Forward contract and forward price). A **forward contract** with maturity T and delivery price K pays $S_T - K$ at T : it is an agreement to buy the underlying at T for the price K fixed today. Its value at $t \leq T$ is denoted $\Pi_t^{\text{fwd}}(K)$. The **forward price** $F_t(T)$ is the delivery price that makes the contract worthless at t : $\Pi_t^{\text{fwd}}(F_t(T)) = 0$.

Forwards are typically traded OTC; currency forwards are widespread, and in interest-rate markets their analogue is the forward rate agreement (Chapter 8).

Theorem 2.2 (Forward price and value of a forward). *If the underlying pays no cash flows over $[t, T]$, then under no arbitrage*

$$\Pi_t^{\text{fwd}}(K) = S_t - KP_t(T), \quad F_t(T) = \frac{S_t}{P_t(T)} = S_t e^{(T-t)R_t(T-t)},$$

and $\Pi_t^{\text{fwd}}(K) = P_t(T)(F_t(T) - K)$.

Proof via the law of one price. The forward's payoff $X = S_T - K$ coincides with that of the portfolio “one unit of the underlying, minus K zero-coupon bonds maturing at T ”, whose value at t is $S_t - KP_t(T)$. Hence $\Pi_t^{\text{fwd}}(K) = S_t - KP_t(T)$ for every K , and setting it to zero gives $F_t(T) = S_t/P_t(T)$. $\square$

Direct arbitrage proof. If $F_t(T) < S_t e^{(T-t)R_t(T-t)}$: at t go long the forward (free), sell the underlying short and deposit S_t ; at T receive $S_T - F_t(T)$ from the forward, buy back the underlying at S_T , and collect $S_t e^{(T-t)R_t(T-t)}$ from the bank. The net cash flow $S_t e^{(T-t)R_t(T-t)} - F_t(T) > 0$ is an arbitrage. Reversing all positions excludes the opposite inequality.

If the underlying pays cash flows (dividends, coupons) with present value I_t over $[t, T]$, the buyer of the forward does not receive them, and $F_t(T) = (S_t - I_t)/P_t(T)$.

2.3.1. Cost of carry

The argument above extends to underlyings that generate income or require costs while they are held. In each case replicate the forward by buying the asset with borrowed money and account for what holding it brings in or costs.

Proposition 2.3 (Forward prices with income, yields and storage). *For a forward with maturity T entered at $t = 0$, with continuously compounded rate r :*

- **known cash income** with present value I (dividends, coupons): $F_0 = (S_0 - I)e^{rT}$;
- **known continuous yield** q (dividend yield of an index): $F_0 = S_0 e^{(r-q)T}$;
- **foreign currency** with foreign rate r_f (covered interest parity): $F_0 = S_0 e^{(r-r_f)T}$;
- **commodity with storage costs** of present value U : $F_0 \leq (S_0 + U)e^{rT}$, with equality for investment assets such as gold.

Proof for the yield case. Buy e^{-qT} units of the asset and reinvest the yield continuously: the holding grows to exactly one unit at T . The cost is $S_0 e^{-qT}$; financing it by borrowing, the debt at T is $S_0 e^{-qT} e^{rT}$. Holding

2. Financial derivatives

the unit and paying the debt replicates a forward with delivery price $S_0 e^{(r-q)T}$. Any other forward price is an arbitrage. A foreign currency is an asset with yield r_f . $\square$

Summarising with the **cost of carry** c — interest paid to finance the asset, plus storage, minus income — $F_0 = S_0 e^{cT}$: $c = r$ for a non-dividend stock, $r - q$ for an index, $r - r_f$ for a currency, $r + u$ for a commodity with proportional storage cost u . For **consumption commodities** (oil, copper) only the inequality holds, because holders do not sell inventory to exploit a low forward price: the gap is measured by the **convenience yield** y , $F_0 = S_0 e^{(c-y)T}$.

Forward price and expected spot price. The forward price is *not* the market's forecast of S_T : since the forward costs nothing, $F_0 = \mathbb{E}[S_T]$ only if investors require no risk premium on the asset. For assets with positive systematic risk, $\mathbb{E}[S_T] = S_0 e^{\mu T} > F_0 = S_0 e^{rT}$.

2.4. Futures

A **futures contract** is, like a forward, an agreement to buy the underlying at a future date for a price fixed today — the **futures price** $f_t(T)$, set so that entering the contract costs nothing. The differences: futures are standardised and exchange-traded; gains and losses are **marked to market daily** (Chapter 1); most futures are closed before maturity by taking the opposite position.

Theorem 2.4 (Futures and forward prices). *If zero rates are deterministic, then $f_t(T) = F_t(T)$.*

Proof. Let $t = t_0 < t_1 < \dots < t_n = T$ be the trading days. Holding θ_i futures over $[t_{i-1}, t_i]$ generates the cash flow $\theta_i(f_{t_i}(T) - f_{t_{i-1}}(T))$ at t_i , which we invest until T , where it is worth $\theta_i(f_{t_i}(T) - f_{t_{i-1}}(T))/P_{t_i}(T)$. The terminal wealth of the **dynamic futures strategy** $\theta = (\theta_1, \dots, \theta_n)$ is

$$V_T(\theta) = \sum_{i=1}^n \theta_i (f_{t_i}(T) - f_{t_{i-1}}(T)) \frac{1}{P_{t_i}(T)}.$$

Choose $\bar{\theta}_i = P_{t_i}(T)$ — possible only if $P_{t_i}(T)$ is known at t_{i-1} , which holds when rates are deterministic. Then the sum telescopes:

$$V_T(\bar{\theta}) = f_T(T) - f_t(T) = S_T - f_t(T),$$

the payoff of a forward with delivery price $f_t(T)$. Since entering futures positions costs nothing, $V_t(\bar{\theta}) = 0$, and by the law of one price $0 = \Pi_t^{\text{fwd}}(f_t(T))$, i.e. $f_t(T) = F_t(T)$. $\square$

With random rates the daily settlement creates a correlation effect and the two prices differ; for short maturities (a few months) the difference is usually negligible.

2.5. Options

Forwards and futures are **linear** derivatives. Options are **non-linear**: they give the right, not the obligation, to do something with the underlying. Their price can never be negative. The **exercise date** is when the right is used; the **strike** K is the fixed price appearing in the payoff.

Definition 2.4 (European and American options). Let g be the payoff function. A **European** option pays $g(S_T)$ at maturity T . An **American** option can be exercised at any time $\tau \leq T$ chosen by the holder, paying $g(S_\tau)$ at τ .

An American option is worth at least as much as its European counterpart: the holder can always wait until T .

Definition 2.5 (Call and put options). A **call** with strike K and maturity T gives the right to buy one unit of the underlying at K ; its payoff is $(S_T - K)^+ = \max(S_T - K, 0)$. A **put** gives the right to sell at K ; its payoff is $(K - S_T)^+$. The American versions pay $(S_\tau - K)^+$ or $(K - S_\tau)^+$ at the chosen exercise time τ .

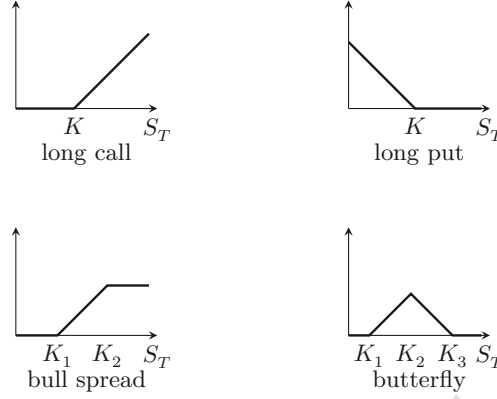


Figure: payoffs at maturity of a long call, a long put, a bull spread and a butterfly spread. Short positions are the mirror images about the horizontal axis.

2.5.1. Exotic options

Vanilla options are the standard calls and puts; all others are **exotic**, typically OTC and possibly very complex. Both kinds come in European and American versions.

- **Asian options** depend on the average price over $[0, T]$: e.g. $(\frac{1}{T} \int_0^T S_t dt - K)^+$ (average price) or $(S_T - \frac{1}{T} \int_0^T S_t dt)^+$ (average strike). In practice the integral is a discrete average over trading days, $\frac{1}{N} \sum_{i=1}^N S_{t_i}$. Geometric-average versions, $(\exp(\frac{1}{T} \int_0^T \log S_u du) - K)^+$, also exist and are useful for computation (Chapter 6). Averaging reduces the impact of price manipulation and volatility near maturity.
- **Lookback options** depend on the extremes of the path: $(S_T - \min_{t \leq T} S_t)^+$ (“buy at the lowest price”) or $(\max_{t \leq T} S_t - S_T)^+$.
- **Barrier options** are activated (“in”) or deactivated (“out”) when the underlying crosses a barrier. A *down-and-out put* pays $\mathbf{1}_{\{\min_{t \leq T} S_t \geq H_1\}} (K - S_T)^+$ with $H_1 < K$: the put payoff, unless the price ever falls below H_1 . An *up-and-out call* pays $\mathbf{1}_{\{\max_{t \leq T} S_t \leq H_2\}} (S_T - K)^+$ with $H_2 > K$. Combining barrier position (up/down) and effect (in/out) gives four types: up-and-out, up-and-in, down-and-out, down-and-in. Barrier options are cheaper than the corresponding vanilla, since the payoff may be cancelled.
- **Compound options** have an option as underlying: a put on a call pays $(K - \Pi_T^{\text{Call}}(K', T'))^+$ at T , the right to sell at K a call with strike K' and maturity $T' > T$.
- **Chooser options** let the holder decide at $T' < T$ whether the option is a call or a put.

2.6. Put–call parity

Theorem 2.5 (Put–call parity). For European options with the same underlying, strike K and maturity T , under no arbitrage,

$$\Pi_t^{\text{Call}}(K) - \Pi_t^{\text{Put}}(K) = S_t - K P_t(T), \quad t \in [0, T].$$

Proof. $(S_T - K)^+ - (K - S_T)^+ = S_T - K$ for every S_T : a long call plus a short put has the payoff of a forward with delivery price K , worth $S_t - KP_t(T)$. Apply the law of one price. $\square$

Parity holds only for European options. It is useful in both directions: a call price gives the put price, and vice versa; deviations in market quotes reveal arbitrage (or transaction costs).

2.7. Model-free bounds

Let $\Pi_t^{EC}, \Pi_t^{EP}, \Pi_t^{AC}, \Pi_t^{AP}$ denote the prices of European and American calls and puts with the same underlying, strike K and maturity T .

Theorem 2.6 (No-arbitrage bounds). *Under no arbitrage:*

1. American options are worth at least their **intrinsic value**: $\Pi_t^{AC} \geq (S_t - K)^+, \Pi_t^{AP} \geq (K - S_t)^+$.
2. American options are worth at least their European versions: $\Pi_t^{EC} \leq \Pi_t^{AC} \leq S_t$ and $\Pi_t^{EP} \leq \Pi_t^{AP} \leq K$.
3. European put: $(KP_t(T) - S_t)^+ \leq \Pi_t^{EP} \leq KP_t(T)$.
4. European call on a non-dividend-paying underlying: $(S_t - KP_t(T))^+ \leq \Pi_t^{EC} = \Pi_t^{AC}$.

All proofs follow the same scheme: assume the opposite inequality, buy what is too cheap, sell what is too expensive, invest the difference in the bank, and show that the result is an arbitrage. Two examples:

Proof that $\Pi_t^{EP} \leq KP_t(T)$. If $\Pi_t^{EP} > KP_t(T)$: buy K ZCBs maturing at T , sell one put, deposit $\Pi_t^{EP} - KP_t(T)$. Zero cost. At T : $K - (K - S_T)^+ + (\Pi_t^{EP} - KP_t(T))/P_t(T) > 0$, since the first two terms are ≥ 0 . Arbitrage.

Proof that $\Pi_t^{EP} \geq KP_t(T) - S_t$. If $\Pi_t^{EP} < KP_t(T) - S_t$: buy a put and one unit of the underlying, and borrow $\Pi_t^{EP} + S_t$ until T . Zero cost. At T : $(K - S_T)^+ + S_T - (\Pi_t^{EP} + S_t)/P_t(T)$; the debt is less than K , while $(K - S_T)^+ + S_T = \max(K, S_T) \geq K$. Arbitrage. Since also $\Pi_t^{EP} \geq 0$, the lower bound $(KP_t(T) - S_t)^+$ follows.

The lower bound for the call, $\Pi_t^{EC} \geq (S_t - KP_t(T))^+$, is proved in the same way (exercise).

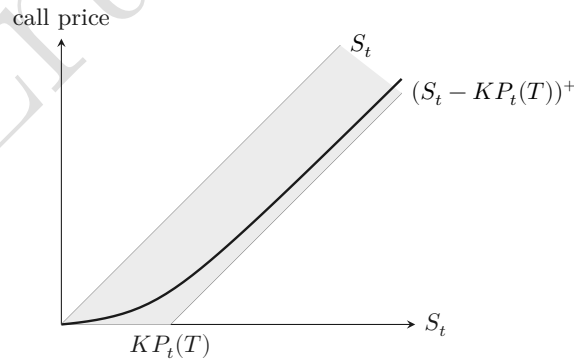


Figure: the price of a European call (bold) lies between the intrinsic lower bound and the price of the underlying.

2.7.1. The American call is never exercised early

Proposition 2.7 (No early exercise of American calls). *If the underlying pays no dividends, $\Pi_t^{EC}(K) = \Pi_t^{AC}(K)$ for all t : it is never optimal to exercise an American call before maturity.*

Proof. We know $\Pi_t^{EC} \leq \Pi_t^{AC}$. Suppose $\Pi_t^{EC} < \Pi_t^{AC}$: buy a European call, sell an American call, deposit the difference — zero cost. If the American call is exercised at some $\tau \in [t, T]$, we must pay $(S_\tau - K)^+$; selling our European call covers it, since by bound 4

$$\Pi_\tau^{EC} - (S_\tau - K)^+ \geq (S_\tau - KP_\tau(T))^+ - (S_\tau - K)^+ \geq (S_\tau - K)^+ - (S_\tau - K)^+ = 0,$$

using $P_\tau(T) \leq 1$. We can always honour the obligation, and the bank deposit is pure profit: an arbitrage. $\square$

Interpretation. Before maturity, a call on a non-dividend stock is always worth more than its intrinsic value: exercising early throws away the time value of money on the strike (paying K now rather than later) and the insurance against a fall of the stock. This is not true for an American put: if the underlying is very low, receiving K immediately can be better than waiting, so American puts may be exercised early (Chapter 3).

2.7.2. Bounds with dividends

If the underlying pays dividends with present value D during the option's life, the holder of an option does not receive them, and the bounds shift by D :

$$\Pi_t^{EC} \geq S_t - D - KP_t(T), \quad \Pi_t^{EP} \geq D + KP_t(T) - S_t, \quad S_t - D - K \leq \Pi_t^{AC} - \Pi_t^{AP} \leq S_t - KP_t(T),$$

and put-call parity becomes $\Pi_t^{EC} - \Pi_t^{EP} = S_t - D - KP_t(T)$. The no-early-exercise argument for American calls breaks down: it can be optimal to exercise **immediately before an ex-dividend date**, when the dividend about to be lost exceeds the remaining time value (approximately, when the dividend exceeds $K(1 - e^{-r\tau})$, τ being the time to the next dividend date or maturity).

2.8. Static strategies with options

Combining options, the underlying and bonds creates payoff profiles tailored to a market view. Profit means payoff minus the initial cost (compounded to T , or ignoring interest for simplicity).

strategy	positions	payoff at T	view
covered call	long stock, short call K	$\min(S_T, K)$	mildly bullish, earn premium
protective put	long stock, long put K	$\max(S_T, K)$	bullish with insurance
bull spread	long call K_1 , short call $K_2 > K_1$	$\min((S_T - K_1)^+, K_2 - K_1)$	moderate rise
bear spread	long put K_2 , short put $K_1 < K_2$	$\min((K_2 - S_T)^+, K_2 - K_1)$	moderate fall
butterfly	long calls K_1, K_3 , short 2 calls $K_2 = \frac{K_1 + K_3}{2}$	tent peaked at K_2	little movement
straddle	long call and put at K	$ S_T - K $	large movement, any direction
strip / strap	1 call + 2 puts / 2 calls + 1 put	asymmetric V	large move, biased down / up
principal-protected note	long call K , K ZCBs	$\max(S_T, K)$	upside with guaranteed principal

Some details:

- **Bull spread.** Profit is capped at $K_2 - K_1 - \Pi_0^{\text{Call}}(K_1) + \Pi_0^{\text{Call}}(K_2)$ and losses at the initial cost $\Pi_0^{\text{Call}}(K_1) - \Pi_0^{\text{Call}}(K_2) > 0$ (call prices decrease in the strike, since $(S_T - K_1)^+ \geq (S_T - K_2)^+$). Compared with a single call, the trader gives up profits above K_2 to reduce the cost.
- **Butterfly.** The payoff is $(S_T - K_1)\mathbf{1}_{\{S_T \in [K_1, K_2]\}} + (K_3 - S_T)\mathbf{1}_{\{S_T \in (K_2, K_3]\}}$; the cost $\Pi_0^{\text{Call}}(K_1) - 2\Pi_0^{\text{Call}}(K_2) + \Pi_0^{\text{Call}}(K_3)$ is small but positive (convexity of the call price in the strike). The same payoff can be built from puts.
- **Principal-protected notes** are the basic structure of many retail products: the worst case is $K - \Pi_0^{\text{Call}}(K) - KP_0(T)$ relative to the initial investment, i.e. the principal K is returned.

These **static** strategies are set up once and held. Chapter 3 introduces **dynamic** strategies, rebalanced over time, which can replicate any payoff — and with them, option prices instead of option bounds.

2.9. Exercises

Problems from Hull (H) and Björk (B), *Arbitrage Theory in Continuous Time*.

Exercise 2.1

(H 5.7) A 1-year long forward on a non-dividend stock is entered when $S_0 = 40$ and $r = 5\%$. (a) Find the forward price and the initial value of the contract. (b) Six months later $S = 45$ and r is unchanged: find the forward price and the value of the contract.

Solution

(a) $F_0 = 40e^{0.05} = 42.05$; the initial value is 0. (b) The new forward price for delivery at $T = 1$ is $45e^{0.025} = 46.14$. The contract's value is $S_t - KP_t(T) = 45 - 42.05e^{-0.025} = 3.99$.

Exercise 2.2

(H 5.10) $r = 6\%$, the dividend yield of an index is 4% , the index is at 400 and the 4-month futures price is 405. What arbitrage exists?

Solution

The theoretical price is $400e^{(0.06-0.04)/3} = 402.68 < 405$: futures are too expensive. Sell futures, borrow to buy the index (reinvesting dividends), deliver at maturity; the profit is about $405 - 402.68 = 2.32$ per unit of the index.

Exercise 2.3

(H 5.12) Two-month rates are 1% in Switzerland and 2% in the US (continuous). The spot CHF is 1.0500 USD and the 2-month futures price is also 1.0500. What arbitrage exists?

Solution

Covered interest parity gives $F = 1.05e^{(0.02-0.01)/6} = 1.0518 > 1.0500$: futures are too cheap. Buy CHF futures, borrow CHF and sell them spot for USD, invest the USD at 2% ; at maturity take delivery of CHF through the futures to repay the CHF loan. Profit ≈ 0.0018 USD per CHF.

Exercise 2.4

(H 5.13) Silver trades at 25 per ounce; storage costs 0.24 per ounce per year, payable quarterly in advance; $r = 5\%$ for all maturities. Find the 9-month futures price.

Solution

Storage payments of 0.06 at 0, 0.25, 0.5: $U = 0.06(1 + e^{-0.0125} + e^{-0.025}) = 0.1778$. Then $F_0 = (25 + 0.1778)e^{0.05 \times 0.75} = 26.14$.

Exercise 2.5

(H 11.1, 11.2) Find a lower bound for (a) a 4-month call with $S = 28$, $K = 25$, $r = 8\%$; (b) a 1-month European put with $S = 12$, $K = 15$, $r = 6\%$ (no dividends).

Solution

(a) $S - Ke^{-rT} = 28 - 25e^{-0.08/3} = 3.66$. (b) $Ke^{-rT} - S = 15e^{-0.005} - 12 = 2.93$.

Exercise 2.6

(H 11.6, 11.11) (a) $S = 19$, a 3-month European call with $K = 20$ costs 1, $r = 4\%$. Price the put with the same strike. (b) A 1-month European put with $K = 50$ trades at 2.50 when $S = 47$ and $r = 6\%$. What arbitrage is available?

Solution

(a) Parity: $P = C + Ke^{-rT} - S = 1 + 20e^{-0.01} - 19 = 1.80$. (b) The lower bound is $50e^{-0.005} - 47 = 2.75 > 2.50$. Borrow 49.50 for one month to buy the put and the stock. At maturity, if $S_T < 50$ exercise the put and receive 50; otherwise sell the stock for more than 50. The loan costs $49.50e^{0.005} = 49.75$, leaving at least 0.25.

Exercise 2.7

(H 11.15) An American call on a non-dividend stock costs 4, $S = 31$, $K = 30$, $T = 3$ months, $r = 8\%$. Find bounds for the American put with the same strike and maturity.

Solution

From $S - K \leq C - P \leq S - Ke^{-rT}$: $1 \leq 4 - P \leq 31 - 30e^{-0.02} = 1.594$. Hence $2.41 \leq P \leq 3$.

Exercise 2.8

(H 11.25) Let c_1, c_2, c_3 be European calls with strikes $K_1 < K_2 < K_3$, equally spaced, and the same maturity. Show that $c_2 \leq \frac{1}{2}(c_1 + c_3)$.

Solution

The butterfly “long K_1 , long K_3 , short two K_2 ” has a non-negative payoff (the tent-shaped payoff of the static strategies above). By no arbitrage its price $c_1 + c_3 - 2c_2 \geq 0$. Call prices are therefore convex in the strike — a fact used by Breeden–Litzenberger in Chapter 7.

Exercise 2.9

(H 12.5) Puts with strikes 30 and 35 cost 4 and 7. Build (a) a bull spread and (b) a bear spread with them, and tabulate payoff and profit.

Solution

(a) Bull spread: buy the 30 put, sell the 35 put — an initial **credit** of 3. Payoff: 0 if $S_T \geq 35$, $-(35 - S_T)$ if $30 < S_T < 35$, -5 if $S_T \leq 30$; profit 3, $S_T - 32$, -2 . (b) Bear spread: buy the 35 put, sell the 30 put, cost 3. Payoff 0, $35 - S_T$, 5; profit -3 , $32 - S_T$, 2.

Exercise 2.10

(B 9.1) A contract pays K if $S_T \leq A$, $K + A - S_T$ if $A < S_T < K + A$, and 0 if $S_T \geq K + A$. Find a static replicating portfolio of bonds and calls, and its price in the Black–Scholes model.

Solution

$X = K - (S_T - A)^+ + (S_T - K - A)^+$ (check the three regions). Replicate with K zero-coupon bonds, a short call struck at A and a long call struck at $K + A$. Price: $Ke^{-r(T-t)} - C(t, S_t; A) + C(t, S_t; K + A)$, with C the Black–Scholes call price. The replication is model-free; only the final price uses Black–Scholes.

2.10. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 2 (slides_2), and the handwritten lecture notes of 14 October 2025 (law of one price, forward and futures prices, exotic payoffs, no-arbitrage bounds, static strategies). Hull, *Options, Futures, and Other Derivatives*, Sections 1.6–1.9, Chapter 2, Chapter 5 (cost of carry), 10.1–10.4, 11.2–11.6, Chapter 12 and Sections 26.7–26.13.

3. The binomial model

No-arbitrage arguments alone give bounds on option prices, not the prices themselves. To compute prices we need a model of how the underlying moves. The binomial model is the simplest one: in each period the stock goes either up or down. Despite — and because of — its simplicity (“extreme complication is contrary to art”, Debussy), it displays every key idea of quantitative finance: arbitrage, hedging, pricing by replication, the risk-neutral probability, martingales, and the optimal exercise of American options. In the limit of many short periods it becomes the Black–Scholes model of Chapter 5.

3.1. The one-period model

Time has two dates: $t = 0$ (today, when strategies are set up and prices computed) and $t = 1$ (maturity, when payoffs are received). With a single period the distinction between European and American options disappears.

Definition 3.1 (One-period binomial market). Two assets are traded:

- a **risk-free asset** (bank account) with $B_0 = 1$ and $B_1 = r_f$, where $r_f > 0$ is the gross risk-free return over the period;
- a **risky asset** (stock) with $S_0 = s > 0$ and $S_1 = sZ$, where the random return Z takes the value u (“up”) with probability p_u and d (“down”) with probability $p_d = 1 - p_u$, with $d < u$ and $p_u, p_d > 0$.

A **portfolio** is a pair $\theta = (\theta^B, \theta^S) \in \mathbb{R}^2$ of units of bank account and stock. Its value is

$$V_0(\theta) = \theta^B + \theta^S s, \quad V_1(\theta) = \theta^B r_f + \theta^S s Z = \begin{cases} \theta^B r_f + \theta^S s u & \text{if } Z = u, \\ \theta^B r_f + \theta^S s d & \text{if } Z = d. \end{cases}$$

$V_0(\theta)$ is the cost of setting up θ , $V_1(\theta)$ its payoff. Allowing any $\theta \in \mathbb{R}^2$ means ideal markets: unlimited short selling, perfect liquidity, no transaction costs.

3.1.1. Absence of arbitrage

Definition 3.2 (Arbitrage). A portfolio θ is an **arbitrage** if $V_0(\theta) = 0$, $\mathbb{P}(V_1(\theta) \geq 0) = 1$ and $\mathbb{P}(V_1(\theta) > 0) > 0$: zero cost, no loss in either scenario, a profit in at least one.

Theorem 3.1 (No arbitrage in one period). *The one-period binomial model is free of arbitrage if and only if $d < r_f < u$.*

Proof. (Necessity.) If $r_f \leq d$, borrow s and buy the stock, $\theta = (-s, 1)$: $V_0 = 0$ and $V_1 = s(Z - r_f) \geq s(d - r_f) \geq 0$, strictly positive if $Z = u$ — an arbitrage. If $r_f \geq u$, the opposite portfolio $(s, -1)$ is an arbitrage. (Sufficiency.) Let $d < r_f < u$ and suppose θ is an arbitrage. From $V_0 = 0$, $\theta^B = -\theta^S s$, so $V_1 = \theta^S s(Z - r_f)$. If $\theta^S > 0$, then $V_1 < 0$ when $Z = d$; if $\theta^S < 0$, then $V_1 < 0$ when $Z = u$; if $\theta^S = 0$, then $V_1 = 0$. None is an arbitrage. $\square$

3. The binomial model

Intuition: no asset may dominate the other in both scenarios.

3.1.2. The pricing problem

The only source of randomness is Z , so every derivative has a payoff $g(Z)$ with $g : \{u, d\} \rightarrow \mathbb{R}$. Its price at maturity is obviously $\Pi_1(g) = g(Z)$. What is $\Pi_0(g)$?

Example 3.1 (The naïve price is wrong)

Let $s = 100$, $r_f = 1$, $u = 1.2$, $d = 0.8$, $p_u = 0.6$, and consider a call with strike $K = 110$: it pays 10 if $Z = u$ and 0 if $Z = d$. The “expected payoff” $0.6 \times 10 = 6$ looks like a natural price. It is not: sell the call at 6, buy 0.25 shares and borrow 19. The cost is $-6 + 25 - 19 = 0$, and at maturity

$$-(100Z - 110)^+ + 25Z - 19 = \begin{cases} -10 + 30 - 19 = 1 & Z = 1.2, \\ 0 + 20 - 19 = 1 & Z = 0.8. \end{cases}$$

A sure profit of 1 from nothing: an arbitrage. The price 6 is too high.

3.1.3. Hedging and pricing by replication

Definition 3.3 (Hedging (replication) problem). Given a payoff $g(Z)$, find a portfolio θ with $V_1(\theta) = g(Z)$ almost surely, i.e.

$$\theta^B r_f + \theta^S s u = g(u), \quad \theta^B r_f + \theta^S s d = g(d).$$

Since $u \neq d$ the linear system has the unique solution

$$\theta^S = \frac{g(u) - g(d)}{s(u - d)}, \quad \theta^B = \frac{1}{r_f} \frac{u g(d) - d g(u)}{u - d}.$$

Every payoff can be replicated: the market is **complete**.

Theorem 3.2 (Pricing by hedging). Under no arbitrage the price of the derivative with payoff $g(Z)$ is the cost of its hedging portfolio:

$$\Pi_0(g) = V_0(\theta) = \frac{1}{r_f} \frac{u g(d) - d g(u)}{u - d} + \frac{g(u) - g(d)}{u - d}.$$

Proof. Derivative and hedge have the same payoff, so this is the law of one price; explicitly, if $\Pi_0(g) < V_0(\theta)$, buy the derivative, sell θ^S shares and invest $\theta^S s - \Pi_0(g)$ in the bank. The cost is zero and the payoff is

$$g(Z) + (\theta^S s - \Pi_0(g))r_f - \theta^S s Z = g(Z) - V_1(\theta) + (V_0(\theta) - \Pi_0(g))r_f = (V_0(\theta) - \Pi_0(g))r_f > 0,$$

an arbitrage; the reverse inequality is excluded symmetrically. $\square$

In the example the formula gives $\Pi_0 = 5$, the unique arbitrage-free price.

Proposition 3.3 (Delta hedging). $\theta^S = \Delta := \frac{g(u) - g(d)}{su - sd}$ is the number of shares that makes the position “short one derivative, long Δ shares” riskless: $\Delta s Z - g(Z)$ takes the same value in both scenarios.

Δ is the sensitivity of the derivative’s payoff to the underlying — the discrete ancestor of the Black–Scholes delta. Pricing can also be derived from this riskless position: if $\Delta s Z - g(Z) = c$ for both Z , its value today must be c/r_f , so $\Pi_0(g) = \Delta s - c/r_f$.

3.1.4. The risk-neutral probability

Rearranging the pricing formula,

$$\Pi_0(g) = \frac{1}{r_f} \left(\underbrace{\frac{r_f - d}{u - d}}_{q_u} g(u) + \underbrace{\frac{u - r_f}{u - d}}_{q_d} g(d) \right).$$

Under no arbitrage ($d < r_f < u$), $q_u, q_d > 0$ and $q_u + q_d = 1$: the pair $Q = (q_u, q_d)$ is a probability on $\{Z = u\}, \{Z = d\}$, the **risk-neutral probability**.

Theorem 3.4 (Risk-neutral valuation).

$$\Pi_0(g) = \frac{1}{r_f} \mathbb{E}^Q[g(Z)] = \frac{1}{r_f} (q_u g(u) + q_d g(d)).$$

Remarks that will recur throughout the course:

- The price is a discounted expectation — what a risk-neutral investor would pay — but under an **artificial** probability Q . The real probabilities p_u, p_d do **not** enter: they only affect the expected return, and the hedge removes all risk.
- With $g(x) = sx$, $\mathbb{E}^Q[S_1] = s(q_u u + q_d d) = sr_f$, i.e. $\frac{S_0}{B_0} = \mathbb{E}^Q[\frac{S_1}{B_1}]$: under Q the stock earns the risk-free rate on average — **no risk premium** — and the discounted price has constant expectation (it is a *martingale*).
- More generally, for any portfolio $\mathbb{E}^Q[V_1(\theta)] = \theta^B r_f + \theta^S s \mathbb{E}^Q[Z] = V_0(\theta) r_f$.

Example 3.2 (*The call, priced twice*)

With $s = 100$, $r_f = 1$, $u = 1.2$, $d = 0.8$, $K = 110$: the hedge is $\theta^S = \frac{10-0}{120-80} = \frac{1}{4}$, $\theta^B = \frac{1.2 \times 0 - 0.8 \times 10}{0.4} = -20$ (check: $-20 + 30 = 10$ and $-20 + 20 = 0$), so $\Pi_0 = -20 + 25 = 5$. With risk-neutral probabilities $q_u = \frac{1-0.8}{0.4} = \frac{1}{2} = q_d$, $\Pi_0 = \frac{1}{2} \times 10 = 5$. Risk-neutral valuation is more direct but tells nothing about the hedge.

3.2. A general one-period market

The binomial model has two states and two assets. What survives with more of each? Let there be N assets with prices $S_0 \in \mathbb{R}^N$ today, and a finite set of M states $\omega_1, \dots, \omega_M$, each with positive probability, in which the assets are worth $S_1(\omega_j)$. Collect the payoffs in the $N \times M$ **payoff matrix** $D = [S_1^i(\omega_j)]$. Asset 1 is the bank account, $S_0^1 = 1$, $S_1^1 = r_f$. A portfolio $h \in \mathbb{R}^N$ costs $h \cdot S_0$ and pays $h^\top D$ (a vector over states).

Definition 3.4 (Arbitrage and state prices). h is an **arbitrage** if $h \cdot S_0 \leq 0$ and $h^\top D \geq 0$ with at least one strict inequality. A vector $\psi \in \mathbb{R}^M$ with $\psi_j > 0$ for all j and

$$S_0 = D \psi$$

is a vector of **state prices**: ψ_j is the price today of an elementary (Arrow–Debreu) security paying 1 in state ω_j and 0 otherwise.

Theorem 3.5 (First fundamental theorem (one period)). *The market is free of arbitrage if and only if strictly positive state prices exist. Equivalently, iff there is a probability Q with $Q(\omega_j) > 0$ for all j such*

3. The binomial model

that

$$S_0^i = \frac{1}{r_f} \mathbb{E}^Q[S_1^i], \quad i = 1, \dots, N.$$

Proof sketch. () If $S_0 = D\psi$ with $\psi > 0$, any h with $h^\top D \geq 0$, not identically zero, has cost $h \cdot S_0 = (h^\top D)\psi > 0$: no arbitrage. () No arbitrage says that the linear subspace $\{(-h \cdot S_0, h^\top D) : h \in \mathbb{R}^N\} \subset \mathbb{R}^{1+N}$ meets the positive orthant only at 0. A separating-hyperplane argument (Farkas' lemma) yields a strictly positive vector orthogonal to that subspace, which, normalised, is $(1, \psi)$. Finally, setting $Q(\omega_j) = r_f \psi_j$: the bank account equation $1 = r_f \sum_j \psi_j$ shows that Q is a probability, and $S_0 = D\psi$ becomes the risk-neutral pricing formula. $\square$

Theorem 3.6 (Second fundamental theorem (one period)). *Assume no arbitrage. The market is **complete** — every payoff $X \in \mathbb{R}^M$ is the payoff of some portfolio — iff $\text{rank}(D) = M$, iff the risk-neutral probability Q is **unique**.*

Completeness means that the rows of D span $\mathbb{R}^M$: one needs at least as many independent assets as states. In the binomial model $N = M = 2$ and $D = \begin{pmatrix} r_f & r_f \\ su & sd \end{pmatrix}$ has rank two since $u \neq d$.

Example 3.3 (An incomplete trinomial market)

Let $r_f = 1$, $S_0 = 100$ and $S_1 \in \{120, 100, 80\}$. Risk-neutral probabilities must satisfy $q_u + q_m + q_d = 1$ and $120q_u + 100q_m + 80q_d = 100$, i.e. $q_u = q_d$, with $q_u \in (0, \frac{1}{2})$: infinitely many. A call with strike 100 pays $(20, 0, 0)$ and would be priced $20q_u$, anywhere in $(0, 10)$. No arbitrage pins down only an **interval** of prices, whose endpoints are the costs of the cheapest super-replicating and the most expensive sub-replicating portfolios: here buying $\frac{1}{2}$ share and borrowing 40 super-replicates the call (payoff $(20, 10, 0) \geq (20, 0, 0)$) at cost 10. Choosing a single price requires extra criteria (preferences, calibration to other traded options) — the situation of the stochastic volatility models of Chapter 7.

3.3. The multi-period model: a two-period example

Rebalancing over time makes the model both more realistic and dynamic. Take two periods, $B_t = r_f^t$, $S_1 = sZ_1$, $S_2 = S_1Z_2$, with Z_1, Z_2 i.i.d. as before. There are four paths: uu , ud , du , dd with probabilities $p_u^2, p_u p_d, p_d p_u, p_d^2$; the middle two lead to the same price sud — the tree **recombines**.

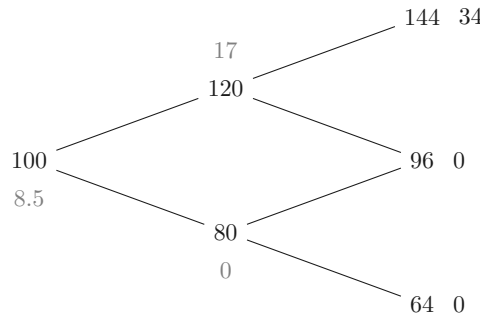


Figure: stock prices ($s = 100$, $u = 1.2$, $d = 0.8$) with, in grey, the value of a call with $K = 110$ at each node ($r_f = 1$); terminal payoffs on the right.

Definition 3.5 (Dynamic strategy and self-financing condition). A dynamic strategy $\theta = (\theta_0, \theta_1)$ consists of a portfolio $\theta_0 = (\theta_0^B, \theta_0^S)$ set up at $t = 0$ and a portfolio $\theta_1(Z_1) = (\theta_1^B(Z_1), \theta_1^S(Z_1))$ chosen at $t = 1$

after observing Z_1 . Its values are $V_0 = \theta_0^B B_0 + \theta_0^S S_0$; $V_1 = \theta_0^B B_1 + \theta_0^S S_1$ (what the initial portfolio is worth at $t = 1$); $V_{1+} = \theta_1^B B_1 + \theta_1^S S_1$ (the cost of the rebalanced portfolio); $V_2 = \theta_1^B B_2 + \theta_1^S S_2$. The strategy is **self-financing** if $V_{1+}(\theta) = V_1(\theta)$: rebalancing neither requires new capital nor withdraws any.

Hedging the call by backward induction. With $K = 110$ the terminal payoffs are 34, 0, 0, 0.

- *Node $S_1 = 120$.* Solve $\theta_1^B + 144\theta_1^S = 34$, $\theta_1^B + 96\theta_1^S = 0$: $\theta_1^S(u) = 0.7083$, $\theta_1^B(u) = -68$. Value at $t = 1$: $-68 + 0.7083 \times 120 = 17$.
- *Node $S_1 = 80$.* The call will surely expire worthless: $\theta_1(d) = (0, 0)$, value 0.
- *Node $S_0 = 100$.* Self-financing requires θ_0 to deliver exactly the cost of the rebalanced portfolio: $\theta_0^B + 120\theta_0^S = 17$, $\theta_0^B + 80\theta_0^S = 0$, so $\theta_0^S = 0.425$, $\theta_0^B = -34$, and $V_0 = -34 + 42.5 = 8.5$.

Starting with 8.5, holding 0.425 shares financed by borrowing 34, and rebalancing at zero cost at $t = 1$, we replicate the call payoff in all four scenarios. By no arbitrage the call price is $\Pi_0 = 8.5$. Risk-neutral valuation gives the same: $q_u = q_d = \frac{1}{2}$ and $\Pi_0 = q_u^2 \times 34 = 8.5$.

3.4. The general multi-period model

Let $B_t = r_f^t$ and $S_t = s \prod_{i=1}^t Z_i$ for $t = 0, \dots, N$, with $Z_1, \dots, Z_N$ i.i.d. taking values $u > d$ (independence under $\mathbb{P}$ is convenient but not essential).

Definition 3.6 (Filtration). The information available at time t is generated by $Z_1, \dots, Z_t$: $\mathcal{F}_t = \sigma(Z_1, \dots, Z_t)$, with $\mathcal{F}_0$ trivial. The family $\mathbb{F} = (\mathcal{F}_t)_{t=0, \dots, N}$ is increasing ($\mathcal{F}_{t-1} \subseteq \mathcal{F}_t$): information accumulates. A random variable is $\mathcal{F}_t$ -measurable if it is determined by $Z_1, \dots, Z_t$.

Definition 3.7 (Self-financing strategy). A dynamic strategy $\theta = (\theta_0, \dots, \theta_{N-1})$ has $\theta_t = (\theta_t^B, \theta_t^S)$ $\mathcal{F}_t$ -measurable (chosen with the information at t , held over $[t, t+1]$). It is **self-financing** if

$$\theta_{t-1}^B B_t + \theta_{t-1}^S S_t = \theta_t^B B_t + \theta_t^S S_t, \quad t = 1, \dots, N-1,$$

and its value process is $V_t(\theta) = \theta_t^B B_t + \theta_t^S S_t$ (with $V_N = \theta_{N-1}^B B_N + \theta_{N-1}^S S_N$). It is an **arbitrage** if $V_0(\theta) = 0$, $V_N(\theta) \geq 0$ a.s. and $\mathbb{P}(V_N(\theta) > 0) > 0$.

Theorem 3.7 (No arbitrage and completeness). *The N -period model is free of arbitrage iff $d < r_f < u$ — iff every one-period sub-tree is. In that case every payoff $g(Z_1, \dots, Z_N)$, $g : \{u, d\}^N \rightarrow \mathbb{R}$, can be replicated by a self-financing strategy, and its arbitrage-free price is $\Pi_t(g) = V_t(\theta)$.*

Construction of the hedge. For a path $(w_1, \dots, w_N) \in \{u, d\}^N$ set $V_N(w_1, \dots, w_N) = g(w_1, \dots, w_N)$ and, going backwards, for $t = N-1, \dots, 0$,

$$\theta_t^S(w_1, \dots, w_t) = \frac{V_{t+1}(w_1, \dots, w_t, u) - V_{t+1}(w_1, \dots, w_t, d)}{s \prod_{i=1}^t w_i (u - d)}, \quad \theta_t^B(w_1, \dots, w_t) = \frac{1}{r_f^{t+1}} \frac{u V_{t+1}(\dots, d) - d V_{t+1}(\dots, u)}{u - d},$$

$$V_t(w_1, \dots, w_t) = \theta_t^B B_t + \theta_t^S S_t.$$

By construction θ_t depends only on the first t returns (it is **adapted**), and one checks directly that $\theta_t^B B_{t+1} + \theta_t^S S_{t+1} = V_{t+1}(w_1, \dots, w_t, Z_{t+1})$ in both cases $Z_{t+1} = u, d$: the strategy is self-financing and ends at $V_N = g$. Each step is a one-period hedging problem.

3. The binomial model

3.4.1. Risk-neutral valuation and martingales

Let $q_u = \frac{r_f - d}{u - d}$, $q_d = \frac{u - r_f}{u - d}$, and let Q be the probability under which $Z_1, \dots, Z_N$ are i.i.d. with $Q(Z_t = u) = q_u$.

Theorem 3.8 (Martingale property of discounted prices). *Under Q , for every $t = 0, \dots, N - 1$,*

$$\frac{S_t}{B_t} = \mathbb{E}^Q \left[\frac{S_{t+1}}{B_{t+1}} \mid \mathcal{F}_t \right], \quad \frac{V_t(\theta)}{B_t} = \mathbb{E}^Q \left[\frac{V_{t+1}(\theta)}{B_{t+1}} \mid \mathcal{F}_t \right]$$

for every self-financing strategy θ : discounted prices and discounted self-financing portfolios are Q -martingales.

Proof. S_t is $\mathcal{F}_t$ -measurable and Z_{t+1} is independent of $\mathcal{F}_t$ under Q , so

$$\mathbb{E}^Q[S_{t+1} \mid \mathcal{F}_t] = S_t \mathbb{E}^Q[Z_{t+1} \mid \mathcal{F}_t] = S_t \mathbb{E}^Q[Z_{t+1}] = S_t(q_u u + q_d d) = S_t r_f.$$

Dividing by $B_{t+1} = B_t r_f$ gives the first claim. For a portfolio, using that θ_t^B, θ_t^S are $\mathcal{F}_t$ -measurable and then self-financing,

$$\mathbb{E}^Q \left[\frac{V_{t+1}}{B_{t+1}} \mid \mathcal{F}_t \right] = \mathbb{E}^Q \left[\theta_t^B + \theta_t^S \frac{S_{t+1}}{B_{t+1}} \mid \mathcal{F}_t \right] = \theta_t^B + \theta_t^S \frac{S_t}{B_t} = \frac{V_t}{B_t}. \quad \square$$

Theorem 3.9 (Risk-neutral valuation formula). *For every payoff $g(Z_1, \dots, Z_N)$ and every t ,*

$$\Pi_t(g) = \frac{1}{r_f^{N-t}} \mathbb{E}^Q[g(Z_1, \dots, Z_N) \mid \mathcal{F}_t].$$

Proof. Let θ replicate g ; then $\Pi_t(g) = V_t(\theta)$, and iterating the martingale property with the tower rule,

$$V_t = \frac{1}{r_f} \mathbb{E}^Q[V_{t+1} \mid \mathcal{F}_t] = \frac{1}{r_f^2} \mathbb{E}^Q[\mathbb{E}^Q[V_{t+2} \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \dots = \frac{1}{r_f^{N-t}} \mathbb{E}^Q[V_N \mid \mathcal{F}_t]. \quad \square$$

Discrete-time versions of the **fundamental theorems of asset pricing** are hidden here: no arbitrage existence of a probability Q (equivalent to $\mathbb{P}$) making discounted prices martingales; completeness Q is unique.

3.4.2. Path-independent payoffs

A payoff is **path-independent** if it depends only on the terminal price $S_N = s \prod_i Z_i$ — i.e. on the *number* of up moves, not their order (calls, puts). Otherwise it is **path-dependent** (Asian, lookback, barrier options). Under Q the number of up moves is binomial, so for $g = G(S_N)$

$$\Pi_0(G) = \frac{1}{r_f^N} \sum_{n=0}^N \binom{N}{n} q_u^n q_d^{N-n} G(su^n d^{N-n}), \quad \Pi_0^{\text{Call}}(K) = \frac{1}{r_f^N} \sum_{n=0}^N \binom{N}{n} q_u^n q_d^{N-n} (su^n d^{N-n} - K)^+.$$

Path-dependent payoffs require working on the full (non-recombining) tree of 2^N paths, or Monte Carlo (Chapter 6).

3.5. Calibration: choosing u and d

In practice the maturity T is split into N steps of length $\Delta t = T/N$ ($N \geq 30$ for a realistic description), with $r_f = e^{R\Delta t}$ for a continuously compounded rate R . The parameters u, d are chosen to match the **volatility**

σ of the stock: the standard deviation of the return over Δt should be $\sigma\sqrt{\Delta t}$, i.e.

$$\text{Var}^Q(Z) = \mathbb{E}^Q[Z^2] - \mathbb{E}^Q[Z]^2 = q_u u^2 + q_d d^2 - r_f^2 = \sigma^2 \Delta t.$$

Substituting q_u, q_d gives $e^{R\Delta t}(u + d) - ud - e^{2R\Delta t} = \sigma^2 \Delta t$. Imposing in addition $u = 1/d$ (a recombining, symmetric tree),

$$e^{R\Delta t}\left(u + \frac{1}{u}\right) - 1 - e^{2R\Delta t} = \sigma^2 \Delta t.$$

Proposition 3.10 (Cox–Ross–Rubinstein parameters (1979)). *The choice $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$ solves the calibration equation up to terms of order higher than Δt .*

Proof. With $e^x \approx 1 + x + x^2/2$, $u + 1/u \approx 2 + \sigma^2 \Delta t$ and $e^{2R\Delta t} \approx 1 + 2R\Delta t$. The left side becomes $(1 + R\Delta t)(2 + \sigma^2 \Delta t) - 1 - 1 - 2R\Delta t = \sigma^2 \Delta t + O(\Delta t^2)$. $\square$

The same u, d also approximately match the volatility under the real-world probability with $\mathbb{E}^{\mathbb{P}}[Z] = e^{\mu\Delta t}$ (exercise): volatility is the same under $\mathbb{P}$ and Q , only the drift changes. As $N \rightarrow \infty$ the CRR price of a European call converges to the Black–Scholes price (Chapter 5): by the central limit theorem, $\log S_N$ becomes Gaussian.

3.6. American options

An American option can be exercised at any $t \in \{0, \dots, N\}$. At each node the holder compares

- the **intrinsic value** — the payoff from exercising now, $g(S_t)$;
- the **continuation value** — the value of keeping the option alive, $\frac{1}{r_f} \mathbb{E}^Q[V_{t+1} \mid \mathcal{F}_t]$.

The option value is the larger of the two,

$$V_t = \max\left(g(S_t), \frac{1}{r_f}(q_u V_{t+1}^u + q_d V_{t+1}^d)\right), \quad V_N = g(S_N),$$

and it is optimal to **exercise at the first time the continuation value falls below the intrinsic value**. The optimal exercise time is random (a *stopping time*): it depends on the path.

Example 3.4 (An American put)

$s = 100$, $u = 1.2$, $d = 0.8$, $r_f = 1.05$, $K = 110$, two periods. Exercising gives 10 at $t = 0$; 0 at node $S_1 = 120$ and 30 at $S_1 = 80$; and 0, 14, 14, 46 at the terminal nodes.

- *Node $S_1 = 120$.* Hedging the terminal payoffs (0, 14): $\theta_1^B \times 1.1025 + 144\theta_1^S = 0$, $\theta_1^B \times 1.1025 + 96\theta_1^S = 14$, so $\theta_1^S = -0.2917$, $\theta_1^B = 38.10$, continuation value $38.10 \times 1.05 - 0.2917 \times 120 = 5$. Intrinsic value 0: **wait**.
- *Node $S_1 = 80$.* Hedging (14, 46): $\theta_1^S = -1$, $\theta_1^B = 99.77$, continuation value $99.77 \times 1.05 - 80 = 24.76$. Intrinsic value 30 > 24.76: **exercise**.
- *Node $S_0 = 100$.* The hedge must deliver 5 if up and 30 if down: $\theta_0^S = -0.625$, $\theta_0^B = 76.19$, so $V_0 = 76.19 - 62.5 = 13.69 > 10$: do not exercise at $t = 0$.

The arbitrage-free price is $\Pi_0^{AP} = 13.69$ (risk-neutral check: $q_u = \frac{1.05-0.8}{0.4} = 0.625$ and $(0.625 \times 5 + 0.375 \times 30)/1.05 = 13.69$). The optimal exercise time is $\tau^* = 1$ after a down move and $\tau^* = 2$ after an up move. The corresponding European put is worth only 11.82 (exercise below): the difference, 1.87, is the **early-exercise premium**.

3. The binomial model

In continuous time the same logic defines, for the American put, a deterministic **exercise boundary** $S^*(t)$: it is optimal to exercise as soon as $S_t \leq S^*(t)$; the boundary increases towards K as $t \rightarrow T$. The binomial tree remains the standard numerical method to compute it. For calls on non-dividend stocks, the continuation value always exceeds the intrinsic value, in agreement with Chapter 2.

3.7. Exercises

Problems from Hull (H), Chapter 13, and Björk (B), Chapters 2–3. Rates are continuously compounded unless stated.

Exercise 3.1

(H 13.4) $S_0 = 50$; in 2 months the stock will be 53 or 48; $r = 10\%$. Value a 2-month European call with $K = 49$ by a no-arbitrage (hedging) argument.

Solution

Payoffs 4 and 0. A position long Δ shares and short one call is riskless if $53\Delta - 4 = 48\Delta$, i.e. $\Delta = 0.8$, worth 38.4 in both states. Its value today is $38.4e^{-0.1/6} = 37.765$, so $50 \times 0.8 - c = 37.765$ and $c = 2.23$. (Check: $q = (50e^{0.1/6} - 48)/5 = 0.568$ and $c = e^{-0.1/6} \times 0.568 \times 4 = 2.23$.)

Exercise 3.2

(H 13.9) $S_0 = 25$; in 2 months the stock will be 23 or 27; $r = 10\%$. Value a derivative paying S_T^2 at that time.

Solution

$q = (25e^{0.1/6} - 23)/4 = 0.6050$. Value $= e^{-0.1/6}(0.6050 \times 729 + 0.3950 \times 529) = 639.26$. Note that $639.26 > S_0^2 = 625$: by convexity, $\mathbb{E}^Q[S_T^2] > (\mathbb{E}^Q S_T)^2$.

Exercise 3.3

(H 13.1, 13.2) $S_0 = 100$; over each of two 6-month periods the stock moves up or down by 10%; $r = 8\%$. Value 1-year European call and put options with $K = 100$ and verify put–call parity.

Solution

$q = (e^{0.04} - 0.9)/0.2 = 0.7041$. Terminal prices 121, 99, 81. Call: only 121 pays 21: $c = e^{-0.08}q^2 \times 21 = 9.61$. Put: 99 pays 1, 81 pays 19: $p = e^{-0.08}(2q(1-q) \times 1 + (1-q)^2 \times 19) = 1.92$. Parity: $c - p = 7.69 = 100 - 100e^{-0.08}$.

Exercise 3.4

(H 13.14) $S_0 = 140$, $\sigma = 25\%$, $r = 4\%$, 3-month steps. Find u , d and the risk-neutral up-probability of the CRR tree, and value 6-month European call and put options with $K = 150$ on a two-step tree.

Solution

$u = e^{0.25\sqrt{0.25}} = 1.1331$ (a 13.3% rise), $d = 1/u = 0.8825$ (an 11.75% fall), $q = (e^{0.01} - d)/(u - d) = 0.5089$. Terminal prices 179.76, 140, 109.03. Call: $e^{-0.02}q^2(29.76) = 7.56$. Put: $e^{-0.02}(2q(1-q) \times 10 + (1-q)^2 \times 40.97) = 14.58$. Parity check: $7.56 - 14.58 = 140 - 150e^{-0.02}$.

Exercise 3.5

(H 13.17) $S_0 = 40$; over each of two 3-month periods the stock moves up or down by 10%; $r = 12\%$. Value 6-month European and American puts with $K = 42$.

Solution

$q = (e^{0.03} - 0.9)/0.2 = 0.6523$. Terminal prices 48.4, 39.6, 32.4 with put payoffs 0, 2.4, 9.6. European: $e^{-0.06}(2q(1-q) \times 2.4 + (1-q)^2 \times 9.6) = 2.12$. American: at $S = 44$ continuation 0.81 > 0 (wait); at $S = 36$ continuation 4.76 < exercise value 6 (exercise). At the root $e^{-0.03}(q \times 0.81 + (1-q) \times 6) = 2.54 > 2$, so wait. The American put is worth 2.54; early-exercise premium 0.42.

Exercise 3.6

(Adapted from B 2.3) In the two-period example of this chapter ($s = 100$, $u = 1.2$, $d = 0.8$, $r_f = 1$, call with $K = 110$), suppose that at $t = 1$, after an up move, the call trades at 20 instead of its arbitrage-free value 17. Construct an arbitrage.

Solution

Sell the call for 20 and set up its replicating portfolio from the node $S_1 = 120$ (0.7083 shares, borrowing 68, cost 17), keeping 3. At $t = 2$ the portfolio pays exactly the call's payoff, so the obligation is covered, and 3 is a riskless profit.

Exercise 3.7

(B 3, finite markets) In the trinomial market of the general one-period section, show that the call with strike 100 cannot be replicated, and that a put with strike 100 (payoff $(0, 0, 20)$) has arbitrage-free prices in $(0, 10)$. Find the unique price if, in addition, the call trades at 6.

Solution

Replicating $(20, 0, 0)$ with bonds and stock requires $a + 120b = 20$, $a + 100b = 0$, $a + 80b = 0$, which is inconsistent: the payoff is not spanned (rank $D = 2 < 3$). The put is priced $20q_d = 20q_u \in (0, 10)$. If the call trades at 6, then $20q_u = 6$ fixes $q_u = q_d = 0.3$ and $q_m = 0.4$: the put is worth 6 too. With the call added as a third asset the market becomes complete.

Exercise 3.8

(B 2.4) Prove the multi-period risk-neutral valuation formula by induction on the number of periods N , using the one-period result.

Solution

For $N = 1$ it is the one-period theorem. Suppose it holds for $N - 1$ periods. Conditionally on Z_1 , the remaining problem is an $(N - 1)$ -period model, so the value at $t = 1$ is $V_1 = r_f^{-(N-1)} \mathbb{E}^Q[g \mid \mathcal{F}_1]$, a function of Z_1 only. Pricing this one-period payoff gives $V_0 = r_f^{-1} \mathbb{E}^Q[V_1] = r_f^{-N} \mathbb{E}^Q[\mathbb{E}^Q[g \mid \mathcal{F}_1]] = r_f^{-N} \mathbb{E}^Q[g]$ by the tower property.

3.8. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 3 (`slides_3`), and the handwritten lecture notes of 21 October 2025 (delta hedging, risk-neutral probability, two-period example, self-financing strategies, martingale property, CRR calibration). Björk, *Arbitrage Theory in Continuous Time*, Chapters 2–3; Turner and Zeindler, *Stochastic Finance*, Chapters 3–4 (finite market models and the fundamental theorems); Hull, *Options, Futures, and Other Derivatives*, Chapter 13. Further reading: H. Föllmer and A. Schied, *Stochastic Finance: An Introduction in Discrete Time*; S. Shreve, *Stochastic Calculus for Finance I*. Original reference: J. Cox, S. Ross and M. Rubinstein, J. Financial Econ. **7**, 229 (1979).

Part II.

Part II: Continuous-time models

4. Brownian motion and stochastic calculus

What happens to the binomial model when the number of periods goes to infinity? Time becomes continuous, and the random walk of the stock becomes a process driven by **Brownian motion** (“the task of art today is to bring chaos into order”, Adorno). Brownian paths are continuous but nowhere differentiable, so ordinary calculus fails: to write differential equations driven by noise we need the **Itô integral** and its chain rule, **Itô’s formula**. This chapter builds these tools, which underlie the Black–Scholes model of Chapter 5.

4.1. Filtrations, processes, martingales

From now on time runs continuously over $[0, T]$.

Definition 4.1 (Filtration and adapted process). A **filtration** $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ is an increasing family of σ -algebras ($\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$): the information available at each time. A **stochastic process** $X = (X_t)_{t \in [0, T]}$ is a family of random variables indexed by time; it is **adapted** if X_t is $\mathcal{F}_t$ -measurable for every t .

Recall the rules of conditional expectation: if ξ is independent of $\mathcal{F}_t$ then $\mathbb{E}[\xi | \mathcal{F}_t] = \mathbb{E}[\xi]$ (the information is useless); if ξ is $\mathcal{F}_t$ -measurable then $\mathbb{E}[\xi | \mathcal{F}_t] = \xi$; and for $s \leq t$ the **tower property** $\mathbb{E}[\mathbb{E}[\xi | \mathcal{F}_t] | \mathcal{F}_s] = \mathbb{E}[\xi | \mathcal{F}_s]$ holds.

Definition 4.2 (Martingale and Markov process). An adapted process X is a **martingale** if $\mathbb{E}[X_t | \mathcal{F}_s] = X_s$ for all $s \leq t$ — constant on average, a “fair game”. It is a **Markov process** if $\mathbb{E}[f(X_t) | \mathcal{F}_s] = \mathbb{E}[f(X_t) | X_s]$ for all $s \leq t$ and bounded f : the future depends on the past only through the present value.

4.2. Brownian motion

Definition 4.3 (Brownian motion). An adapted process $W = (W_t)_{t \geq 0}$ with $W_0 = 0$ is a (standard) **Brownian motion** if

1. $W_t - W_s$ is independent of $\mathcal{F}_s$ for all $s \leq t$ (independent increments);
2. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for all $s \leq t$ (stationary Gaussian increments);
3. $t \mapsto W_t$ is continuous.

A short history: Robert Brown observes the erratic motion of pollen particles in water (1827); Louis Bachelier’s thesis *Théorie de la spéculation* (1900) uses it to model stock prices; Einstein explains it physically (1905); Wiener gives a rigorous construction (1923, hence “Wiener process”); Samuelson rediscovers Bachelier in the 1960s; Black, Scholes and Merton publish their option-pricing theory in 1973.

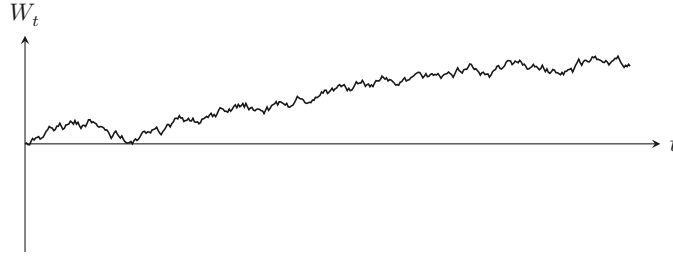


Figure: a simulated Brownian path — continuous, but rough at every scale.

Proposition 4.1 (Basic properties). For $s \leq t$: $\mathbb{E}[W_t] = 0$, $\text{Var}(W_t) = t$, $\text{Cov}(W_s, W_t) = s$, and $\mathbb{E}[e^{\lambda(W_t - W_s)}] = e^{\lambda^2(t-s)/2}$. The processes

$$W_t, \quad W_t^2 - t, \quad M_t = e^{\lambda W_t - \lambda^2 t/2}$$

are martingales, and W is a Markov process.

Proof. Covariance: write $W_t = W_s + (W_t - W_s)$; since W_s is $\mathcal{F}_s$ -measurable and the increment is independent of $\mathcal{F}_s$ with mean zero, $\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_t - W_s]\mathbb{E}[W_s] = s$.

Martingale: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s + \mathbb{E}[W_t - W_s | \mathcal{F}_s] = W_s + \mathbb{E}[W_t - W_s] = W_s$.

$W_t^2 - t$: $\mathbb{E}[W_t^2 | \mathcal{F}_s] = W_s^2 + 2W_s\mathbb{E}[W_t - W_s] + \mathbb{E}[(W_t - W_s)^2] = W_s^2 + t - s$.

Exponential martingale: $\mathbb{E}[e^{\lambda W_t} | \mathcal{F}_s] = e^{\lambda W_s} \mathbb{E}[e^{\lambda(W_t - W_s)}] = e^{\lambda W_s} e^{\lambda^2(t-s)/2}$, using the moment generating function of $\mathcal{N}(0, t-s)$; multiply by $e^{-\lambda^2 t/2}$.

Markov: for bounded f , $\mathbb{E}[f(W_t) | \mathcal{F}_s] = \mathbb{E}[f(x + W_t - W_s)]|_{x=W_s}$, a function of W_s only; taking $\mathbb{E}[\cdot | W_s]$ of both sides gives $\mathbb{E}[f(W_t) | \mathcal{F}_s] = \mathbb{E}[f(W_t) | W_s]$. $\square$

4.2.1. Nowhere differentiability and infinite variation

Brownian paths are continuous but **nowhere differentiable** — “a continuous function made entirely of corners” (think of $|x|$ at 0, everywhere). Heuristically, $W_{t+\Delta t} - W_t \sim \sqrt{\Delta t} Z$ with $Z \sim \mathcal{N}(0, 1)$, so

$$\frac{W_{t+\Delta t} - W_t}{\Delta t} \sim \frac{Z}{\sqrt{\Delta t}} \xrightarrow{\Delta t \rightarrow 0} \pm\infty.$$

Paths also have **infinite variation**: they cover an infinite length in any finite interval (proved below from the quadratic variation). Why should we care? Because we want Brownian motion to drive differential equations.

4.3. Differential equations driven by noise

Suppose a quantity evolves over a short interval Δt through a locally deterministic **drift** proportional to Δt and a Gaussian **noise** with variance proportional to Δt :

$$X_{t+\Delta t} - X_t = \alpha(t, X_t) \Delta t + \beta(t, X_t) (W_{t+\Delta t} - W_t), \quad X_0 = x.$$

Dividing by Δt and letting $\Delta t \rightarrow 0$ fails, because dW_t/dt does not exist. Instead, on a grid $t_k = k\Delta t$ sum the increments,

$$X_{t_n} = x + \sum_{k=0}^{n-1} \alpha(t_k, X_{t_k}) \Delta t + \sum_{k=0}^{n-1} \beta(t_k, X_{t_k}) \Delta W_{t_{k+1}},$$

and let $\Delta t \rightarrow 0$. The first sum converges to an ordinary (Riemann–Stieltjes) integral $\int_0^t \alpha(u, X_u) du$. The second should converge to “ $\int_0^t \beta(u, X_u) dW_u$ ”, but this cannot be defined path by path as a Riemann–Stieltjes integral, precisely because W has infinite variation. Itô (1944) solved the problem by giving up the pathwise definition and constructing the integral in the space of square-integrable random variables.

4.4. The Itô integral

Definition 4.4 (L^2 space). L^2 is the space of random variables with $\mathbb{E}[X^2] < \infty$, with norm $\|X\| = \mathbb{E}[X^2]^{1/2}$. $X_n \rightarrow X$ in L^2 (in mean square) if $\|X_n - X\| \rightarrow 0$. L^2 is **complete**: every Cauchy sequence converges to a unique limit. Moreover, L^2 limits of Gaussian random variables are Gaussian, and L^2 convergence implies $\mathbb{E}[X_n^2] \rightarrow \mathbb{E}[X^2]$ and $\mathbb{E}[X_n | \mathcal{G}] \rightarrow \mathbb{E}[X | \mathcal{G}]$ in L^2 for any σ -algebra $\mathcal{G}$.

The plan: define $\int_0^T \beta_u dW_u$ for simple integrands, prove its properties, then extend by approximation. Integrands are always adapted.

Step 0. For a deterministic indicator $\beta_t = c \mathbf{1}_{[a,b)}(t)$ the only reasonable definition is $\int_0^T \beta_u dW_u = c(W_b - W_a)$.

Definition 4.5 (Simple process and its integral). β is **simple** if there are times $0 = t_0 < t_1 < \dots < t_n = T$ such that

$$\beta_t = \sum_{i=1}^n \beta_{t_{i-1}} \mathbf{1}_{[t_{i-1}, t_i)}(t),$$

with each $\beta_{t_{i-1}}$ square-integrable and $\mathcal{F}_{t_{i-1}}$ -measurable: a piecewise-constant function of time with random heights. Its stochastic integral is

$$\int_0^T \beta_u dW_u := \sum_{i=1}^n \beta_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}),$$

and $\int_0^t \beta_u dW_u = \sum_i \beta_{t_{i-1}} (W_{t_i \wedge t} - W_{t_{i-1} \wedge t})$ for $t \leq T$.

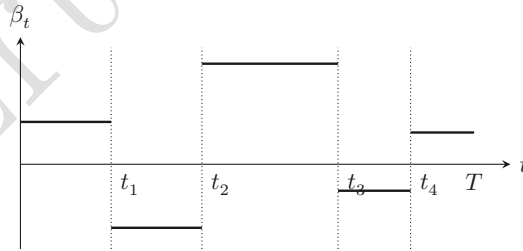


Figure: a simple integrand — constant on each interval, with a random height known at the left endpoint.

It is essential that the integrand is evaluated at the **left endpoint** of each interval, against the **forward** increment of W : the height is decided before the noise of that interval is revealed. (In finance: you choose your portfolio before prices move.)

Proposition 4.2 (Properties for simple integrands). For simple β :

1. **Linearity:** $\int (\beta + \gamma) dW = \int \beta dW + \int \gamma dW$.
2. **Martingale property:** $\mathbb{E}[\int_0^t \beta_u dW_u | \mathcal{F}_s] = \int_0^s \beta_u dW_u$ for $s \leq t$; in particular $\mathbb{E}[\int_0^T \beta_u dW_u] = 0$.

3. **Itô isometry:** $\mathbb{E}\left[\left(\int_0^T \beta_u dW_u\right)^2\right] = \mathbb{E}\left[\int_0^T \beta_u^2 du\right]$; hence $\text{Var}\left(\int_0^T \beta_u dW_u\right) = \mathbb{E}\left[\int_0^T \beta_u^2 du\right]$.
4. If β is **deterministic**, $\int_0^T \beta_u dW_u \sim \mathcal{N}(0, \int_0^T \beta_u^2 du)$.

Proof of 2. Each term contributes $\beta_{t_{i-1}}(W_{t_i \wedge t} - W_{t_{i-1} \wedge t})$. Terms entirely before s are $\mathcal{F}_s$ -measurable and stay. For the interval $[t_j, t_{j+1})$ containing s , β_{t_j} is $\mathcal{F}_s$ -measurable and $\mathbb{E}[W_{t_{j+1} \wedge t} | \mathcal{F}_s] = W_s$, giving $\beta_{t_j}(W_s - W_{t_j})$. For later intervals, condition first on $\mathcal{F}_{t_{i-1}} \supseteq \mathcal{F}_s$: $\mathbb{E}[\beta_{t_{i-1}}(W_{t_i \wedge t} - W_{t_{i-1} \wedge t}) | \mathcal{F}_{t_{i-1}}] = \beta_{t_{i-1}} \cdot 0$, and the tower rule kills them. What remains is exactly $\int_0^s \beta_u dW_u$.

Proof of 3. Expand the square: $\sum_i \beta_{t_{i-1}}^2 (\Delta W_i)^2 + 2 \sum_{i < j} \beta_{t_{i-1}} \beta_{t_{j-1}} \Delta W_i \Delta W_j$. For the diagonal terms, conditioning on $\mathcal{F}_{t_{i-1}}$, $\mathbb{E}[\beta_{t_{i-1}}^2 (\Delta W_i)^2] = \mathbb{E}[\beta_{t_{i-1}}^2] (t_i - t_{i-1})$, whose sum is $\mathbb{E} \int_0^T \beta_u^2 du$. For $i < j$ everything but ΔW_j is $\mathcal{F}_{t_{j-1}}$ -measurable and $\mathbb{E}[\Delta W_j | \mathcal{F}_{t_{j-1}}] = 0$, so cross terms vanish.

Proof of 4. The integral is a finite sum of independent Gaussians $\beta_{t_{i-1}} \Delta W_i \sim \mathcal{N}(0, \beta_{t_{i-1}}^2 (t_i - t_{i-1}))$. $\square$

Definition 4.6 (Integrable processes). An adapted process β is **integrable** if $\|\beta\|_T^2 := \mathbb{E}[\int_0^T \beta_u^2 du] < \infty$. ($\|\cdot\|_T$ is a norm on processes and satisfies the triangle inequality.)

Theorem 4.3 (Construction of the Itô integral). If β is integrable, there exist simple processes β_n with $\|\beta - \beta_n\|_T \rightarrow 0$. The integrals $X_n = \int_0^T \beta_{n,u} dW_u$ form a Cauchy sequence in L^2 , and $\int_0^T \beta_u dW_u$ is **defined** as their L^2 limit, which does not depend on the approximating sequence.

Proof. (β_n) is Cauchy for $\|\cdot\|_T$: $\|\beta_m - \beta_n\|_T \leq \|\beta_m - \beta\|_T + \|\beta - \beta_n\|_T$. By linearity and the Itô isometry for simple processes,

$$\|X_m - X_n\|_{L^2} = \left\| \int_0^T (\beta_{m,u} - \beta_{n,u}) dW_u \right\|_{L^2} = \|\beta_m - \beta_n\|_T,$$

so (X_n) is Cauchy in L^2 , and completeness gives a unique limit. $\square$

The limit is a random variable, not built path by path; letting the upper limit vary gives the **stochastic integral process** $(\int_0^t \beta_u dW_u)_{t \in [0, T]}$. All four properties pass to the limit: the Itô isometry because L^2 convergence implies convergence of second moments; the martingale property because conditional expectations converge in L^2 and limits are unique; Gaussianity for deterministic integrands because L^2 limits of Gaussians are Gaussian.

Note

Executive summary. For an adapted β with $\mathbb{E} \int_0^T \beta_u^2 du < \infty$: the Itô integral exists; $(\int_0^t \beta_u dW_u)_t$ is a martingale with zero mean; its variance is $\mathbb{E} \int_0^T \beta_u^2 du$ (Itô isometry); for deterministic β it is Gaussian.

4.5. Stochastic differential equations

We can now make sense of

$$X_t = x + \int_0^t \alpha(u, X_u) du + \int_0^t \beta(u, X_u) dW_u, \quad t \in [0, T],$$

written in short-hand as the **stochastic differential equation** (SDE)

$$dX_t = \alpha(t, X_t) dt + \beta(t, X_t) dW_t, \quad X_0 = x.$$

α is the **drift** and β the **diffusion** (volatility) coefficient. If α and β are Lipschitz in x with linear growth, a unique solution exists (and in many more general cases); the solution is a **Markov process**.

4.6. Quadratic variation

For a partition $\pi = \{0 = t_0 < \dots < t_n = t\}$ with mesh $|\pi| = \max_i(t_i - t_{i-1})$, define

$$[W]_t := \lim_{|\pi| \rightarrow 0} \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 \quad (\text{limit in } L^2).$$

Theorem 4.4 (Quadratic variation of Brownian motion). $[W]_t = t$ for every $t \geq 0$.

Proof. Write $\Delta W_i = W_{t_i} - W_{t_{i-1}}$, $\Delta t_i = t_i - t_{i-1}$ and $Z_i = \Delta W_i / \sqrt{\Delta t_i}$, i.i.d. $\mathcal{N}(0, 1)$. Then $\sum_i (\Delta W_i)^2 - t = \sum_i \Delta t_i (Z_i^2 - 1)$, a sum of independent zero-mean terms, so

$$\left\| \sum_i (\Delta W_i)^2 - t \right\|_{L^2}^2 = \sum_i \Delta t_i^2 \mathbb{E}[(Z_i^2 - 1)^2] = 2 \sum_i \Delta t_i^2 \leq 2|\pi| \sum_i \Delta t_i = 2|\pi|t \rightarrow 0. \quad \square$$

Note that $[W]_t = t = \text{Var}(W_t)$. Locally: $\mathbb{E}[(\Delta W_t)^2] = \Delta t$ while $\text{Var}((\Delta W_t)^2) = 2\Delta t^2$, so as $\Delta t \rightarrow 0$ the squared increment becomes deterministic, $(\Delta W_t)^2 \approx \Delta t$. In differential form,

$$d[W]_t = dt, \quad (dW_t)^2 = dt.$$

Corollary 4.5 (Infinite variation). *Brownian paths have infinite first variation* $V_W(t) = \sup_\pi \sum_i |W_{t_i} - W_{t_{i-1}}|$ on every interval $[0, t]$.

Proof. $\sum_i (\Delta W_i)^2 \leq \max_i |\Delta W_i| \sum_i |\Delta W_i| \leq \max_i |\Delta W_i| V_W(t)$. As $|\pi| \rightarrow 0$ the left side tends to $t > 0$, while $\max_i |\Delta W_i| \rightarrow 0$ by continuity of the path. This forces $V_W(t) = +\infty$. $\square$

(For a C^1 function f , $V_f(t) = \int_0^t |f'(u)| du < \infty$ and its quadratic variation is zero.)

4.7. Itô's formula

If X solves an SDE, what SDE does $f(t, X_t)$ solve? Write f_t, f_x, f_{xx} for the partial derivatives.

Theorem 4.6 (Itô's formula). *Let $f \in C^{1,2}([0, T] \times \mathbb{R})$ and $dX_t = \alpha(t, X_t) dt + \beta(t, X_t) dW_t$. Then*

$$df(t, X_t) = \left(f_t + \alpha f_x + \frac{1}{2} \beta^2 f_{xx} \right) (t, X_t) dt + \beta(t, X_t) f_x(t, X_t) dW_t.$$

The extra term $\frac{1}{2} \beta^2 f_{xx}$, the **Itô correction**, is absent from ordinary calculus. It exists because Brownian motion has non-zero quadratic variation.

Heuristic derivation. Over $[t, t + \Delta t]$, $\Delta X \approx \alpha \Delta t + \beta \Delta W$. A second-order Taylor expansion gives

$$f(t + \Delta t, X_t + \Delta X) - f(t, X_t) \approx f_t \Delta t + f_x \Delta X + \frac{1}{2} f_{tt} \Delta t^2 + \frac{1}{2} f_{xx} (\Delta X)^2 + f_{tx} \Delta t \Delta X.$$

Substituting ΔX : $(\Delta X)^2 = \alpha^2 \Delta t^2 + 2\alpha\beta \Delta t \Delta W + \beta^2 (\Delta W)^2$. All terms of order higher than Δt — Δt^2 , $\Delta t \Delta W \sim \Delta t^{3/2}$ — vanish in the limit, but $(\Delta W)^2 \approx \Delta t$ does **not**: it contributes $\frac{1}{2} f_{xx} \beta^2 \Delta t$. Keeping the

first-order terms gives the formula. The rigorous proof makes this precise with the quadratic variation. A convenient mnemonic is the multiplication table $dt \cdot dt = 0$, $dt \cdot dW_t = 0$, $dW_t \cdot dW_t = dt$.

Example 4.1 (*Integration by parts*)

With $f(t, x) = tx$ and $X = W$: $f_t = x$, $f_x = t$, $f_{xx} = 0$, so $d(tW_t) = W_t dt + t dW_t$, i.e. $\int_0^t s dW_s = tW_t - \int_0^t W_s ds$. With $f(x) = x^2$: $d(W_t^2) = dt + 2W_t dW_t$, i.e. $\int_0^t W_s dW_s = \frac{1}{2}(W_t^2 - t)$ — not $\frac{1}{2}W_t^2$ as ordinary calculus would suggest.

4.8. Geometric Brownian motion

Definition 4.7 (Geometric Brownian motion with drift). The SDE

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad X_0 = x_0 > 0,$$

with **drift** μ and **volatility** $\sigma > 0$ defines **geometric Brownian motion** (GBM). It is the case $\alpha(t, x) = \mu x$, $\beta(t, x) = \sigma x$ of the general SDE, and the starting point of the Black–Scholes model.

Theorem 4.7 (Explicit solution).

$$X_t = x_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right), \quad t \geq 0.$$

First derivation (separation of variables). Since the solution is Markov, look for $X_t = f(t, W_t)$ with $f(t, x) = g(t)h(x)$. Itô's formula with $X = W$ gives

$$dX_t = \left(g'(t)h(W_t) + \frac{1}{2}g(t)h''(W_t)\right)dt + g(t)h'(W_t)dW_t.$$

Matching with $dX_t = \mu gh dt + \sigma gh dW_t$: the dW_t terms give $h' = \sigma h$, so $h(x) = e^{\sigma x}$; the dt terms give $g' + \frac{1}{2}\sigma^2 g = \mu g$, so $g(t) = g(0)e^{(\mu - \sigma^2/2)t}$; and $g(0) = x_0$.

Second derivation (logarithm). If $X > 0$, set $Y_t = \log X_t$. Itô's formula with $f(x) = \log x$ ($f' = 1/x$, $f'' = -1/x^2$):

$$dY_t = \frac{1}{X_t} dX_t - \frac{1}{2X_t^2} \sigma^2 X_t^2 dt = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma dW_t,$$

which integrates directly: $Y_t = \log x_0 + (\mu - \sigma^2/2)t + \sigma W_t$. $\square$

Consequences: $X_t > 0$; $\log X_t \sim \mathcal{N}(\log x_0 + (\mu - \sigma^2/2)t, \sigma^2 t)$, i.e. X_t is **log-normal**; and $\mathbb{E}[X_t] = x_0 e^{\mu t}$ (by the exponential martingale), so μ is the expected growth rate. The term $-\sigma^2/2$ in the exponent is the Itô correction: the *median* grows at rate $\mu - \sigma^2/2 < \mu$.

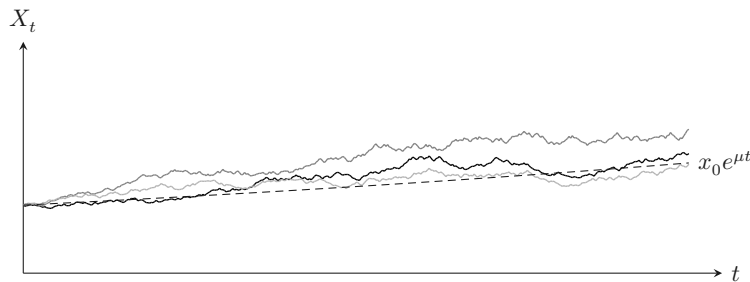


Figure: three simulated paths of geometric Brownian motion and the mean $x_0 e^{\mu t}$ (dashed). Fluctuations grow with the level of the process.

4.9. Diffusions and partial differential equations

Itô's formula connects SDEs with PDEs; this connection is the bridge between the probabilistic (expectation) and analytic (PDE) views of pricing used from Chapter 5 on.

Proposition 4.8 (Martingales and drift). *If $dX_t = \mu_t dt + \sigma_t dW_t$ with $\mathbb{E} \int_0^T \sigma_t^2 dt < \infty$, then X is a martingale iff $\mu_t = 0$ (a.s., for a.e. t); if $\mu_t \geq 0$ it is a submartingale.*

Proof. $X_t - X_s = \int_s^t \mu_u du + \int_s^t \sigma_u dW_u$; the stochastic integral has zero conditional expectation, so $\mathbb{E}[X_t - X_s | \mathcal{F}_s] = \mathbb{E}[\int_s^t \mu_u du | \mathcal{F}_s]$, which is zero for all $s < t$ iff $\mu = 0$, and non-negative if $\mu \geq 0$. $\square$

Consider a diffusion $dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dW_s$ started at $X_t = x$, and its **infinitesimal generator**

$$\mathcal{A}F(t, x) = \mu(t, x) F_x(t, x) + \frac{1}{2} \sigma^2(t, x) F_{xx}(t, x).$$

By Itô's formula $dF(s, X_s) = (F_t + \mathcal{A}F)(s, X_s) ds + \sigma F_x dW_s$.

Theorem 4.9 (Feynman–Kac representation). *Let F solve the boundary-value problem*

$$F_t + \mathcal{A}F - r(t, x)F = 0, \quad F(T, x) = \Phi(x).$$

Under integrability conditions,

$$F(t, x) = \mathbb{E}_{t,x} \left[e^{-\int_t^T r(s, X_s) ds} \Phi(X_T) \right],$$

where $\mathbb{E}_{t,x}$ denotes expectation for the diffusion started at $X_t = x$. With a source term, $F_t + \mathcal{A}F + k = 0$ and $F(T, \cdot) = \Phi$ (and $r = 0$), $F(t, x) = \mathbb{E}_{t,x}[\Phi(X_T)] + \int_t^T \mathbb{E}_{t,x}[k(s, X_s)] ds$.

Proof. Apply Itô's formula to $Z_s = e^{-\int_t^s r(u, X_u) du} F(s, X_s)$ on $[t, T]$: the ds -term is $e^{-\int_t^s r} (F_t + \mathcal{A}F - rF) = 0$, so Z is a martingale (given integrability), and $F(t, x) = Z_t = \mathbb{E}_{t,x}[Z_T]$. For the source term, $F(T, X_T) - F(t, x) = -\int_t^T k ds + \int_t^T \sigma F_x dW$; take expectations. $\square$

The Black–Scholes PDE is the case $\mu = rx$, $\sigma(x) = \sigma x$, constant r . Two further PDEs describe the **transition density** $p(s, y; t, x)$ of X_t given $X_s = y$:

- **Kolmogorov backward equation** (in the initial variables): $\partial_s p + \mu(s, y) \partial_y p + \frac{1}{2} \sigma^2(s, y) \partial_{yy} p = 0$, with $p(s, \cdot; t, x) \rightarrow \delta_x$ as $s \rightarrow t$. It is Feynman–Kac with $\Phi = \delta_x$.
- **Kolmogorov forward (Fokker–Planck) equation** (in the final variables): $\partial_t p = -\partial_x(\mu(t, x)p) + \frac{1}{2} \partial_{xx}(\sigma^2(t, x)p)$, with $p \rightarrow \delta_y$ as $t \rightarrow s$. It follows from $\frac{d}{dt} \mathbb{E}[h(X_t)] = \mathbb{E}[\mathcal{A}h(X_t)]$ for test functions h , after integrating by parts.

For Brownian motion with drift ($\mu = \alpha$, σ constant) the solution is the Gaussian kernel $p = \frac{1}{\sqrt{2\pi\sigma^2(t-s)}} \exp(-\frac{(x-y-\alpha(t-s))^2}{2\sigma^2(t-s)})$; Dupire's equation of Chapter 7 is a forward equation for call prices.

4.10. Exercises

Problems from Björk (B), Chapters 4–5.

Exercise 4.1

(B 4.1) Compute dZ for (a) $Z_t = e^{\alpha t}$; (b) $Z_t = \int_0^t g_s dW_s$ with g adapted; (c) $Z_t = e^{\alpha W_t}$; (d) $Z_t = e^{\alpha X_t}$ with $dX = \mu dt + \sigma dW$; (e) $Z_t = X_t^2$ with $dX = \alpha X dt + \sigma X dW$.

Solution

(a) $dZ = \alpha Z dt$. (b) $dZ = g_t dW_t$. (c) $dZ = \frac{1}{2}\alpha^2 Z dt + \alpha Z dW$. (d) $dZ = Z(\alpha\mu + \frac{1}{2}\alpha^2\sigma^2)dt + \alpha\sigma Z dW$. (e) $dZ = 2X dX + (dX)^2 = (2\alpha + \sigma^2)Z dt + 2\sigma Z dW$: the square of a GBM is a GBM.

Exercise 4.2

(B 4.2) If $dX = \alpha X dt + \sigma X dW$, find the SDE satisfied by $Z = 1/X$.

Solution

With $f(x) = 1/x$, $f' = -1/x^2$, $f'' = 2/x^3$: $dZ = -\frac{dX}{X^2} + \frac{\sigma^2 X^2}{X^3} dt = (\sigma^2 - \alpha)Z dt - \sigma Z dW$. Z is again a GBM — relevant for exchange rates, where $1/S$ is the rate quoted the other way round.

Exercise 4.3

(B 4.3) Let $X_t = \int_0^t \sigma(s) dW_s$ with σ deterministic. Using Itô's formula on e^{iuX_t} , show that $\mathbb{E}[e^{iuX_t}] = \exp(-\frac{u^2}{2} \int_0^t \sigma^2(s) ds)$, so that $X_t \sim \mathcal{N}(0, \int_0^t \sigma^2)$.

Solution

$de^{iuX} = iue^{iuX} \sigma dW - \frac{1}{2}u^2 \sigma^2 e^{iuX} dt$. Taking expectations, $m(t) = \mathbb{E}[e^{iuX_t}]$ satisfies $m' = -\frac{1}{2}u^2 \sigma^2(t)m$, $m(0) = 1$, so $m(t) = \exp(-\frac{u^2}{2} \int_0^t \sigma^2)$, the characteristic function of the stated Gaussian.

Exercise 4.4

(B 4.7) Let W^1, W^2 be independent Brownian motions and $Q_n = \sum_{k=1}^n \Delta W_k^1 \Delta W_k^2$ on an equidistant partition of $[0, t]$. Show that $Q_n \rightarrow 0$ in L^2 , justifying $dW^1 dW^2 = 0$.

Solution

The products are independent with mean 0 and variance $\mathbb{E}[(\Delta W^1)^2] \mathbb{E}[(\Delta W^2)^2] = (t/n)^2$. Hence $\mathbb{E}[Q_n] = 0$ and $\text{Var}(Q_n) = n(t/n)^2 = t^2/n \rightarrow 0$.

Exercise 4.5

(B 5.5) If $dX = \alpha X dt + \sigma X dW$ and $Y = X^\beta$, show that Y is a GBM and identify its drift and volatility.

Solution

$dY = \beta X^{\beta-1} dX + \frac{1}{2}\beta(\beta-1)X^{\beta-2}\sigma^2 X^2 dt = Y(\beta\alpha + \frac{1}{2}\beta(\beta-1)\sigma^2)dt + \beta\sigma Y dW$.

Exercise 4.6

(B 5.1) Show that $X_t = e^{\alpha t} x_0 + \sigma \int_0^t e^{\alpha(t-s)} dW_s$ solves $dX = \alpha X dt + \sigma dW$, $X_0 = x_0$, and find its mean and variance.

Solution

Write $X_t = e^{\alpha t}(x_0 + \sigma R_t)$ with $R_t = \int_0^t e^{-\alpha s} dW_s$. By the product rule, $dX = \alpha X dt + e^{\alpha t} \sigma e^{-\alpha t} dW = \alpha X dt + \sigma dW$. Mean $e^{\alpha t} x_0$; variance $\sigma^2 \int_0^t e^{2\alpha(t-s)} ds = \frac{\sigma^2}{2\alpha}(e^{2\alpha t} - 1)$. For $\alpha = -\kappa < 0$ this is the Ornstein–Uhlenbeck process, with stationary variance $\sigma^2/(2\kappa)$.

Exercise 4.7

(B 5.11) Solve $F_t + \frac{1}{2}x^2 F_{xx} + x = 0$, $F(T, x) = \log(x^2)$.

Solution

Feynman–Kac with source $k = x$ and the diffusion $dX = X dW$, so $X_s = x \exp(W_s - W_t - (s - t)/2)$ and $\mathbb{E}_{t,x}[X_s] = x$. Then $\mathbb{E}_{t,x}[\log X_T^2] = 2 \log x - (T - t)$ and

$$F(t, x) = \log x^2 - (T - t) + x(T - t).$$

Check: $F_t = 1 - x$, $\frac{1}{2}x^2 F_{xx} = \frac{1}{2}x^2(-2/x^2) = -1$, and $1 - x - 1 + x = 0$.

Exercise 4.8

(B 5.15) For $dX = \alpha dt + \sigma dW$, compute the transition density $p(s, y; t, x)$ and verify the Fokker–Planck equation.

Solution

$X_t = y + \alpha(t - s) + \sigma(W_t - W_s)$, Gaussian: $p = (2\pi\sigma^2\tau)^{-1/2} e^{-(x-y-\alpha\tau)^2/(2\sigma^2\tau)}$, $\tau = t - s$. With $z = x - y - \alpha\tau$: $\partial_x p = -\frac{z}{\sigma^2\tau} p$, $\partial_{xx} p = (\frac{z^2}{\sigma^4\tau^2} - \frac{1}{\sigma^2\tau}) p$, $\partial_t p = (-\frac{1}{2\tau} + \frac{z^2}{2\sigma^2\tau^2} + \frac{\alpha z}{\sigma^2\tau}) p$, and indeed $\partial_t p = -\alpha \partial_x p + \frac{1}{2}\sigma^2 \partial_{xx} p$.

4.11. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 4 (slides_4), and the handwritten lecture notes of 4 November 2025 (properties of Brownian motion, construction of the Itô integral, Itô isometry, quadratic variation, Itô's formula, geometric Brownian motion). Björk, *Arbitrage Theory in Continuous Time*, Chapters 4–5 (including the Feynman–Kac and Kolmogorov equations, Sections 5.5–5.6). Further reading: S. Shreve, *Stochastic Calculus for Finance II*, Chapters 3–4; A. Turner and D. Zeindler, *Stochastic Finance: An Introduction with Examples*, Chapters 7–8.

5. The Black–Scholes–Merton model

The Black–Scholes–Merton (BSM) model, published in 1973 by Black and Scholes and by Merton (Nobel prize 1997 to Scholes and Merton), is the most successful stochastic model of a financial market (“imagination creates reality”, Wagner). It is the continuous-time limit of the binomial model, and the same logic applies: replicate the derivative with a self-financing portfolio, and its price follows by no arbitrage. This chapter derives the hedge, the Black–Scholes PDE, its probabilistic solution through the Feynman–Kac formula and the risk-neutral probability, and the explicit formula for calls and puts. It then turns to practice: numerical methods, historical and implied volatility, the Greeks, discrete hedging errors, Delta–Gamma hedging, and the model’s limitations.

5.1. The market

Trading takes place continuously over $[0, T]$. Two assets are traded:

- a **bank account** with continuously compounded rate $r > 0$: $dB_t = rB_t dt$, $B_0 = 1$, so $B_t = e^{rt}$. It is riskless — no Brownian motion appears;
- a **stock** (paying no dividends) following a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 = s > 0, \quad S_t = s \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right),$$

with **drift** μ (the expected rate of return) and **volatility** $\sigma > 0$.

S is a Markov process and S_t is log-normal with $\mathbb{E}[S_t] = se^{\mu t}$ and $\text{Var}(S_t) = s^2 e^{2\mu t} (e^{\sigma^2 t} - 1)$.

5.2. Self-financing strategies

Definition 5.1 (Dynamic portfolio). A dynamic portfolio is a pair $\theta = (\theta^B, \theta^S)$ of adapted processes: θ_t^B and θ_t^S are the units of bank account and stock held over $[t, t + dt]$. Its value is $V_t(\theta) = \theta_t^B B_t + \theta_t^S S_t$.

In the multi-period binomial model the self-financing condition $\theta_t^B B_{t+1} + \theta_t^S S_{t+1} = \theta_{t+1}^B B_{t+1} + \theta_{t+1}^S S_{t+1}$ implies

$$\Delta V_{t+1} = V_{t+1} - V_t = \theta_t^B \Delta B_{t+1} + \theta_t^S \Delta S_{t+1} :$$

the value changes only through price changes, never through injections or withdrawals of money. Letting the time step go to zero motivates:

Definition 5.2 (Self-financing strategy). θ is **self-financing** if

$$V_t(\theta) = V_0(\theta) + \int_0^t \theta_u^B dB_u + \int_0^t \theta_u^S dS_u, \quad \text{i.e.} \quad dV_t(\theta) = \theta_t^B dB_t + \theta_t^S dS_t.$$

Substituting the dynamics, $dV_t = (\theta_t^B r B_t + \theta_t^S \mu S_t) dt + \theta_t^S \sigma S_t dW_t$. The last term is a stochastic integral, so we require $\mathbb{E} \int_0^T (\theta_t^S S_t)^2 dt < \infty$.

5. The Black–Scholes–Merton model

An **arbitrage** is a self-financing θ with $V_0(\theta) = 0$, $V_T(\theta) \geq 0$ a.s. and $\mathbb{P}(V_T(\theta) > 0) > 0$. The BSM model is free of arbitrage for all values of r , μ , σ , provided strategies are restricted to a suitable class (excluding, for instance, “doubling strategies”); this follows from the existence of the risk-neutral probability below.

5.3. Hedging and the Black–Scholes PDE

Consider a European, path-independent derivative with payoff $G(S_T)$. Since S is Markov, it is natural to look for its price in the form $\Pi_t(G) = F(t, S_t)$ for a **pricing function** $F : [0, T] \times (0, \infty) \rightarrow \mathbb{R}$: derivatives are priced *relative* to their underlying.

Theorem 5.1 (The hedging portfolio). *Let $F \in C^{1,2}$ satisfy $F(T, x) = G(x)$. The self-financing portfolio*

$$\theta_t^S = F_x(t, S_t), \quad \theta_t^B = \frac{F_t(t, S_t) + \frac{1}{2}\sigma^2 S_t^2 F_{xx}(t, S_t)}{rB_t},$$

started from $V_0(\theta) = F(0, s)$, has $dV_t(\theta) = dF(t, S_t)$; hence $V_T(\theta) = G(S_T)$ almost surely.

Derivation. For a self-financing θ ,

$$dV_t = (\theta_t^B rB_t + \theta_t^S \mu S_t) dt + \theta_t^S \sigma S_t dW_t. \quad (1)$$

By Itô’s formula,

$$dF(t, S_t) = \left(F_t + \mu S_t F_x + \frac{1}{2}\sigma^2 S_t^2 F_{xx} \right) dt + \sigma S_t F_x dW_t. \quad (2)$$

Matching the dW_t terms of (1) and (2) gives $\theta_t^S = F_x(t, S_t)$. Matching the dt terms with this choice, the μ -terms cancel and $\theta_t^B rB_t = F_t + \frac{1}{2}\sigma^2 S_t^2 F_{xx}$. If $dV_t = dF(t, S_t)$ and $V_0 = F(0, s)$, then $V_T = F(T, S_T) = G(S_T)$. $\square$

$\theta_t^S = F_x(t, S_t)$ is the **Delta** of the derivative: exactly as in the binomial model, holding Δ shares offsets a short position in the derivative.

By the law of one price, $F(t, S_t) = V_t(\theta) = \theta_t^B B_t + \theta_t^S S_t$. Substituting the hedge,

$$F(t, S_t) = \frac{F_t + \frac{1}{2}\sigma^2 S_t^2 F_{xx}}{r} + F_x S_t,$$

which must hold with probability one, i.e. for every possible value of S_t — every $x \in (0, \infty)$.

Theorem 5.2 (Black–Scholes PDE). *Under no arbitrage, the pricing function of the derivative with payoff $G(S_T)$ solves*

$$F_t(t, x) + rx F_x(t, x) + \frac{1}{2}\sigma^2 x^2 F_{xx}(t, x) - rF(t, x) = 0, \quad (t, x) \in [0, T] \times (0, \infty),$$

$$F(T, x) = G(x), \quad x \in (0, \infty).$$

The pricing problem has become a deterministic one: randomness has disappeared. Some PDEs can be solved explicitly; otherwise numerical schemes apply.

! Important

The drift μ does not appear in the PDE. The arbitrage-free price does not depend on the expected return of the stock — the continuous-time analogue of the irrelevance of p_u, p_d in the binomial model. Hence the price would be the same if μ were replaced by r , i.e. in a world where investors are

risk-neutral, where prices are discounted expectations. This is risk-neutral valuation.

5.4. The Feynman–Kac formula and the risk-neutral probability

Theorem 5.3 (Feynman–Kac formula). *Let $F \in C^{1,2}$ solve the Black–Scholes PDE with terminal condition G . Under suitable integrability conditions,*

$$F(t, S_t) = e^{-r(T-t)} \mathbb{E}^Q[G(S_T) \mid S_t],$$

where Q is a probability under which

$$dS_t = rS_t dt + \sigma S_t dW_t^Q$$

with W^Q a Q -Brownian motion. In particular, the price today is $F(0, s) = e^{-rT} \mathbb{E}^Q[G(S_T)]$.

Proof. Under Q , consider $Y_t = e^{-rt} F(t, S_t) = f(t, S_t)$ with $f(t, x) = e^{-rt} F(t, x)$. By Itô's formula, with $f_t = -re^{-rt} F + e^{-rt} F_t$, $f_x = e^{-rt} F_x$, $f_{xx} = e^{-rt} F_{xx}$,

$$dY_t = e^{-rt} \left(F_t + rS_t F_x + \frac{1}{2} \sigma^2 S_t^2 F_{xx} - rF \right) dt + e^{-rt} \sigma S_t F_x dW_t^Q = e^{-rt} \sigma S_t F_x(t, S_t) dW_t^Q,$$

since F solves the PDE. So $Y_t = Y_0 + \int_0^t e^{-ru} \sigma S_u F_x(u, S_u) dW_u^Q$; if $\mathbb{E}^Q \int_0^T e^{-2ru} (\sigma S_u F_x)^2 du < \infty$, the stochastic integral is a Q -martingale and

$$e^{-rt} F(t, S_t) = Y_t = \mathbb{E}^Q[Y_T \mid \mathcal{F}_t] = e^{-rT} \mathbb{E}^Q[G(S_T) \mid \mathcal{F}_t],$$

using the terminal condition $F(T, S_T) = G(S_T)$. By the Markov property, conditioning on $\mathcal{F}_t$ is the same as conditioning on S_t . $\square$

This is the **risk-neutral valuation formula** of the BSM model. The model is formulated under the real-world (statistical) probability $\mathbb{P}$, where $dS_t = \mu S_t dt + \sigma S_t dW_t$; under the risk-neutral probability Q , $dS_t = rS_t dt + \sigma S_t dW_t^Q$. Equating the two,

$$dW_t^Q = dW_t + \frac{\mu - r}{\sigma} dt.$$

The quantity $\lambda = \frac{\mu - r}{\sigma}$ is the **market price of risk** (analogous to the Sharpe ratio): passing from $\mathbb{P}$ to Q , the risk premium of the stock is absorbed into the Brownian motion. **Girsanov's theorem** makes this rigorous: there is a probability Q equivalent to $\mathbb{P}$ under which $W_t + \lambda t$ is a Brownian motion. Under Q the discounted stock price $e^{-rt} S_t$ is a martingale, and so is every discounted self-financing portfolio — the continuous-time version of the binomial martingale property and the source of absence of arbitrage.

5.5. The Black–Scholes formula

Theorem 5.4 (Black–Scholes formula). *The price of a European call with strike K and maturity T is*

$$\Pi_t^{\text{Call}}(K, T) = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

$$d_1 = \frac{\log(S_t/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma \sqrt{T-t}}, \quad d_2 = d_1 - \sigma \sqrt{T-t},$$

where Φ is the standard normal distribution function.

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Proof. Under Q , $S_T = S_t \exp((r - \frac{\sigma^2}{2})(T-t) + \sigma(W_T^Q - W_t^Q))$ and $Z = (W_T^Q - W_t^Q)/\sqrt{T-t} \sim \mathcal{N}(0, 1)$ is independent of $\mathcal{F}_t$. Split

$$e^{-r(T-t)} \mathbb{E}^Q[(S_T - K)^+ | S_t] = e^{-r(T-t)} \underbrace{\mathbb{E}^Q[S_T \mathbf{1}_{\{S_T > K\}} | S_t]}_B - K e^{-r(T-t)} \underbrace{\mathbb{E}^Q[\mathbf{1}_{\{S_T > K\}} | S_t]}_A.$$

(A) $S_T > K$ iff $\log S_t + (r - \frac{\sigma^2}{2})(T-t) + \sigma\sqrt{T-t}Z > \log K$, iff $Z > -d_2$. Hence $A = Q(Z > -d_2) = \Phi(d_2)$ by symmetry.

(B) With the same event,

$$B = S_t e^{r(T-t)} \int_{-d_2}^{\infty} e^{-\frac{\sigma^2}{2}(T-t) + \sigma\sqrt{T-t}z} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz = S_t e^{r(T-t)} \int_{-d_2}^{\infty} \frac{e^{-(z - \sigma\sqrt{T-t})^2/2}}{\sqrt{2\pi}} dz,$$

by completing the square. Substituting $y = z - \sigma\sqrt{T-t}$, the integral is $Q(Z > -d_2 - \sigma\sqrt{T-t}) = \Phi(d_2 + \sigma\sqrt{T-t}) = \Phi(d_1)$. Putting the pieces together gives the formula. $\square$

Interpretation. $\Phi(d_2) = Q(S_T > K)$ is the risk-neutral probability that the call is exercised; $S_t \Phi(d_1)$ is the value at t of receiving one unit of the underlying at T on the event $\{S_T > K\}$.

Corollary 5.5 (Put price).

$$\Pi_t^{\text{Put}}(K, T) = K e^{-r(T-t)} \Phi(-d_2) - S_t \Phi(-d_1).$$

Proof. By put–call parity, $\Pi_t^{\text{Put}} = \Pi_t^{\text{Call}} - S_t + K e^{-r(T-t)} = K e^{-r(T-t)}(1 - \Phi(d_2)) - S_t(1 - \Phi(d_1))$, and $1 - \Phi(x) = \Phi(-x)$. $\square$

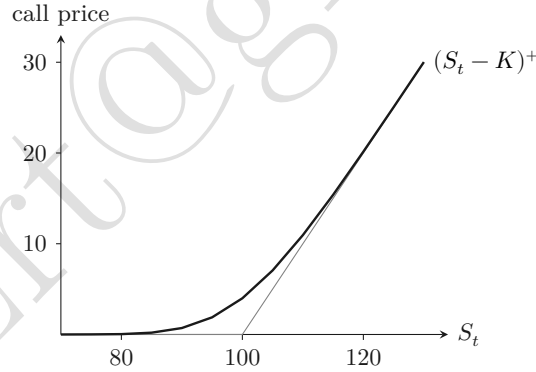


Figure: Black–Scholes call price (bold) as a function of the spot, with $K = 100$, $\sigma = 0.2$, $r = 0$, $T - t = 0.25$; grey: intrinsic value.

Moneyness. At time t a call is *in the money* if $S_t > K$, *out of the money* if $S_t < K$, *at the money* if $S_t = K$.

Proposition 5.6 (Comparative statics of the call price). *The BS call price is*

- decreasing and convex in K (clear from $e^{-r(T-t)} \mathbb{E}^Q[(S_T - K)^+]$);
- increasing and convex in S_t ;
- increasing in r (under Q , r is the growth rate of the underlying and the discount rate of the strike);
- increasing in σ (an option is a right: more dispersion raises the upside, while the downside is capped);
- increasing in the maturity T (same intuition as volatility).

5.6. Extensions: dividends and other European claims

Continuous dividend yield. If the stock pays dividends continuously at rate q (a good model for an index), the self-financing condition includes the dividends received, and under Q the *total* return — price change plus dividends — is r :

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t^Q.$$

Repeating the computation of the Black–Scholes formula with $\mathbb{E}^Q[S_T] = S_t e^{(r-q)\tau}$, $\tau = T - t$, gives **Merton's formula**

$$\Pi_t^{\text{Call}} = S_t e^{-q\tau} \Phi(d_1) - K e^{-r\tau} \Phi(d_2), \quad d_{1,2} = \frac{\log(S_t/K) + (r - q \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}},$$

i.e. Black–Scholes with S_t replaced by $S_t e^{-q\tau}$. Special cases: a currency ($q = r_f$, the Garman–Kohlhagen formula of Chapter 10) and a futures price ($q = r$, **Black's formula** $e^{-r\tau}[F\Phi(d_1) - K\Phi(d_2)]$, used for interest-rate options in Chapter 8). With **discrete** dividends of known present value D , a standard approximation replaces S_t by $S_t - D$.

Other path-independent claims. For a payoff $\Phi(S_T)$ the price is $e^{-r\tau} \mathbb{E}^Q[\Phi(S_T)]$ with $\log S_T \sim \mathcal{N}(\log S_t + (r - \sigma^2/2)\tau, \sigma^2\tau)$. Useful building blocks:

claim	payoff	price at t
cash-or-nothing (digital) call	$\mathbf{1}_{\{S_T > K\}}$	$e^{-r\tau} \Phi(d_2)$
asset-or-nothing call	$S_T \mathbf{1}_{\{S_T > K\}}$	$S_t \Phi(d_1)$
power claim	S_T^β	$S_t^\beta \exp((\beta - 1)(r + \frac{1}{2}\beta\sigma^2)\tau)$
log contract	$\log S_T$	$e^{-r\tau} (\log S_t + (r - \frac{1}{2}\sigma^2)\tau)$

A call is an asset-or-nothing call minus K digital calls — which *is* the Black–Scholes formula. Any payoff can be decomposed this way: binary options, capped payoffs, and (Chapter 7) log contracts underlie variance swaps.

5.7. Numerical methods

For standard payoffs the price is explicit; more complex products, or standard products in more complex models, need numerical methods:

- **Numerical integration.** If the risk-neutral law of S_T is known — in BSM, $\log S_T | S_t \sim \mathcal{N}(\log S_t + (r - \frac{\sigma^2}{2})(T - t), \sigma^2(T - t))$ — then $\Pi_t(G) = e^{-r(T-t)} \int_{-\infty}^{\infty} G(e^x) \varphi(x) dx$ with φ that Gaussian density, a one-dimensional integral. The law is explicit only in some models.
- **Trees.** Approximate the SDE by a binomial tree (Chapter 3); hard with several stochastic factors.
- **PDE methods.** Finite-difference schemes for the pricing PDE; complexity grows quickly with the number of factors (machine-learning methods are promising here).
- **Monte Carlo.** Simulate many paths of the underlying under Q and average discounted payoffs; universal and simple, requires no knowledge of the law of S_T (Chapter 6).

5.8. Historical and implied volatility

The volatility σ is the crucial — and unobservable — parameter.

Historical volatility estimates σ from a time series of log-returns $Z_i = \log(S_{t_i}/S_{t_{i-1}})$ with $t_i - t_{i-1} = \Delta$:

$$\hat{\sigma} = \frac{1}{\sqrt{\Delta}} \left(\frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})^2 \right)^{1/2}.$$

Implied volatility is calibrated from option prices: if a call trades at $\Pi_{\text{mkt}}^{\text{Call}}(K, T)$, its implied volatility σ_{imp} solves

$$\Pi_{\text{mkt}}^{\text{Call}}(K, T) = \Pi_{BS}^{\text{Call}}(K, T; \sigma_{\text{imp}}).$$

Since the BS price is strictly increasing in σ , the solution is unique (found by Newton’s method or bisection). In the BSM model σ_{imp} would be the same for all strikes and maturities. In reality it is not: the **implied-volatility surface** $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$ shows a *skew* (higher implied volatility for low strikes, especially on equity indices such as the SPX) and a term structure. As the saying goes, implied volatility is “the wrong number to put in the wrong formula to get the right price”. Chapter 7 develops models that fit the surface.

5.9. The Greeks

The **Greeks** measure the sensitivity of a price $F(t, S_t)$ to the underlying and to the parameters; they are the basic tools of risk management for derivative portfolios. Since differentiation is linear, the Greek of a portfolio is the sum of the Greeks of its components.

Greek	definition	European call (BS)	European put (BS)
Delta	$\Delta = \partial F / \partial S$	$\Phi(d_1) \in [0, 1]$	$-\Phi(-d_1) \in [-1, 0]$
Gamma	$\Gamma = \partial^2 F / \partial S^2$	$\frac{\varphi(d_1)}{S\sigma\sqrt{T-t}}$	same as call
Vega	$\mathcal{V} = \partial F / \partial \sigma$	$S\varphi(d_1)\sqrt{T-t}$	same as call
Theta	$\Theta = \partial F / \partial t$	$-\frac{S\varphi(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}\Phi(d_2)$	$-\frac{S\varphi(d_1)\sigma}{2\sqrt{T-t}} + rKe^{-r(T-t)}\Phi(-d_2)$
Rho	$\rho = \partial F / \partial r$	$K(T-t)e^{-r(T-t)}\Phi(d_2)$	$-K(T-t)e^{-r(T-t)}\Phi(-d_2)$

Here φ is the standard normal density, $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$. One unit of the underlying has $\Delta = 1$ and $\Gamma = 0$. The derivation of Delta uses the identity $S\varphi(d_1) = Ke^{-r(T-t)}\varphi(d_2)$, which makes the terms coming from $\partial d_1 / \partial S$ and $\partial d_2 / \partial S$ cancel. The Black–Scholes PDE itself reads $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rF$.

5.10. Hedging in practice

In theory, holding Δ_t shares and rebalancing continuously replicates the derivative perfectly:

$$dV_t = (V_t - \Delta_t S_t) r dt + \Delta_t dS_t = dF(t, S_t).$$

In practice continuous rebalancing is impossible (transaction costs); the portfolio is rebalanced at discrete dates, every Δ units of time (say, daily). Over one step,

$$V_{t+\Delta} - V_t \approx (V_t - \Delta_t S_t)r\Delta + \Delta_t(S_{t+\Delta} - S_t),$$

while by Taylor’s expansion, keeping the second-order (Itô) term,

$$F(t + \Delta, S_{t+\Delta}) - F(t, S_t) \approx \Theta_t \Delta + \Delta_t(S_{t+\Delta} - S_t) + \frac{\Gamma_t}{2}(S_{t+\Delta} - S_t)^2.$$

The Delta terms cancel, and the **hedging error** over one step is

$$(V_{t+\Delta} - F(t + \Delta, S_{t+\Delta})) - (V_t - F(t, S_t)) \approx (V_t - F(t, S_t))r\Delta + \frac{\Gamma_t}{2}(\sigma^2 S_t^2 \Delta - (S_{t+\Delta} - S_t)^2),$$

where we used $rF - \Delta_t rS_t - \Theta_t = \frac{1}{2}\sigma^2 S_t^2 \Gamma_t$ (the BS PDE). The first term propagates the error already accumulated. The second compares the variance the model expects over Δ with the realised squared move: it has approximately zero mean and vanishes as $\Delta \rightarrow 0$ (continuous-time perfect hedging), but over finite steps it is a source of P&L, proportional to **Gamma**. Simulations of a Delta-hedged call show final P&Ls of a few cents with frequent rebalancing, and larger ones (tens of cents) with daily or static hedging.

To reduce the error one wants $\Gamma = 0$ as well, which requires a second traded derivative, since the underlying has zero Gamma.

Definition 5.3 (Delta–Gamma hedging). To hedge a short position in a derivative F (with Δ^F, Γ^F) using the underlying and a traded derivative G on the same underlying (with Δ^G, Γ^G), choose π^S units of the underlying and π^G units of G such that

$$\Delta^F = \pi^S + \pi^G \Delta^G, \quad \Gamma^F = \pi^G \Gamma^G.$$

The solution is found **in this order**: first $\pi^G = \Gamma^F / \Gamma^G$ (Gamma neutrality uses only G), then $\pi^S = \Delta^F - \pi^G \Delta^G$. Making the position Delta-neutral first and then adding G would destroy Delta neutrality. In practice the procedure is applied at the level of whole portfolios of derivatives.

5.11. Criticisms of the model

Despite its success, the BSM model is contradicted by data in several ways:

- **log-returns are not normal**: they are skewed and fat-tailed;
- **large moves over short periods** (jumps, crashes) are far more frequent than the model allows;
- **volatility is not constant**: it clusters (heteroskedasticity) and is negatively correlated with returns;
- **markets are not complete**: not every payoff can be perfectly hedged.

These shortcomings motivate local and stochastic volatility models (Chapter 7), jump and Lévy models, and beyond. As Carlo Rovelli put it, between absolute certainty and total uncertainty there is a precious intermediate space — and that is where models live.

5.12. Exercises

Problems from Hull (H), Chapters 15 and 19, and Björk (B), Chapters 7 and 9.

Exercise 5.1

(H 15.3, 15.4) Price a 3-month European put with $S = K = 50$, $r = 10\%$, $\sigma = 30\%$. How does the price change if a dividend of 1.50 is expected in 2 months?

Solution

$d_1 = \frac{(0.1+0.045) \times 0.25}{0.3 \times 0.5} = 0.2417$, $d_2 = 0.0917$; $p = 50e^{-0.025} \Phi(-0.0917) - 50\Phi(-0.2417) = 2.38$. With the dividend, replace S by $50 - 1.5e^{-0.1 \times 2/12} = 48.52$: the put rises to 3.03, since the dividend lowers the expected stock price.

Exercise 5.2

(H 15.6) A stock follows GBM with $\mu = 16\%$, $\sigma = 35\%$, $S_0 = 38$. What is the (real-world) probability that a 6-month European call with $K = 40$ is exercised? And the put?

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Solution

$\mathbb{P}(S_T > K) = \Phi\left(\frac{\log(38/40) + (0.16 - 0.35^2/2) \times 0.5}{0.35\sqrt{0.5}}\right) = \Phi(-0.0077) = 0.497$; the put is exercised with probability 0.503. These use μ ; risk-neutral probabilities would use r instead.

Exercise 5.3

(H 15.14) A 3-month call with $K = 13$ on a stock at 15 costs 2.50; $r = 5\%$. Find the implied volatility.

Solution

Solving $C_{BS}(\sigma) = 2.50$ (e.g. by bisection, the price being increasing in σ) gives $\sigma_{\text{imp}} \approx 39.6\%$.

Exercise 5.4

(B 7.2, H 15.9) In the Black–Scholes model, find the price of the “golden logarithm”, which pays $\log S_T$ at T , and verify that it solves the Black–Scholes PDE.

Solution

$F(t, s) = e^{-r\tau} \mathbb{E}^Q[\log S_T] = e^{-r\tau}(\log s + (r - \frac{1}{2}\sigma^2)\tau)$, $\tau = T - t$. Then $F_t = rF - e^{-r\tau}(r - \frac{1}{2}\sigma^2)$, $sF_s = e^{-r\tau}$, $s^2F_{ss} = -e^{-r\tau}$, and $F_t + rsF_s + \frac{1}{2}\sigma^2s^2F_{ss} - rF = e^{-r\tau}[-(r - \frac{1}{2}\sigma^2) + r - \frac{1}{2}\sigma^2] = 0$.

Exercise 5.5

(B 7.4, H 15.10) Price the claim S_T^β and show that its price has the form $h(t, T)S_t^\beta$.

Solution

Under Q , $S_T^\beta = S_t^\beta \exp(\beta(r - \frac{1}{2}\sigma^2)\tau + \beta\sigma(W_T^Q - W_t^Q))$, so $\mathbb{E}^Q[S_T^\beta | S_t] = S_t^\beta e^{\beta(r - \frac{1}{2}\sigma^2)\tau + \beta^2\sigma^2\tau/2}$ and the price is $S_t^\beta \exp((\beta - 1)(r + \frac{1}{2}\beta\sigma^2)\tau)$. Checks: $\beta = 1$ gives S_t ; $\beta = 0$ gives the bond $e^{-r\tau}$.

Exercise 5.6

(B 7.5, H 15.20) Price a binary option paying K if $S_T \in [\alpha, \beta]$ and nothing otherwise. In particular, what is a claim paying 100 if $S_T > K$ worth?

Solution

$Ke^{-r\tau}Q(\alpha \leq S_T \leq \beta) = Ke^{-r\tau}[\Phi(d_2(\alpha)) - \Phi(d_2(\beta))]$, where $d_2(x) = \frac{\log(S_t/x) + (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}$. The digital paying 100 above K is worth $100e^{-r\tau}\Phi(d_2)$: $\Phi(d_2)$ is the risk-neutral exercise probability.

Exercise 5.7

(H 15.21) A perpetual American put with strike K is exercised when the stock first hits $H < K$. Using that $S^{-\gamma}$ with $\gamma = 2r/\sigma^2$ solves the time-independent Black–Scholes equation, find its value for $S > H$, the optimal H , and the option value.

Solution

For a perpetual claim $F = F(S)$, the PDE is $rSF' + \frac{1}{2}\sigma^2 S^2 F'' - rF = 0$, with solutions S and $S^{-\gamma}$. A bounded solution equal to $K - H$ at $S = H$ is $V(S) = (K - H)(S/H)^{-\gamma}$. Maximising over H , $\frac{d}{dH}[(K - H)H^\gamma] = 0$ gives $H^* = \frac{\gamma K}{1+\gamma}$. Hence $V(S) = \frac{K}{1+\gamma}(\frac{S}{H^*})^{-\gamma}$ for $S > H^*$, and $V = K - S$ below.

Exercise 5.8

(H 19.2) Compute the Delta of an at-the-money 6-month European call when $r = 10\%$ and $\sigma = 25\%$.

Solution

$$d_1 = \frac{(0.1+0.03125)\times 0.5}{0.25\sqrt{0.5}} = 0.3712, \text{ so } \Delta = \Phi(0.3712) = 0.645.$$

Exercise 5.9

(B 9.5, 9.6) A portfolio has $\Delta_P = 2$, $\Gamma_P = 3$. (a) Make it Delta- and Gamma-neutral with two derivatives F ($\Delta_F = -1$, $\Gamma_F = 2$) and G ($\Delta_G = 5$, $\Gamma_G = -2$). (b) Do the same using F and the underlying.

Solution

(a) Holdings x of F , y of G : $2 - x + 5y = 0$, $3 + 2x - 2y = 0$, giving $x = -2.375$, $y = -0.875$ (sell both). (b) Gamma first: $3 + 2x = 0$, $x = -1.5$; then Delta: $2 + 1.5 + y = 0$, $y = -3.5$ units of the underlying.

Exercise 5.10

(B 9.8, 9.9) Show that for any self-financing portfolio of derivatives on the same underlying, $\Theta_P + rS\Delta_P + \frac{1}{2}\sigma^2 S^2 \Gamma_P = rP$, and deduce that a Delta- and Gamma-neutral portfolio earns the risk-free rate.

Solution

Each derivative's pricing function solves the Black–Scholes PDE, which is linear, so the portfolio value P solves it too: this is the identity. If $\Delta_P = \Gamma_P = 0$, it reduces to $\Theta_P = rP$: the value changes only deterministically, at rate r — the portfolio replicates the bank account.

5.13. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 5 (slides_5), and the handwritten lecture notes of 19 November 2025 (self-financing strategies in continuous time, hedging, Black–Scholes PDE, Feynman–Kac, change of drift and market price of risk, derivation of the Black–Scholes formula, numerical integration, discrete Delta hedging error). Björk, *Arbitrage Theory in Continuous Time*, Chapter 7, Sections 8.1–8.2 and Chapter 10; Hull, *Options, Futures, and Other Derivatives*, Chapters 15 and 19; Shreve, *Stochastic Calculus for Finance II*, Section 4.5; Turner and Zeindler, *Stochastic Finance*, Chapter 9. Original references: F. Black and M. Scholes, J. Polit. Econ. **81**, 637 (1973); R. Merton, Bell J. Econ. **4**, 141 (1973).

6. Monte Carlo methods

Risk-neutral valuation expresses every price as an expectation, $\Pi_0(G) = e^{-rT} \mathbb{E}^Q[G(S_T)]$. When the payoff or the model is complicated, the expectation cannot be computed in closed form — but it can be **estimated by simulation**. The Monte Carlo method is simple, universal, and indifferent to the dimension of the problem; its weakness is slow convergence, of order $1/\sqrt{M}$ in the number of simulations. This chapter presents the method, analyses its two sources of error, and develops the three classical variance-reduction techniques: antithetic variates, control variates and importance sampling.

6.1. The basic method

In the BSM model, $\log S_T \sim \mathcal{N}(\log S_0 + (r - \sigma^2/2)T, \sigma^2 T)$ under Q . To price $G(S_T)$:

1. simulate M i.i.d. random variables ξ^k with the law of $\log S_T$ (equivalently $S_T^k = S_0 \exp((r - \sigma^2/2)T + \sigma\sqrt{T}Z^k)$ with Z^k i.i.d. $\mathcal{N}(0,1)$);
2. approximate the price by the discounted sample mean

$$\Pi_0(G) \approx e^{-rT} \frac{1}{M} \sum_{k=1}^M G(S_T^k).$$

The justification is the **law of large numbers**: $\frac{1}{M} \sum_k G(S_T^k) \rightarrow \mathbb{E}^Q[G(S_T)]$ almost surely as $M \rightarrow \infty$.

6.1.1. Path-dependent payoffs

Monte Carlo handles path-dependence naturally. For an average-strike Asian option, $\Pi_0 = e^{-rT} \mathbb{E}^Q[(S_T - \frac{1}{T} \int_0^T S_u du)^+]$, which has no closed form in the BSM model:

1. discretise $[0, T]$ with $t_j = jT/N$, $j = 0, \dots, N$;
2. simulate M discretised paths $(\hat{S}_{t_j}^k)_{j=0, \dots, N}$;
3. approximate each payoff by $(\hat{S}_T^k - \frac{1}{N} \sum_{j=1}^N \hat{S}_{t_j}^k)^+$ and average.

In general, with $g(S_t; t \leq T)$ the exact (possibly path-dependent) payoff, $\hat{S}$ the discretised process and $\hat{S}^k$ its k -th simulated path,

$$\Pi_0(g) = e^{-rT} \mathbb{E}^Q[g(S)] \approx e^{-rT} \frac{1}{M} \sum_{k=1}^M g(\hat{S}^k).$$

6.2. Error analysis

The error splits into two parts:

$$\Pi_0(g) - e^{-rT} \frac{1}{M} \sum_k g(\hat{S}^k) = \underbrace{e^{-rT} (\mathbb{E}^Q[g(S)] - \mathbb{E}^Q[g(\hat{S})])}_{\text{discretisation error (bias)}} + \underbrace{e^{-rT} \left(\mathbb{E}^Q[g(\hat{S})] - \frac{1}{M} \sum_k g(\hat{S}^k) \right)}_{\text{statistical error}}.$$

Statistical error. By the central limit theorem,

$$\frac{\sqrt{M}(\frac{1}{M} \sum_k g(\hat{S}^k) - \mathbb{E}^Q[g(\hat{S})])}{\sqrt{\text{Var}^Q(g(\hat{S}))}} \xrightarrow{d} \mathcal{N}(0, 1),$$

which gives the asymptotic confidence interval of level $1 - \alpha$

$$\frac{1}{M} \sum_k g(\hat{S}^k) \pm \frac{q_{1-\alpha/2}}{\sqrt{M}} \sqrt{\text{Var}^Q(g(\hat{S}))},$$

with $q_{0.975} = 1.96$ for 95%. In practice the variance is replaced by the sample variance. The error decays like $1/\sqrt{M}$: **one more correct digit costs 100 times more simulations.**

Discretisation error (bias). Bias = $\mathbb{E}^Q[g(S)] - \mathbb{E}^Q[g(\hat{S})]$. For geometric Brownian motion compare the **Euler scheme**

$$\hat{S}_{t+\Delta} = \hat{S}_t(1 + r\Delta + \sigma\Delta W_t^Q)$$

with the **exact scheme**

$$\hat{S}_{t+\Delta} = \hat{S}_t \exp((r - \sigma^2/2)\Delta + \sigma\Delta W_t^Q).$$

Already after one step, $\mathbb{E}^Q[\hat{S}_\Delta] = S_0(1 + r\Delta) < S_0 e^{r\Delta} = \mathbb{E}^Q[S_\Delta]$ for Euler (e.g. 101.25 against 101.258 with $S_0 = 100$, $r = 5\%$, $\Delta = 0.25$), while the exact scheme has no bias: **simulating from the exact distribution, when it is known, eliminates the discretisation error.**

Mean square error. Accuracy is measured by

$$\text{MSE} = \mathbb{E} \left[\left(\mathbb{E}[g(S)] - \frac{1}{M} \sum_k g(\hat{S}^k) \right)^2 \right] = \text{Bias}^2 + \text{Var} \left(\frac{1}{M} \sum_k g(\hat{S}^k) \right),$$

so a good computation balances bias (a finer time grid, $N \uparrow$) against sampling error ($M \uparrow$) for a given budget. Monte Carlo parallelises trivially.

6.3. Generating random numbers

Every simulation starts from **pseudo-random** uniform numbers $U \sim \mathcal{U}(0, 1)$, produced by deterministic generators (linear congruential generators, the Mersenne twister) that must have a long period and pass statistical tests. Other distributions are obtained by transformation:

- **Inversion.** If F is a continuous distribution function, $X = F^{-1}(U)$ has distribution F , since $\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$. Example: $X = -\log(U)/\lambda$ is exponential. For the normal distribution Φ^{-1} is computed by accurate rational approximations (Moro's algorithm).
- **Box-Muller.** If U_1, U_2 are independent uniforms,

$$Z_1 = \sqrt{-2 \log U_1} \cos(2\pi U_2), \quad Z_2 = \sqrt{-2 \log U_1} \sin(2\pi U_2)$$

are independent $\mathcal{N}(0, 1)$. (In polar coordinates the squared radius of a standard 2D Gaussian is exponential with mean 2 and the angle is uniform.) Marsaglia's **polar method** avoids the trigonometric functions by rejection sampling on the unit disc.

- **Correlated normals.** For a covariance matrix $\Sigma = LL^\top$ (Cholesky factorisation), $X = LZ$ with Z standard normal has covariance Σ . In two dimensions, Z_1 and $\rho Z_1 + \sqrt{1 - \rho^2} Z_2$ — the construction of correlated Brownian motions in Chapter 7.

6.4. Discretisation schemes for SDEs

When the law of the solution is not explicit, the SDE $dX_t = a(X_t)dt + b(X_t)dW_t$ is integrated numerically on a grid of step $h = \Delta t$.

Definition 6.1 (Euler and Milstein schemes).

$$\text{Euler--Maruyama: } \hat{X}_{j+1} = \hat{X}_j + a(\hat{X}_j)h + b(\hat{X}_j)\Delta W_j,$$

$$\text{Milstein: } \hat{X}_{j+1} = \hat{X}_j + a(\hat{X}_j)h + b(\hat{X}_j)\Delta W_j + \frac{1}{2}b(\hat{X}_j)b'(\hat{X}_j)((\Delta W_j)^2 - h),$$

with $\Delta W_j \sim \mathcal{N}(0, h)$ independent.

The Milstein correction comes from the next term of the Itô–Taylor expansion, the double integral $\iint dW dW = \frac{1}{2}((\Delta W)^2 - h)$. For GBM ($b(x) = \sigma x$) it reads $\hat{X}_{j+1} = \hat{X}_j(1 + rh + \sigma\Delta W_j + \frac{1}{2}\sigma^2((\Delta W_j)^2 - h))$.

Definition 6.2 (Strong and weak convergence). A scheme converges **strongly** with order γ if $\mathbb{E}|X_T - \hat{X}_T| = O(h^\gamma)$ (pathwise accuracy), and **weakly** with order β if $|\mathbb{E}[g(X_T)] - \mathbb{E}[g(\hat{X}_T)]| = O(h^\beta)$ for smooth g (accuracy of expectations).

Euler has strong order $\frac{1}{2}$ and weak order 1; Milstein has strong order 1 (weak order still 1). For **pricing**, only weak convergence matters, so Euler is often enough; strong convergence matters for path-dependent quantities and for multilevel Monte Carlo. A simulation of GBM ($\mu = 6\%$, $\sigma = 30\%$, $T = 1$, 2×10^4 paths) illustrates the orders:

steps N	Euler strong error	Milstein strong error
16	0.0133	0.00106
64	0.0068	0.00028
256	0.0034	0.00007

Quadrupling N halves the Euler error (order $\frac{1}{2}$) and divides the Milstein error by four (order 1).

Bias–variance budget. With a weak order β the MSE behaves like $\text{MSE} \approx c_1 h^{2\beta} + c_2/N$, while the cost grows like N/h . Minimising the MSE for a fixed budget C gives $\text{MSE} = O(C^{-2\beta/(1+2\beta)})$: for Euler ($\beta = 1$), $O(C^{-2/3})$, slower than the $O(C^{-1})$ of exact simulation.

6.5. Variance reduction

Rather than increasing M , reduce the variance of what is being averaged. The three standard techniques exploit increasing amounts of problem-specific information. All the numbers below are for $S_0 = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ and $M = 10^4$ (95% half-widths).

6.5.1. Antithetic variates

If $Z \sim \mathcal{N}(0, 1)$, then $-Z$ has the same law: the sample $\{-Z^k\}$ is as good as $\{Z^k\}$, and large values in one are paired with small values in the other, so errors tend to cancel.

Definition 6.3 (Antithetic pair). Two identically distributed random variables X, Y are **antithetic** if $\text{Cov}(X, Y) \leq 0$.

Proposition 6.1 (Monotone functions give antithetic pairs). Let g be monotone. Then $\text{Cov}(g(X), g(-X)) \leq 0$ for any random variable X , and $\text{Cov}(g(U), g(1 - U)) \leq 0$ for any U . (The first is useful when X is symmetric about 0, the second for $U \sim \text{Unif}(0, 1)$.)

Proof. Let $f(x) = -g(-x)$; f and g are monotone in the same direction. With Y an independent copy of X , $(g(X) - g(Y))(f(X) - f(Y)) \geq 0$. Taking expectations and using independence,

$$0 \leq 2 \mathbb{E}[g(X)f(X)] - 2 \mathbb{E}[g(X)] \mathbb{E}[f(X)] = 2 \text{Cov}(g(X), f(X)) = -2 \text{Cov}(g(X), g(-X)). \quad \square$$

For symmetric X , the estimator $\frac{g(X) + g(-X)}{2}$ is unbiased for $\mathbb{E}[g(X)]$ and

$$\text{Var}\left(\frac{g(X) + g(-X)}{2}\right) = \frac{\text{Var}(g(X))}{2}(1 + \rho), \quad \rho = \text{Corr}(g(X), g(-X)) \leq 0.$$

Definition 6.4 (Antithetic Monte Carlo). Generate Z^k i.i.d. $\mathcal{N}(0, 1)$, set $S_T^k = S_0 e^{(r - \sigma^2/2)T + \sigma\sqrt{T}Z^k}$ and $S_T^{-,k} = S_0 e^{(r - \sigma^2/2)T - \sigma\sqrt{T}Z^k}$, and estimate

$$\Pi_0(G) \approx e^{-rT} \frac{1}{M} \sum_{k=1}^M \frac{G(S_T^k) + G(S_T^{-,k})}{2}.$$

For path-dependent payoffs, each path built from increments Z_j^k is paired with the path built from $-Z_j^k$.

The estimator is unbiased and consistent, and confidence intervals follow from the CLT applied to the pairs. Its variance is $\frac{\text{Var}(G(S_T))}{2M}(1 + \rho)$: compared with $2M$ independent samples (the same number of payoff evaluations) it wins iff $\rho < 0$, which is guaranteed when G is monotone.

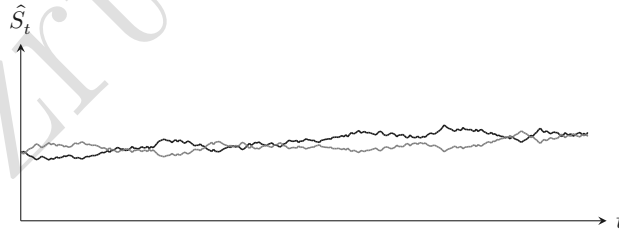


Figure: a simulated path (black) and its antithetic path (grey), driven by the opposite Gaussian increments.

6.5.2. Control variates

Use the known error on a quantity we *can* compute to correct the estimate of one we cannot.

Proposition 6.2 (Control variate). Let Y be a random variable with known mean, and for $c \in \mathbb{R}$ let $\tilde{X} = X + c(Y - \mathbb{E}[Y])$. Then $\mathbb{E}[\tilde{X}] = \mathbb{E}[X]$ and

$$\text{Var}(\tilde{X}) = \text{Var}(X) + c^2 \text{Var}(Y) + 2c \text{Cov}(X, Y),$$

minimised by $c^* = -\text{Cov}(X, Y)/\text{Var}(Y)$, which gives $\text{Var}(\tilde{X}) = \text{Var}(X)(1 - \rho_{XY}^2)$.

The stronger the correlation between X and the control Y , the larger the reduction; any c with $c^2\text{Var}(Y) + 2c\text{Cov}(X, Y) < 0$ still helps. Intuition (for $\rho > 0$): if the simulated control came out above its known mean, the target was probably overestimated too, and is corrected downwards.

Definition 6.5 (Control-variate Monte Carlo). To price $G(S_T)$ using a payoff $H(S_T)$ whose price is known,

$$\Pi_0(G) \approx e^{-rT} \frac{1}{M} \sum_{k=1}^M \left[G(S_T^k) + c(H(S_T^k) - \mathbb{E}^Q[H(S_T)]) \right].$$

The cost is that of plain Monte Carlo. Since c^* is unknown, one uses the regression estimate

$$\hat{c} = - \frac{\sum_k (G(S_T^k) - \bar{G})(H(S_T^k) - \bar{H})}{\sum_k (H(S_T^k) - \bar{H})^2},$$

at the price of a small bias. Choice of control: the underlying itself is universal ($\mathbb{E}^Q[S_T] = S_0 e^{rT}$); for exotic payoffs, a tractable relative — for **arithmetic Asian options**, the **geometric Asian option**, whose price is explicit because the geometric average of log-normals is log-normal.

6.5.3. Importance sampling

Put more probability on the outcomes that matter.

Let $Z > 0$ with $\mathbb{E}[Z] = 1$ and define the probability $\tilde{\mathbb{P}}$ by $d\tilde{\mathbb{P}} = Z d\mathbb{P}$. Then

$$\mathbb{E}[X] = \tilde{\mathbb{E}}\left[\frac{X}{Z}\right], \quad \text{Var}_{\tilde{\mathbb{P}}}\left(\frac{X}{Z}\right) = \mathbb{E}\left[\frac{X^2}{Z}\right] - \mathbb{E}[X]^2,$$

which can be much smaller than $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. Any $Y > 0$ with finite mean gives a valid $Z = Y/\mathbb{E}[Y]$. For $X > 0$, the choice $Z = X/\mathbb{E}[X]$ gives *zero* variance — useless, since it needs the answer — but it shows that Z should be roughly proportional to X .

In the BSM model a convenient family is

$$Z(\nu) = \exp\left(-\nu W_T^Q - \frac{\nu^2}{2}T\right), \quad \nu \in \mathbb{R},$$

which is positive with $\mathbb{E}^Q[Z(\nu)] = 1$.

Theorem 6.3 (Girsanov's theorem (Brownian case)). *The probability $d\tilde{Q}_\nu = Z(\nu) dQ$ is equivalent to Q , and $\tilde{W}_t^\nu = W_t^Q + \nu t$ is a $\tilde{Q}_\nu$ -Brownian motion.*

Proof. $\tilde{Q}_\nu(\Omega) = \mathbb{E}^Q[Z(\nu)] = 1$, and equivalence holds because $Z(\nu) > 0$. For $s \leq t$, $u \in \mathbb{R}$ and $A \in \mathcal{F}_s$, using the Q -martingale property of $e^{-\nu W_t - \frac{\nu^2}{2}t}$ and the independence of increments,

$$\mathbb{E}^{\tilde{Q}_\nu}[e^{u(\tilde{W}_t^\nu - \tilde{W}_s^\nu)} \mathbf{1}_A] = \mathbb{E}^Q\left[e^{(u-\nu)(W_t - W_s)}\right] e^{-\frac{\nu^2}{2}(t-s) + u\nu(t-s)} \mathbb{E}^Q[e^{-\nu W_s - \frac{\nu^2}{2}s} \mathbf{1}_A] = e^{\frac{u^2}{2}(t-s)} \tilde{Q}_\nu(A).$$

So under $\tilde{Q}_\nu$ the increment $\tilde{W}_t^\nu - \tilde{W}_s^\nu$ is $\mathcal{N}(0, t-s)$ and independent of $\mathcal{F}_s$. $\square$

Under $\tilde{Q}_\nu$, since $dW_t^Q = d\tilde{W}_t^\nu - \nu dt$,

$$dS_t = (r - \sigma\nu)S_t dt + \sigma S_t d\tilde{W}_t^\nu :$$

choosing ν shifts the drift and makes high (or low) values of S more likely. (The course slides write the new drift as $r - \nu$; with the density $Z(\nu)$ above it is $r - \sigma\nu$.)

Definition 6.6 (Importance-sampling Monte Carlo). Choose ν ; generate Z^k i.i.d. $\mathcal{N}(0, 1)$ and set

$$S_T^k = S_0 \exp((r - \sigma\nu - \frac{\sigma^2}{2})T + \sigma\sqrt{T}Z^k), \quad Z^k(\nu) = \exp(-\nu\sqrt{T}Z^k + \frac{\nu^2}{2}T),$$

and estimate $\Pi_0(G) \approx e^{-rT} \frac{1}{M} \sum_k G(S_T^k) / Z^k(\nu)$.

($Z^k(\nu)$ is $Z(\nu)$ evaluated on the simulated path, using $W_T^Q = \tilde{W}_T^\nu - \nu T$.) The estimator is unbiased and consistent, with variance

$$\frac{1}{M} \left(\mathbb{E}^Q \left[\frac{G(S_T)^2}{Z(\nu)} \right] - \mathbb{E}^Q[G(S_T)]^2 \right),$$

so ν must make $\mathbb{E}^Q[G^2/Z(\nu)] < \mathbb{E}^Q[G^2]$. For a **deep out-of-the-money call** ($K \gg S_0$) almost all plain samples pay zero; choosing $\nu < 0$ such that $\mathbb{E}^{\tilde{Q}_\nu}[S_T] = K$, i.e. $\nu = \frac{1}{\sigma}(r - \frac{1}{T} \log \frac{K}{S_0})$, centres the simulation on the exercise region. Importance sampling is the tool for **rare events**: deep out-of-the-money options, credit losses, ruin probabilities in insurance.

6.5.4. Comparison

case	method	estimate	95% half-width
call $K = 100$ (exact 10.451)	plain	10.304	0.286
	antithetic	10.444	0.144
	control variate S_T ($\rho = 0.92$)	10.455	0.110
call $K = 160$ (exact 0.159)	plain	0.167	0.040
	importance sampling ($\nu = -2.10$)	0.159	0.004
arithmetic Asian, $K = 100$, 12 dates	plain	6.181	0.167
	geometric Asian control ($\rho = 0.9996$)	6.150	0.004

method	complexity	effectiveness	best for
antithetic variates	low	moderate	symmetric distributions, monotone payoffs
control variates	average	high with a good control	payoffs with a correlated, priceable relative
importance sampling	high	very high with a good Z	rare events

Antithetic variates need almost no information but rarely give large gains. Control variates require knowing something about the problem and can be very efficient. Importance sampling is the most delicate: with a good change of measure it gains orders of magnitude, with a bad one it can make things worse. Whenever possible combine techniques, and always validate against a benchmark.

6.6. Exercises

Problems from Seydel (S), *Tools for Computational Finance*, Chapters 2–3, and Glasserman (G), Chapter 4.

Exercise 6.1

(S 2, Box–Muller) Show that if U_1, U_2 are independent uniforms on $(0, 1)$, then $Z_1 = \sqrt{-2 \log U_1} \cos(2\pi U_2)$ and $Z_2 = \sqrt{-2 \log U_1} \sin(2\pi U_2)$ are independent standard normals.

Solution

Set $R^2 = -2 \log U_1$ and $\Theta = 2\pi U_2$. Then R^2 is exponential with mean 2 (density $\frac{1}{2}e^{-\rho/2}$) and Θ is uniform on $[0, 2\pi)$, independent. The joint density of $(Z_1, Z_2) = (R \cos \Theta, R \sin \Theta)$ is obtained with $dz_1 dz_2 = \frac{1}{2} d\rho d\theta$ (with $\rho = r^2$): $\frac{1}{2\pi} \cdot \frac{1}{2} e^{-\rho/2} \cdot 2 = \frac{1}{2\pi} e^{-(z_1^2 + z_2^2)/2}$, the product of two standard normal densities.

Exercise 6.2

(S 3.13) Implement Monte Carlo for European options in the Black–Scholes model and test it on: (a) a vanilla put with $S_0 = 5$, $K = 10$, $r = 0.06$, $\sigma = 0.3$, $T = 1$; (b) an up-and-out call with $S_0 = 5$, $K = 6$, $r = 0.05$, $\sigma = 0.3$, $T = 1$ and barrier $B = 8$. Reference values: 4.43046 and 0.0983.

Solution

(a) The exact Black–Scholes put value is 4.43046; plain Monte Carlo with 10^5 samples of S_T reproduces it within about ± 0.005 . (b) Simulating paths with 1000 monitoring dates and killing those that reach B gives ≈ 0.102 , *above* the continuous-barrier value 0.0983: with discrete monitoring the path can cross the barrier between dates unnoticed. The bias decays only like $\sqrt{\Delta t}$; it can be removed by the Brownian-bridge crossing probability (next exercise) or by shifting the barrier by $e^{-0.5826 \sigma \sqrt{\Delta t}}$ (Broadie–Glasserman–Kou).

Exercise 6.3

(S 3.11) For a Brownian motion, let $X_t = W_t - \frac{t}{T} W_T$ on $[0, T]$ (the **Brownian bridge** from 0 to 0). Compute $\text{Var}(X_t)$ and give a way to simulate W_t conditionally on W_T .

Solution

$\text{Var}(X_t) = t - 2\frac{t}{T}t + \frac{t^2}{T^2}T = t(1 - \frac{t}{T})$. X_t is independent of W_T (zero covariance, jointly Gaussian), so given $W_T = w$, $W_t = \frac{t}{T}w + \sqrt{t(1 - t/T)} Z$ with $Z \sim \mathcal{N}(0, 1)$. This allows generating paths “from the end” (useful with quasi-Monte Carlo) and computing barrier-crossing probabilities between monitoring dates.

Exercise 6.4

(S 3.5) Write the Milstein scheme for GBM, and show that for $dX = a dt + b dW$ with constant b it coincides with Euler.

Solution

For GBM, $b(x) = \sigma x$, $b' = \sigma$: $\hat{X}_{j+1} = \hat{X}_j(1 + \mu h + \sigma \Delta W_j + \frac{1}{2}\sigma^2((\Delta W_j)^2 - h))$. If b is constant, $b' = 0$ and the correction vanishes: Euler already has strong order 1 for additive noise.

Exercise 6.5

(Weak error of Euler) For GBM with drift μ , compute $\mathbb{E}[\hat{X}_T]$ under the Euler scheme with N steps and compare with $\mathbb{E}[X_T] = X_0 e^{\mu T}$.

Solution

Each step multiplies the mean by $1 + \mu h$, so $\mathbb{E}[\hat{X}_T] = X_0(1 + \mu T/N)^N$. Since $(1 + x/N)^N = e^x(1 - x^2/(2N) + O(N^{-2}))$, the error is $X_0 e^{\mu T} \mu^2 T^2/(2N) = O(h)$: weak order one. With $\mu = 6\%$, $T = 1$, $N = 16$: 1.06172 against 1.06184.

Exercise 6.6

(S 3.14) Assume $\text{MSE}(h, N) = \alpha_1^2 h^{2\beta} + \alpha_2^2/N$ and cost $C = \alpha_3 N/h$. Minimise the MSE for a given budget C .

Solution

Substituting $N = Ch/\alpha_3$: $\text{MSE}(h) = \alpha_1^2 h^{2\beta} + \frac{\alpha_2^2 \alpha_3}{Ch}$. Setting the derivative to zero gives $h^{2\beta+1} = \frac{\alpha_2^2 \alpha_3}{2\beta \alpha_1^2 C}$, so $h^* \propto C^{-1/(1+2\beta)}$ and $\text{MSE}^* \propto C^{-2\beta/(1+2\beta)}$, i.e. $\sqrt{\text{MSE}} \propto C^{-\beta/(1+2\beta)}$: $C^{-1/3}$ for Euler, against $C^{-1/2}$ for unbiased simulation.

Exercise 6.7

(S 3.15) For the control variate $V_\alpha = \hat{V} + \alpha(\hat{V}^* - \mathbb{E}\hat{V}^*)$, find the optimal α and the optimal variance ratio $q_0 = \text{Var}(V_{\alpha_0})/\text{Var}(\hat{V})$ as a function of the correlation ρ . Evaluate for $\rho = 0.95$ and $\rho = 0.8$.

Solution

$\alpha_0 = -\text{Cov}(\hat{V}, \hat{V}^*)/\text{Var}(\hat{V}^*)$ and $q_0 = 1 - \rho^2$: 0.0975 for $\rho = 0.95$ (a tenfold reduction, worth ten times more samples) and 0.36 for $\rho = 0.8$. The gain is very sensitive to ρ near 1.

Exercise 6.8

(G 4) Derive the closed-form price of the discretely monitored geometric Asian call with payoff $((\prod_{j=1}^N S_{t_j})^{1/N} - K)^+$, $t_j = jT/N$, in the Black–Scholes model.

Solution

$\frac{1}{N} \sum_j \log S_{t_j}$ is Gaussian with mean $m = \log S_0 + (r - \sigma^2/2)\Delta \frac{N+1}{2}$ and variance $v = \sigma^2 \Delta \frac{(N+1)(2N+1)}{6N}$, with $\Delta = T/N$ (using $\text{Var}(\sum_j W_{t_j}) = \Delta \sum_{i,j} \min(i, j) = \Delta \frac{N(N+1)(2N+1)}{6}$). For a log-normal e^Y , $Y \sim \mathcal{N}(m, v)$: $\mathbb{E}[(e^Y - K)^+] = e^{m+v/2} \Phi(d_1) - K \Phi(d_1 - \sqrt{v})$ with $d_1 = (m - \log K + v)/\sqrt{v}$; multiply by e^{-rT} . With the parameters of the comparison table this gives 5.940.

6.7. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 6 (slides_6). P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer (2004), Chapter 4; R. Seydel, *Tools for Computational Finance*, 6th ed., Chapter 2 (random numbers) and Chapter 3 (SDE discretisation, Section 3.5 on variance reduction); G. H. Choe, *Stochastic Analysis for Finance with Simulations*, Chapter 28. The numerical results in the comparison

table were computed for these notes (seed-dependent).

7. Volatility modelling

The Black–Scholes model assumes constant volatility and Gaussian log-returns, but market option prices imply a volatility surface that is far from flat, and returns are skewed and fat-tailed. This chapter introduces the two main families of models that address these shortcomings. **Local volatility** models make volatility a deterministic function of time and price, and can reproduce all quoted option prices exactly (Dupire’s formula). **Stochastic volatility** models add a second random factor, giving realistic volatility dynamics at the cost of market incompleteness; Heston’s model is the reference example. Local-stochastic volatility models combine the two, and volatility derivatives trade volatility itself.

7.1. Reading the smile

Before modelling the smile it helps to understand what it encodes (Hull, Chapter 20).

Calls and puts share the same implied volatility. Put–call parity $C - P = S_0 - Ke^{-rT}$ holds for market prices (by no arbitrage) and for Black–Scholes prices (as a model identity). Subtracting, $C_{\text{mkt}} - C_{BS} = P_{\text{mkt}} - P_{BS}$ for any volatility: the volatility that matches the call also matches the put. The smile is therefore a property of the strike and maturity, not of the option type.

Smiles are statements about the risk-neutral distribution. By Breeden–Litzenberger (below), call prices across strikes determine $f^Q(T, \cdot)$. Compared with the log-normal:

- **equity indices** show a downward-sloping *skew*: the left tail is heavier and the right tail lighter (crash risk, “crashophobia” after 1987, and the leverage effect: volatility rises when prices fall);
- **currencies** show a more symmetric *smile*: both tails heavier than log-normal;
- **jumps** in the underlying create a pronounced smile at short maturities that flattens with maturity, whereas **stochastic volatility** creates smiles that build up over longer horizons;
- the **term structure** of at-the-money volatility tends to be upward-sloping when volatility is historically low, and downward-sloping when it is high (mean reversion of volatility).

Smile dynamics. How the smile moves when the spot moves matters for hedging. Under *sticky strike* the implied volatility of a given strike is unchanged; under *sticky delta* (or sticky moneyness) the smile moves with the spot. Local volatility models imply smile dynamics that contradict both in the wrong direction; stochastic volatility models give more realistic behaviour.

7.2. Local volatility

Under the risk-neutral probability Q , postulate

$$dS_t = rS_t dt + \sigma(t, S_t) S_t dW_t^Q, \quad S_0 = s,$$

with a **local volatility function** $\sigma : [0, T] \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$. Volatility now varies with time and price, but no new source of randomness is introduced: the market is still complete, and the hedging argument of Chapter 5 goes through unchanged.

Proposition 7.1 (Pricing PDE under local volatility). *The pricing function F of a European payoff $G(S_T)$ solves*

$$F_t + rx F_x + \frac{1}{2} \sigma(t, x)^2 x^2 F_{xx} - rF = 0, \quad F(T, x) = G(x),$$

which in general must be solved numerically.

How should $\sigma(t, x)$ be chosen? So that the model reproduces the call prices observed on the market for **all strikes and maturities**. The answer rests on a link between call prices and the risk-neutral distribution.

7.2.1. Breeden–Litzenberger

Recall the **Dirac delta** δ , a generalised function used under integrals: $\int f(x) \delta(x - a) dx = f(a)$. Intuitively $\delta(x - a)$ is the derivative of $\mathbf{1}_{[a, \infty)}(x)$.

Theorem 7.2 (Breeden–Litzenberger formulas). *Let $C(T, K)$ be the time-0 price of a call with maturity T and strike K , and $f^Q(T, \cdot)$ the risk-neutral density of S_T . Then*

$$\frac{\partial C}{\partial K}(T, K) = -e^{-rT} Q(S_T > K), \quad \frac{\partial^2 C}{\partial K^2}(T, K) = e^{-rT} f^Q(T, K) = e^{-rT} \mathbb{E}^Q[\delta(S_T - K)].$$

Proof. Writing $(x - K)^+ = \int_K^x dy$ for $x > K$ and exchanging the order of integration,

$$C(T, K) = e^{-rT} \int_K^\infty (x - K) f^Q(T, x) dx = e^{-rT} \int_K^\infty \left(\int_y^\infty f^Q(T, x) dx \right) dy.$$

Differentiating in K gives $-e^{-rT} \int_K^\infty f^Q(T, x) dx = -e^{-rT} Q(S_T > K)$, and again $e^{-rT} f^Q(T, K)$. Finally $f^Q(T, K) = \int f^Q(T, x) \delta(x - K) dx = \mathbb{E}^Q[\delta(S_T - K)]$. $\square$

Message: the risk-neutral distribution of S_T is encoded in call prices for all strikes at maturity T — equivalently, in the implied volatility smile at T . Can we find $\sigma(t, x)$ such that the model generates exactly these distributions? Yes.

7.2.2. Dupire's formula

Theorem 7.3 (Dupire (1994)). *If call prices $C(T, K)$ are known for all $T, K > 0$, the local volatility function that reproduces them is uniquely given by*

$$\sigma^2(T, K) = \frac{\frac{\partial C}{\partial T}(T, K) + rK \frac{\partial C}{\partial K}(T, K)}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}(T, K)}.$$

(This does not say that the local volatility model is the only model reproducing all call prices: stochastic volatility models can do so too.)

Proof. We would like to apply Itô's formula to $(S_t - K)^+$, which is not C^2 . For convex functions there is the **Itô–Tanaka–Meyer formula**: if $dX_t = \alpha dt + \beta dW_t$ and F is convex,

$$dF(X_t) = F'(X_t) \alpha dt + \frac{1}{2} \int_{-\infty}^{\infty} dL_t^x F''(dx) + F'(X_t) \beta dW_t,$$

where $F''(dx)$ is the second derivative in the sense of distributions and L_t^x is the **local time** of X at

level x — roughly, the time X spends near x up to t — characterised by the occupation-time formula $\int g(x) L_t^x dx = \int_0^t g(X_s) \beta^2(s, X_s) ds$.

For $F(x) = (x - K)^+$, $F'(x) = \mathbf{1}_{(K, \infty)}(x)$ and $F''(dx) = \delta(x - K) dx$, so $\int dL_t^x \delta(x - K) = \delta(S_t - K) S_t^2 \sigma^2(t, S_t) dt$ and

$$d(S_t - K)^+ = \mathbf{1}_{\{S_t > K\}} r S_t dt + \frac{1}{2} \delta(S_t - K) S_t^2 \sigma^2(t, S_t) dt + \mathbf{1}_{\{S_t > K\}} \sigma(t, S_t) S_t dW_t^Q.$$

Discounting, $d(e^{-rt}(S_t - K)^+) = e^{-rt} [rK \mathbf{1}_{\{S_t > K\}} + \frac{1}{2} \delta(S_t - K) S_t^2 \sigma^2(t, S_t)] dt + (\text{martingale})$, since $-r(S_t - K)^+ + rS_t \mathbf{1}_{\{S_t > K\}} = rK \mathbf{1}_{\{S_t > K\}}$. Integrating over $[0, T]$ and taking Q -expectations (the stochastic integral has zero mean), with Breeden–Litzenberger,

$$C(T, K) = (S_0 - K)^+ + \int_0^T \left[rK e^{-rt} Q(S_t > K) + \frac{1}{2} \frac{\partial^2 C}{\partial K^2}(t, K) K^2 \sigma^2(t, K) \right] dt.$$

Differentiating in T and using $e^{-rT} Q(S_T > K) = -\partial_K C$,

$$\frac{\partial C}{\partial T} = -rK \frac{\partial C}{\partial K} + \frac{1}{2} K^2 \sigma^2(T, K) \frac{\partial^2 C}{\partial K^2},$$

which is Dupire's formula. $\square$

The last equation is the **Dupire forward PDE**: a PDE in (T, K) satisfied by call prices as functions of strike and maturity. Dupire's formula can also be rewritten in terms of implied volatilities.

In practice only finitely many calls are quoted. One first calibrates, for each maturity, a smooth parametric function of the strike to the quoted prices or implied volatilities, then applies Dupire's formula to it. Limitations: the result is very sensitive to the interpolation; the second derivative $\partial^2 C / \partial K^2$ in the denominator causes numerical instability; and local volatility models predict unrealistic dynamics of the smile (the smile “moves the wrong way” when the spot moves), so they are inadequate for managing volatility risk.

7.3. Interlude: multi-dimensional Itô formula

Definition 7.1 (Multi-dimensional and correlated Brownian motions). $W = (W^1, \dots, W^d)$ is a d -dimensional Brownian motion if each W^j is a Brownian motion and the components are independent. From a two-dimensional W , the process $\hat{W}_t = \rho W_t^1 + \sqrt{1 - \rho^2} W_t^2$ is a Brownian motion with $\text{Cov}(W_t^1, \hat{W}_t) = \rho t$, $\rho \in [-1, 1]$. More generally, RW_t with $R \in \mathbb{R}^{d \times d}$ having unit-norm rows has correlated Brownian components.

Consider an $\mathbb{R}^n$ -valued process with $dX_t = \alpha(t, X_t) dt + \beta(t, X_t) dW_t$, $\alpha : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\beta : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$, i.e. $dX_t^i = \alpha_i dt + \sum_{j=1}^d \beta_{ij} dW_t^j$. Components driven by common Brownian motions are correlated.

Theorem 7.4 (Multi-dimensional Itô formula). *Let $f \in C^{1,2}$ and $B = \beta \beta^\top$. Then*

$$df(t, X_t) = \left(f_t + \sum_{i=1}^n f_{x_i} \alpha_i + \frac{1}{2} \sum_{i,j=1}^n f_{x_i x_j} B_{ij} \right) dt + \sum_{i=1}^n \sum_{j=1}^d f_{x_i} \beta_{ij} dW_t^j.$$

Mnemonic: $dW_t^i dW_t^j = \delta_{ij} dt$, and for correlated Brownian motions $dW_t d\hat{W}_t = \rho dt$.

7.4. Stochastic volatility

To capture the dynamics of volatility, make it random. Under the real-world probability $\mathbb{P}$,

$$dS_t = \mu(t, S_t)S_t dt + \sigma_t S_t dW_t, \quad d\sigma_t = a(t, \sigma_t) dt + b(t, \sigma_t) d\hat{W}_t,$$

with $W, \hat{W}$ Brownian motions with correlation ρ . Features:

- a second random factor; S alone is no longer Markov, but the pair (S, σ) is;
- usually $\rho < 0$, reproducing the **leverage effect**: volatility rises when prices fall;
- these models are continuous-time limits of GARCH models; a and b may also depend on S .

7.4.1. Hedging and the pricing PDE

With two sources of randomness and only one risky asset, perfect hedging is impossible: the market is **incomplete**. Add a second traded instrument, an arbitrary derivative with price $H(t, S_t, \sigma_t)$, and look for the price of a payoff $G(S_T)$ as $F(t, S_t, \sigma_t)$. A self-financing portfolio $\theta = (\theta^B, \theta^S, \theta^H)$ has $dV_t = \theta_t^B dB_t + \theta_t^S dS_t + \theta_t^H dH_t$, and by the two-dimensional Itô formula (subscripts x for S , v for σ ; arguments omitted)

$$dH = \left(H_t + \frac{1}{2} H_{xx} S^2 \sigma^2 + \frac{1}{2} H_{vv} b^2 + \rho H_{xv} S \sigma b \right) dt + H_x dS_t + H_v d\sigma_t,$$

and the same for F . Matching $dV = dF$: the $d\sigma_t$ terms give $\theta^H = F_v / H_v$; the dS_t terms give $\theta^S = F_x - \frac{F_v}{H_v} H_x$; the dt terms fix θ^B . Imposing $V_t = F$ and rearranging yields

$$\frac{F_t + rSF_x + \frac{1}{2}S^2\sigma^2F_{xx} + \frac{1}{2}b^2F_{vv} + \rho S\sigma bF_{xv} - rF}{F_v} = \frac{H_t + rSH_x + \frac{1}{2}S^2\sigma^2H_{xx} + \frac{1}{2}b^2H_{vv} + \rho S\sigma bH_{xv} - rH}{H_v}.$$

Since F and H are arbitrary derivatives, both sides must equal a common function $q(t, S_t, \sigma_t)$.

Theorem 7.5 (Stochastic volatility PDE).

$$F_t + rxF_x + \frac{1}{2}x^2v^2F_{xx} + \frac{1}{2}b^2F_{vv} + \rho xvbF_{xv} - rF = qF_v, \quad F(T, x, v) = G(x).$$

Writing $q = \varphi b - a$,

$$F_t + rxF_x + aF_v + \frac{1}{2}x^2v^2F_{xx} + \frac{1}{2}b^2F_{vv} + \rho xvbF_{xv} - rF = \varphi bF_v,$$

where $\varphi(t, S_t, \sigma_t)$ is the **market price of volatility risk**.

Who determines φ ? The market. To see its meaning, Delta-hedge a short derivative with the underlying only: $\Pi_t = F_x S_t - F$. Using the PDE,

$$d\Pi_t = r\Pi_t dt - F_v b(\varphi dt + d\hat{W}_t).$$

Delta hedging leaves **volatility risk**, and φ is the extra expected return that compensates for bearing it: a volatility risk premium. By Girsanov's theorem there is a probability $Q \sim \mathbb{P}$ absorbing both the stock's risk premium (so that S earns r) and φ (so that the volatility drift becomes $-q$). Assuming q does not depend on S , the PDE involves q only, so one may as well start directly under Q . Unlike in BSM, Q is **not unique**: it depends on φ , and in practice it is selected by calibration to option prices.

|

Definition 7.2 (Stochastic volatility, risk-neutral form).

$$dS_t = rS_t dt + \sigma_t S_t dW_t^Q, \quad d\sigma_t = a^Q(t, \sigma_t) dt + b(t, \sigma_t) d\hat{W}_t^Q,$$

with $a^Q = -q$ and $W^Q, \hat{W}^Q$ correlated Q -Brownian motions.

7.4.2. The Heston model

Definition 7.3 (Heston (1993)). With v_t the instantaneous **variance**,

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^Q, \quad dv_t = \theta(\kappa - v_t) dt + \gamma\sqrt{v_t} d\hat{W}_t^Q, \quad d\langle W^Q, \hat{W}^Q \rangle_t = \rho dt,$$

where θ is the speed of mean reversion, κ the long-run variance, γ the volatility of volatility (“vol of vol”).

The variance follows a Cox–Ingersoll–Ross process (Chapter 8), positive if $2\theta\kappa \geq \gamma^2$ (Feller condition).

The pricing PDE becomes

$$F_t + rxF_x + \theta(\kappa - v)F_v + \frac{1}{2}x^2vF_{xx} + \frac{1}{2}\gamma^2vF_{vv} + \rho\gamma xvF_{xv} - rF = 0, \quad F(T, x, v) = G(x).$$

Semi-explicit pricing. Heston’s key property: the characteristic function of $\log S_T$,

$$f_{(t,x,v)}(u) = \mathbb{E}^Q[e^{iu \log S_T} \mid \log S_t = x, v_t = v],$$

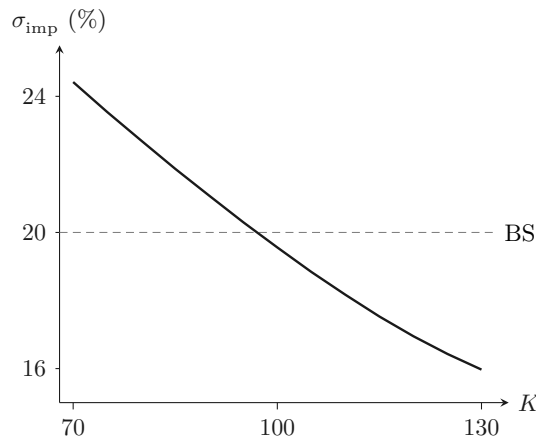
is explicit. The process $t \mapsto f_{(t, \log S_t, v_t)}(u)$ is a Q -martingale, so applying Itô’s formula its dt part must vanish, which gives a PDE for f ; trying the **affine** form

$$f_{(t,x,v)}(u) = \exp(A(T-t) + B(T-t)v + iux)$$

reduces it to Riccati ODEs for A and B , which can be solved in closed form. Call prices then follow by Fourier inversion. This tractability explains the model’s popularity.

Example 7.1 (*A Heston smile*)

With $S_0 = K = 100$, $r = 3\%$, $T = 1$, $v_0 = \kappa = 0.04$ (i.e. 20% volatility), $\theta = 2$, $\gamma = 0.3$ and $\rho = -0.7$, Fourier inversion gives an at-the-money call price of 9.24 (a Monte Carlo simulation with full truncation of the variance, 10^5 paths and daily steps, gives 9.26 ± 0.07). The corresponding Black–Scholes implied volatilities decrease with the strike: the negative correlation produces a **skew**, as observed in equity markets.



7. Volatility modelling

Figure: implied volatility across strikes in the Heston example (bold) against the flat Black–Scholes volatility of 20% (dashed).

7.5. Local-stochastic volatility

Local volatility fits the market exactly but has unrealistic dynamics; stochastic volatility has realistic dynamics but may not fit all quotes exactly. **Local-stochastic volatility** (LSV) models combine them:

$$dS_t = rS_t dt + \ell(t, S_t) \sigma_t S_t dW_t^Q,$$

with σ_t from a stochastic volatility model (e.g. Heston) and a **leverage function** $\ell(t, x)$ calibrated so that the model reproduces the market's local volatilities (via Dupire's formula). Pricing requires numerical methods (usually PDEs). LSV models are standard for exotic equity options and in foreign-exchange markets (Chapter 10).

7.6. Volatility derivatives

Since the late 1990s volatility itself has been traded.

Definition 7.4 (Variance and volatility swaps). With business days $t_0, \dots, t_N$ in $[0, T]$, the annualised **realised variance** is

$$V_T = \frac{252}{N} \sum_{i=1}^N \left(\log \frac{S_{t_i}}{S_{t_{i-1}}} \right)^2.$$

A **variance swap** pays $V_T - K_{\text{var}}$ at T ; a **volatility swap** pays $\sqrt{V_T} - K_\sigma$. Both have zero initial value when the strike is the fair one, $K = \mathbb{E}^Q[\cdot]$.

They give pure exposure to realised volatility. In the Heston model the variance swap is easy: V_T approximates $\frac{1}{T} \int_0^T v_t dt$, and since $\mathbb{E}^Q[v_t] = \kappa + (v_0 - \kappa)e^{-\theta t}$,

$$K_{\text{var}} = \mathbb{E}^Q \left[\frac{1}{T} \int_0^T v_t dt \right] = \kappa + (v_0 - \kappa) \frac{1 - e^{-\theta T}}{\theta T}.$$

For the volatility swap, Jensen's inequality gives $\mathbb{E}^Q[\sqrt{V_T}] \leq \sqrt{\mathbb{E}^Q[V_T]}$ (a convexity adjustment); in Heston it is semi-explicit. Model-free, a variance swap can be replicated statically by a portfolio of out-of-the-money options weighted by $1/K^2$ plus dynamic trading in the underlying — the construction behind the VIX index.

7.7. Exercises

Problems from Hull (H), Chapter 20, Gatheral (Ga), and the course material.

Exercise 7.1

(H 20.1, 20.4) What smile is observed when (a) both tails of the distribution of S_T are lighter than log-normal; (b) the right tail is heavier and the left tail lighter? Explain why a heavy left tail produces a downward-sloping smile.

Solution

- (a) Both wings are cheaper than log-normal: implied volatility is *lower* away from the money (an inverted smile, a “frown”). (b) Out-of-the-money calls (high strikes) are expensive and out-of-the-money puts cheap: implied volatility *increases* with the strike. With a heavy left tail, low-strike puts — and, by parity, low-strike calls — are worth more than in the log-normal model, so their implied volatility is higher: the skew slopes down.

Exercise 7.2

(H 20.5, 20.3) (a) A call’s market price is 3.00 while Black–Scholes with $\sigma = 30\%$ gives 3.50; Black–Scholes with 30% gives 1.00 for the put with the same strike and maturity. What should the put’s market price be? (b) A call and a put with the same strike and maturity have implied volatilities 30% and 25%. What trade would you make?

Solution

- (a) $C_{\text{mkt}} - C_{BS} = P_{\text{mkt}} - P_{BS}$, so $P_{\text{mkt}} = 1.00 - 0.50 = 0.50$. (b) The call is expensive relative to the put: by parity this is an arbitrage. Sell the call, buy the put, buy the stock, borrow the present value of the strike. The position is riskless at maturity and earns the price discrepancy. The argument uses only parity, so it works whatever the true distribution.

Exercise 7.3

(H 20.13) A stock trades at 60; tomorrow a court ruling will move it to 75 (favourable) or 50 (unfavourable). What is the risk-neutral probability of a favourable ruling? What shape of smile do you expect for short-dated options?

Solution

Ignoring one day of interest, $75p + 50(1 - p) = 60$ gives $p = 0.4$. The distribution of S_T is a mixture of two log-normals centred near 75 and 50: bimodal, with little mass near 60. At-the-money options are then *cheap* relative to the wings (a “frown”): the opposite of the usual smile, typical before binary events such as earnings or elections.

Exercise 7.4

(H 20.14) An exchange rate is 0.80 with quoted volatility 12%, equal interest rates. Under the log-normal assumption, estimate the probabilities that in 3 months the rate is below 0.70, in $[0.70, 0.75]$, $[0.75, 0.80]$, $[0.80, 0.85]$, $[0.85, 0.90]$, above 0.90. Which are likely too low given the FX smile?

Solution

With $\log S_T \sim \mathcal{N}(\log 0.8 - \frac{1}{2}\sigma^2 T, \sigma^2 T)$, $\sigma\sqrt{T} = 0.06$: 1.41%, 13.38%, 36.41%, 33.90%, 12.59%, 2.31%. FX smiles indicate heavier tails than log-normal: the extreme buckets (< 0.70 and > 0.90) are underestimated, and the central ones slightly overestimated.

Exercise 7.5

(Breedeen–Litzenberger in BSM) Show that $e^{rT} \partial^2 C / \partial K^2$ is the log-normal risk-neutral density of S_T when C is the Black–Scholes price.

Solution

From $C = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$, $\partial_K C = -e^{-rT} \Phi(d_2)$ (the terms from $\partial_K d_{1,2}$ cancel since $S_0 \varphi(d_1) = K e^{-rT} \varphi(d_2)$), matching $-e^{-rT} Q(S_T > K)$. Then $\partial_K^2 C = e^{-rT} \frac{\varphi(d_2)}{K \sigma \sqrt{T}}$, and $\frac{\varphi(d_2)}{K \sigma \sqrt{T}}$ is the density of $S_0 e^{(r-\sigma^2/2)T + \sigma \sqrt{T}Z}$ at K .

Exercise 7.6

(Dupire consistency) Show that Dupire’s formula returns σ^2 when call prices are given by the Black–Scholes formula with constant σ .

Solution

$\partial_T C = \frac{S_0 \varphi(d_1) \sigma}{2\sqrt{T}} + r K e^{-rT} \Phi(d_2)$, $r K \partial_K C = -r K e^{-rT} \Phi(d_2)$, and $\frac{1}{2} K^2 \partial_K^2 C = \frac{S_0 \varphi(d_1)}{2\sigma \sqrt{T}}$. The ratio is σ^2 .

Exercise 7.7

(Ga 11, variance swaps) In the Heston model with $\theta = 2$, $\kappa = 0.04$, find the fair 1-year variance-swap strike when $v_0 = 0.09$. Why is the fair volatility-swap strike lower than $\sqrt{K_{\text{var}}}$?

Solution

$K_{\text{var}} = 0.04 + 0.05 \frac{1-e^{-2}}{2} = 0.0616$, a volatility of 24.8%. Since $\sqrt{\cdot}$ is concave, Jensen gives $\mathbb{E}[\sqrt{V_T}] \leq \sqrt{\mathbb{E}[V_T]}$: the difference (the *convexity adjustment*) grows with the vol of vol.

Exercise 7.8

(Two-dimensional Itô) In the Heston model compute $d(S_t v_t)$.

Solution

$d(S_t v_t) = v_t dS_t + S_t dv_t + d\langle S, v \rangle_t$ with $d\langle S, v \rangle_t = \rho \gamma v_t S_t dt$, so $d(S_t v_t) = S_t(rv_t + \theta(\kappa - v_t) + \rho \gamma v_t) dt + S_t v_t^{3/2} dW_t^Q + \gamma S_t \sqrt{v_t} d\hat{W}_t^Q$.

7.8. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 7 (`slides_7`). Hull, *Options, Futures, and Other Derivatives*, Chapter 20 (volatility smiles). J. Gatheral, *The Volatility Surface*, Wiley (2006), Chapters 1–2 and 11; Musiela and Rutkowski, *Martingale Methods in Financial Modelling*, Chapter 7. Original references: D. Breedeen and R. Litzenberger, *J. Business* **51**, 621 (1978); B. Dupire, *Pricing with a smile*, *Risk* **7**, 18 (1994); S. Heston, *Rev. Financial Stud.* **6**, 327 (1993). The Heston numbers were computed for these notes.

8. Interest rate modelling

The Black–Scholes model takes the interest rate as a known constant. In reality rates move unpredictably, creating **interest rate risk** — and fixed-income derivatives make up about 80% of the OTC derivatives market. This chapter defines the various interest rates and how they are built from zero-coupon bond prices, prices the linear products (FRAs, floating-rate bonds, swaps) by pure no-arbitrage arguments, introduces the optional ones (caps, floors, swaptions), and then models the short rate with an SDE. Hedging arguments lead to the term-structure PDE and its Feynman–Kac solution; affine models (Vasiček, Cox–Ingersoll–Ross, Hull–White) make bond prices explicit.

8.1. Interest rates in the market

Several rates coexist:

- **Treasury rates:** the rate at which a government borrows in its own currency; traditionally “risk-free”, but not really (sovereign spreads).
- **Interbank rates** (Euribor; formerly Libor, discontinued on 1 July 2023): the rate at which major banks lend to each other for a given **tenor** (1, 3, 6, 12 months); reference rates for an enormous volume of loans and derivatives.
- **Overnight rates:** banks adjust their central-bank reserves by lending and borrowing overnight (€STR in the euro area, the Federal Funds rate in the US).
- **Repo rates:** rates on secured borrowing (repurchase agreements), e.g. SOFR in the US.

The rate applicable to a loan depends on **credit risk**, the risk that the borrower defaults. A truly risk-free rate does not exist, but derivative pricing needs one; **overnight rates are used as proxies** (Chapter 9).

8.2. Zero-coupon bonds and forward rates

The fundamental purpose of the risk-free rate is to measure the **value of time**. Let $P_t(T)$ be the value at t of one unit of currency delivered at $T \geq t$ without risk — the price of a **zero-coupon bond** (ZCB), with $P_T(T) = 1$. The **term structure** at t is the curve $T \mapsto P_t(T)$; modelling it means describing its random evolution over time.

Definition 8.1 (Interest rates from bond prices). For $t \leq S < T$:

- **simple spot rate** for $[S, T]$: $L_S(S, T) = \frac{1}{T - S} \left(\frac{1}{P_S(T)} - 1 \right)$, the rate for lending at S until T ;
- **simple forward rate** for $[S, T]$ contracted at t : $L_t(S, T) = \frac{1}{T - S} \left(\frac{P_t(S)}{P_t(T)} - 1 \right)$;
- **continuously compounded forward rate**: $F_t(S, T) = -\frac{\log P_t(T) - \log P_t(S)}{T - S}$;
- **instantaneous forward rate**: $f_t(T) = -\frac{\partial}{\partial T} \log P_t(T) = \lim_{S \rightarrow T} F_t(S, T)$, the rate contracted at t for $[T, T + dT]$;
- **short rate**: $r_t = f_t(t)$, the rate for $[t, t + dt]$.

Consequently $P_t(T) = \exp(-\int_t^T f_t(u) du)$.

Why the forward rate formula. At t , buy $P_t(S)/P_t(T)$ ZCBs maturing at T and sell one ZCB maturing at S : zero cost. At S pay 1; at T receive $P_t(S)/P_t(T)$. The rate for $[S, T]$ has been locked in at t : $1 + (T - S)L_t(S, T) = P_t(S)/P_t(T)$.

The **bank account** pays the short rate continuously: $dB_t = r_t B_t dt$, $B_0 = 1$, i.e. $B_t = \exp(\int_0^t r_u du)$ — the value of rolling over, at each instant, an investment in ZCBs maturing at $t + dt$. It is *locally* riskless. One can model the ZCB prices $P_t(T)$, the forward rates $f_t(T)$ (the Heath–Jarrow–Morton approach), the simple forward rates $L_t(S, T)$ (LIBOR market models), or the short rate r_t and deduce everything else by no arbitrage. This chapter follows the last route.

8.3. Linear interest rate products

8.3.1. Forward rate agreements

Rates change: the 1-year Euribor fixed in one year, $L_1(1, 2)$, may differ from today's $L_0(0, 1)$. A borrower paying a floating rate faces interest rate risk.

Definition 8.2 (Forward rate agreement). A **FRA** for the period $[S, T]$ with fixed rate K and notional N exchanges at T the floating rate $L_S(S, T)$ (received, the *floating leg*) against K (paid, the *fixed leg*):

$$\text{payoff at } T = N(T - S)(L_S(S, T) - K).$$

Example 8.1 (Hedging a floating loan)

Today you arrange to borrow 10 000 € in one year, repaying in two years at the 1-year Euribor fixed in one year. A FRA on $[1, 2]$ with notional 10 000 € and rate K pays 10 000 $L_1(1, 2)$ against 10 000 K : the net cost of the loan is 10 000 K , fixed today.

Proposition 8.1 (Value of a FRA). For $t \leq S$ (unit notional),

$$\Pi_t^{FRA} = P_t(S) - (1 + (T - S)K)P_t(T) = (T - S)(L_t(S, T) - K)P_t(T).$$

The FRA has zero value iff $K = L_t(S, T)$: the forward rate plays the role of the forward price.

Proof. The payoff $(T - S)(L_S(S, T) - K) = \frac{1}{P_S(T)} - 1 - (T - S)K$ is known at S , so its value at S is $P_S(T)$ times it: $1 - (1 + (T - S)K)P_S(T)$ — one ZCB maturing at S minus $1 + (T - S)K$ ZCBs maturing at T . By the law of one price this holds at every $t \leq S$. $\square$

8.3.2. Floating-rate bonds and swaps

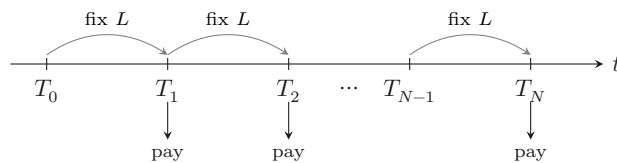


Figure: payment schedule. The floating rate for $[T_{k-1}, T_k]$ is fixed at T_{k-1} and paid at T_k .

Definition 8.3 (Floating-rate bond). A floating-rate bond with dates $0 \leq T_0 < T_1 < \dots < T_N$ pays at each T_k the coupon $c_k = (T_k - T_{k-1})(L_{T_{k-1}}(T_{k-1}, T_k) + \alpha)$, with a fixed **spread** α (in basis points), plus the principal 1 at T_N .

Proposition 8.2 (Price of a floating-rate bond).

$$\Pi_0(\alpha) = P_0(T_0) + \alpha \sum_{k=1}^N (T_k - T_{k-1}) P_0(T_k).$$

In particular, a floater with zero spread is worth exactly $P_0(T_0)$: par at the fixing date.

Proof. Split each floating coupon as $(T_k - T_{k-1})(L_{T_{k-1}}(T_{k-1}, T_k) - L_0(T_{k-1}, T_k)) + (T_k - T_{k-1})L_0(T_{k-1}, T_k)$. The first part is a FRA struck at the forward rate, worth zero. The second is a known amount paid at T_k , worth $(\frac{P_0(T_{k-1})}{P_0(T_k)} - 1)P_0(T_k) = P_0(T_{k-1}) - P_0(T_k)$. Summing telescopes to $P_0(T_0) - P_0(T_N)$. The spreads are worth $\alpha \sum_k (T_k - T_{k-1})P_0(T_k)$, and the principal $P_0(T_N)$. $\square$

Definition 8.4 (Interest rate swap). An **IRS** with dates $T_0 < \dots < T_N$ and fixed rate S exchanges at each T_k the net amount $(T_k - T_{k-1})(L_{T_{k-1}}(T_{k-1}, T_k) - S)$ (payer swap: receive floating, pay fixed). It is a portfolio of N FRAs with common rate $K = S$.

Proposition 8.3 (Swap rate). The fixed rate making the swap worthless at $t = 0$ is

$$S(T_0, \dots, T_N) = \frac{P_0(T_0) - P_0(T_N)}{\sum_{k=1}^N (T_k - T_{k-1}) P_0(T_k)}.$$

The denominator $A_0 = \sum_k (T_k - T_{k-1}) P_0(T_k)$ is the **annuity** (or **PVBP**).

Proof. The swap's cash flows are those of a floater with spread $\alpha = -S$ minus the principal, worth $\Pi_0(-S) - P_0(T_N) = P_0(T_0) - P_0(T_N) - S A_0$; set it to zero. $\square$

For example, with a flat zero curve at 3% (continuous) and annual payments over 5 years, $S = (1 - e^{-0.15}) / \sum_{k=1}^5 e^{-0.03k} = 3.045\%$ — the annually compounded equivalent of 3%.

8.4. Options on rates: caps, floors, swaptions

Definition 8.5 (Caplets, caps, floorlets, floors). A **caplet** on $[S, T]$ with strike K pays $(T - S)(L_S(S, T) - K)^+$ at T ; a **floorlet** pays $(T - S)(K - L_S(S, T))^+$. A **cap (floor)** with dates $T_0, \dots, T_N$ is the sum of the caplets (floorlets) on $[T_{k-1}, T_k]$.

A borrower paying a floating rate who buys a cap never pays more than K ; a lender receiving floating who buys a floor never receives less than K . A caplet is an option on a FRA, a cap a portfolio of caplets.

Proposition 8.4 (Caplets are puts on bonds). The caplet is worth, at time S ,

$$\Pi_S^{cpl} = (1 + (T - S)K) \left(\frac{1}{1 + (T - S)K} - P_S(T) \right)^+,$$

i.e. $1 + (T - S)K$ European **puts** with maturity S on the ZCB maturing at T , with strike $1/(1 + (T - S)K)$. Similarly a floorlet is a portfolio of **calls** on a ZCB.

Proof. $(T - S)(L_S(S, T) - K)^+ = (\frac{1}{P_S(T)} - (1 + (T - S)K))^+$ is known at S ; multiplying by $P_S(T)$ gives the value at S : $(1 - (1 + (T - S)K)P_S(T))^+$. $\square$

Definition 8.6 (Swaption). A (payer) **swaption** gives the right to enter at T_0 into the swap with fixed rate S . Its payoff at T_0 is

$$(S(T_0, \dots, T_N)_{T_0} - S)^+ \sum_{k=1}^N (T_k - T_{k-1}) P_{T_0}(T_k),$$

where $S(T_0, \dots, T_N)_{T_0}$ is the swap rate prevailing at T_0 : the swap is entered only if the market rate exceeds S .

Equivalently a swaption is an option on a portfolio of ZCBs, i.e. on a coupon bond. Their market pricing is described below.

8.4.1. Market practice: Black's formula

Traders quote caplets and swaptions through **Black's (1976) formula**, a log-normal model for the *underlying rate* rather than for the short rate. For a caplet on $[T_{k-1}, T_k]$ with accrual τ_k , strike K , forward rate $F_k = L_0(T_{k-1}, T_k)$ and volatility σ_k :

$$\text{Cpl}_0 = N\tau_k P_0(T_k) [F_k \Phi(d_1) - K \Phi(d_2)], \quad d_{1,2} = \frac{\log(F_k/K) \pm \frac{1}{2}\sigma_k^2 T_{k-1}}{\sigma_k \sqrt{T_{k-1}}}.$$

For a payer swaption on a swap starting at T_0 , with forward swap rate $S_0 = S(T_0, \dots, T_N)$ and annuity $A_0 = \sum_k \tau_k P_0(T_k)$:

$$\text{Swpt}_0 = N A_0 [S_0 \Phi(d_1) - K \Phi(d_2)], \quad d_{1,2} = \frac{\log(S_0/K) \pm \frac{1}{2}\sigma^2 T_0}{\sigma \sqrt{T_0}}.$$

These are not ad hoc. Under the T_k -**forward measure** Q^{T_k} — the probability with numeraire $P_t(T_k)$ instead of the bank account — the forward rate $L_t(T_{k-1}, T_k) = \frac{1}{\tau_k} (\frac{P_t(T_{k-1})}{P_t(T_k)} - 1)$ is a ratio of a traded asset to the numeraire, hence a martingale, and

$$\text{Cpl}_0 = N\tau_k P_0(T_k) \mathbb{E}^{Q^{T_k}} [(L_{T_{k-1}}(T_{k-1}, T_k) - K)^+].$$

If the forward rate is log-normal under Q^{T_k} , this is exactly Black's formula (the LIBOR market model makes this assumption for all forward rates simultaneously). Likewise the swap rate is a martingale under the **annuity measure** with numeraire A_t . Caps are then quoted by a single **flat volatility** applied to all their caplets, or by caplet-specific **spot volatilities**. When rates can be negative, the **Bachelier (normal) model**, $dF = \sigma_N dW$, replaces the log-normal one.

8.5. Short-rate models

Model the short rate under $\mathbb{P}$ by

$$dr_t = \mu(t, r_t) dt + \sigma(t, r_t) dW_t.$$

ZCBs are taken as the fundamental traded assets, for every maturity T . Since r is Markov, look for prices of the form $P_t(T) = F^T(t, r_t)$.

8.5.1. The term-structure PDE

1. **Portfolio of two bonds.** Fix maturities $S < T$ and a self-financing $\theta = (\theta^S, \theta^T)$ in the two ZCBs: $V_t = \theta_t^S P_t(S) + \theta_t^T P_t(T)$, $dV_t = \theta_t^S dP_t(S) + \theta_t^T dP_t(T)$, where by Itô's formula

$$dF^T(t, r_t) = \left(F_t^T + \mu F_r^T + \frac{1}{2} \sigma^2 F_{rr}^T \right) dt + \sigma F_r^T dW_t.$$

2. **Make it riskless.** Choose $\theta_t^S = -\theta_t^T F_r^T / F_r^S$, which kills the dW_t term.
3. **No arbitrage.** A riskless portfolio must earn the short rate: $dV_t = r_t V_t dt$. After simplification,

$$\frac{F_t^T + \frac{1}{2} \sigma^2 F_{rr}^T - r F^T}{F_r^T} = \frac{F_t^S + \frac{1}{2} \sigma^2 F_{rr}^S - r F^S}{F_r^S}.$$

4. **Separation.** Since S and T are arbitrary, both sides equal a function $q(t, r_t)$ independent of maturity. Writing $q = \lambda\sigma - \mu$:

Theorem 8.5 (Term-structure PDE). *For every $T > 0$,*

$$F_t^T + (\mu - \lambda\sigma) F_r^T + \frac{1}{2} \sigma^2 F_{rr}^T - r F^T = 0, \quad F^T(T, r) = 1,$$

where $\lambda(t, r)$ is the **market price of interest rate risk**.

Why does λ appear? In BSM there is one source of risk and one exogenously given risky asset to hedge it; here the only exogenous asset is the bank account, and the short rate is *not* a traded asset. The real-world dynamics of r alone do not determine bond prices: one must also know λ , which is set by the market — in practice, by calibration.

Theorem 8.6 (Feynman–Kac with stochastic rates). *The solution of the term-structure PDE is*

$$P_t(T) = F^T(t, r_t) = \mathbb{E}^Q \left[e^{-\int_t^T r_u du} \mid r_t \right],$$

where under Q the short rate follows $dr_t = (\mu - \lambda\sigma)(t, r_t) dt + \sigma(t, r_t) dW_t^Q$.

Under this **risk-neutral** probability every ZCB has expected return r_t . Prices depend on μ and λ only through the **risk-neutral drift** $\mu^Q = \mu - \lambda\sigma = -q$. Hence the usual practice of **risk-neutral modelling**: specify r directly under Q and calibrate its parameters to market prices.

8.5.2. Affine models

Definition 8.7 (Affine short-rate model). A model $dr_t = \mu^Q(t, r_t) dt + \sigma(t, r_t) dW_t^Q$ is **affine** if

$$\mu^Q(t, r) = \alpha_0 + \alpha_1 r, \quad \sigma^2(t, r) = \beta_0 + \beta_1 r.$$

Try the exponential-affine ansatz $F^T(t, r) = \exp(A(T-t) - B(T-t)r)$. Substituting into the PDE (with $\tau = T-t$, so $\partial_t = -\partial_\tau$),

$$-A'(\tau) - (1 - B'(\tau))r - (\alpha_0 + \alpha_1 r)B(\tau) + \frac{1}{2}(\beta_0 + \beta_1 r)B(\tau)^2 = 0.$$

This must hold for all r , so the constant and the coefficient of r vanish separately:

$$B'(\tau) = 1 + \alpha_1 B(\tau) - \frac{1}{2} \beta_1 B(\tau)^2, \quad A'(\tau) = -\alpha_0 B(\tau) + \frac{1}{2} \beta_0 B(\tau)^2, \quad A(0) = B(0) = 0.$$

The equation for B is a **Riccati ODE**; A then follows by integration. The PDE has been reduced to ODEs, solvable explicitly in the classical cases.

8.5.3. Vasiček

Definition 8.8 (Vasiček (1977)).

$$dr_t = \theta(\kappa - r_t) dt + \gamma dW_t^Q,$$

with speed of mean reversion θ , long-run mean κ , volatility γ .

This is an Ornstein–Uhlenbeck process (Chapter 4):

$$r_t = r_0 e^{-\theta t} + \kappa(1 - e^{-\theta t}) + \gamma \int_0^t e^{-\theta(t-u)} dW_u^Q,$$

Gaussian with $\mathbb{E}^Q[r_t] = r_0 e^{-\theta t} + \kappa(1 - e^{-\theta t})$ and $\text{Var}^Q(r_t) = \frac{\gamma^2}{2\theta}(1 - e^{-2\theta t})$. The rate can become negative — once seen as a defect, but a feature after 2014. Here $\alpha_0 = \theta\kappa$, $\alpha_1 = -\theta$, $\beta_0 = \gamma^2$, $\beta_1 = 0$, and

$$B(\tau) = \frac{1 - e^{-\theta\tau}}{\theta}, \quad A(\tau) = \left(\kappa - \frac{\gamma^2}{2\theta^2}\right)(B(\tau) - \tau) - \frac{\gamma^2}{4\theta}B(\tau)^2.$$

(The course slides print $\kappa - \gamma^2/2$ in the first factor; the correct factor is $\kappa - \gamma^2/(2\theta^2)$.) Caplets and floorlets have Black–Scholes-like closed forms, since bond prices are log-normal.

8.5.4. Cox–Ingersoll–Ross

Definition 8.9 (Cox–Ingersoll–Ross (1985)).

$$dr_t = \theta(\kappa - r_t) dt + \gamma\sqrt{r_t} dW_t^Q.$$

The rate stays non-negative, and strictly positive if $2\kappa\theta \geq \gamma^2$ (**Feller condition**). r_t follows a non-central chi-square distribution, with the same mean as in Vasiček and variance

$$\text{Var}^Q(r_t) = r_0 \frac{\gamma^2}{\theta}(e^{-\theta t} - e^{-2\theta t}) + \frac{\kappa\gamma^2}{2\theta}(1 - e^{-\theta t})^2.$$

It is the same SDE as the variance in the Heston model. With $h = \sqrt{\theta^2 + 2\gamma^2}$,

$$B(\tau) = \frac{2(e^{h\tau} - 1)}{2h + (\theta + h)(e^{h\tau} - 1)}, \quad A(\tau) = \frac{2\kappa\theta}{\gamma^2} \log \frac{2he^{(\theta+h)\tau/2}}{2h + (\theta + h)(e^{h\tau} - 1)}.$$

Caplets and floorlets are semi-explicit.

Example 8.2 (Two models, one bond)

With $r_0 = 3\%$, $\theta = 0.5$, $\kappa = 4\%$ and $T = 5$: Vasiček with $\gamma = 1\%$ gives $P_0(5) = 0.8343$ (yield 3.62%); CIR with $\gamma = 5\%$ (Feller holds: $0.04 > 0.0025$) gives $P_0(5) = 0.8342$, confirmed by a Monte Carlo simulation (0.8342). The yield lies between r_0 and κ , pulled down slightly by the convexity term $-\gamma^2/(2\theta^2)$.

8.5.5. Hull–White: fitting today’s curve

Vasiček and CIR have three or four parameters and cannot reproduce an arbitrary observed term structure. The **Hull–White** model (extended Vasiček) makes the drift time-dependent:

$$dr_t = (\alpha(t) - \theta r_t)dt + \gamma dW_t^Q,$$

still Gaussian and mean-reverting, with $\mathbb{E}^Q[r_t] = r_0 e^{-\theta t} + \int_0^t e^{-\theta(t-u)} \alpha(u) du$. It is affine with $B(\tau) = (1 - e^{-\theta\tau})/\theta$ as in Vasiček and

$$A(T-t) = \int_t^T \left(\frac{\gamma^2}{2} B(T-u)^2 - \alpha(u) B(T-u) \right) du.$$

The model forward curve is $f_0(T) = -\partial_T \log P_0(T) = -\partial_T A(T) + \partial_T B(T) r_0$. Requiring $f_0(T) = f_0^{\text{mkt}}(T)$ for all T determines α uniquely:

$$\alpha(T) = \frac{\partial}{\partial T} f_0^{\text{mkt}}(T) + \theta f_0^{\text{mkt}}(T) + \frac{\gamma^2}{2\theta} (1 - e^{-2\theta T}).$$

The model then reproduces today’s curve exactly by construction, and θ, γ are calibrated to option prices (caplets, swaptions), which also have closed forms.

i Note

Short-rate models are the natural first step but have limitations: one factor implies perfectly correlated rates of all maturities. Richer approaches model the whole forward curve (Heath–Jarrow–Morton) or the market forward rates directly (LIBOR/forward market models). After 2008 the spread between interbank and overnight rates forced a **multi-curve** framework, and the 2023 end of Libor moved markets to backward-looking risk-free rates — the subject of Chapter 9.

8.6. Exercises

Problems from Hull (H), Chapters 7, 29 and 31, and Björk (B), Chapters 22–23.

Exercise 8.1

(B 22.1) In a FRA with continuous compounding, the lender pays K at S and receives $Ke^{R^*(T-S)}$ at T . (a) Value the contract at $t < S$ in terms of ZCB prices. (b) Show that it is worth zero iff R^* equals the forward rate.

Solution

(a) $\Pi_t = -KP_t(S) + Ke^{R^*(T-S)}P_t(T)$. (b) $\Pi_t = 0$ iff $e^{R^*(T-S)} = P_t(S)/P_t(T)$, i.e. $R^* = -\frac{\log P_t(T) - \log P_t(S)}{T-S} = F_t(S, T)$.

Exercise 8.2

(H 7.2) A swap on 100 million has 10 months to run: six-month Libor is received against 4% (semiannual) paid. Forward Libor rates for all maturities are 3% (semiannual); the Libor fixed two months ago was 2.4%; OIS rates are 2.7% (continuous) for discounting. Value the swap for both parties.

8. Interest rate modelling

Solution

Payments in 4 and 10 months. Fixed leg: 2m each. Floating: $0.5 \times 2.4\% \times 100 = 1.2\text{m}$ (already fixed) and 1.5m (at the forward rate). To the party paying floating and receiving fixed: $(2 - 1.2)e^{-0.027/3} + (2 - 1.5)e^{-0.027 \times 10/12} = 1.282\text{m}$. To the party paying fixed: -1.282m .

Exercise 8.3

(H 29.1) A company caps 3-month Libor at 2% on 20 million. At a reset date the rate is 4%. What payment results, and when?

Solution

$20\,000\,000 \times 0.25 \times (0.04 - 0.02) = 100\,000$, paid three months after the reset date (at the end of the accrual period).

Exercise 8.4

(H 29.5) Price a caplet on the 3-month rate for the period from 15 to 18 months, struck at 13% (quarterly compounding), on a principal of 1000. The forward rate is 12%, the 18-month zero rate is 11.5% (continuous), and the forward-rate volatility is 12%.

Solution

Black's formula with $F = 0.12$, $K = 0.13$, $T_{k-1} = 1.25$, $\sigma = 0.12$: $d_1 = \frac{\log(12/13) + \frac{1}{2} \cdot 0.12^2 \times 1.25}{0.12\sqrt{1.25}} = -0.5295$, $d_2 = -0.6637$, so $F\Phi(d_1) - K\Phi(d_2) = 0.12 \times 0.2982 - 0.13 \times 0.2534 = 0.00284$. The price is $1000 \times 0.25 \times e^{-0.115 \times 1.5} \times 0.00284 = 0.60$.

Exercise 8.5

(H 29.14) Value a European swaption giving the right to enter in 4 years a 3-year annual-pay swap paying 5% fixed on 10 million. All swap rates are 5%, OIS rates 4.7% (continuous), swap-rate volatility 20%.

Solution

Annuity $A = e^{-0.047 \times 5} + e^{-0.047 \times 6} + e^{-0.047 \times 7} = 2.2645$. At the money ($S_0 = K = 5\%$): $d_{1,2} = \pm \frac{1}{2} \times 0.2 \times 2 = \pm 0.2$, and the swaption is worth $10^7 \times 2.2645 \times 0.05 (\Phi(0.2) - \Phi(-0.2)) = 179\,500$.

Exercise 8.6

(H 29.11, B 23.3) (a) Derive a put-call parity for swaptions. (b) Find the fixed rate R that makes a contract exchanging $K(e^{RT} - 1)$ against $K(e^{\int_0^T r_s ds} - 1)$ at T worthless.

Solution

(a) Payer minus receiver swaption with the same strike K pays $(S_{T_0} - K)^+ A_{T_0} - (K - S_{T_0})^+ A_{T_0} = (S_{T_0} - K)A_{T_0}$, the value of a forward-starting payer swap at rate K : Payer - Receiver = $A_0(S_0 - K)$. (b) The value is $K(e^{RT} P_0(T) - \mathbb{E}^Q[e^{-\int_0^T r_s ds}]) = K(e^{RT} P_0(T) - 1)$, zero iff $R = -\log P_0(T)/T$: the T -year zero rate, whatever the short-rate model.

Exercise 8.7

(H 31.4) In the risk-neutral Vasiček model with speed 0.1, level 0.03 and volatility 0.01, price a 5-year ZCB with principal 1 when $r_0 = 2\%$.

Solution

$B(5) = (1 - e^{-0.5})/0.1 = 3.935$, $A(5) = (0.03 - 0.01^2/0.02)(3.935 - 5) - 0.01^2 \times 3.935^2/0.4 = -0.03050$, so $P = e^{A - Br_0} = e^{-0.03050 - 0.0787} = 0.8966$.

Exercise 8.8

(H 31.9) In both Vasiček and CIR take speed 0.1, level 0.1, $r_0 = 10\%$, and an initial short-rate volatility of 2% (so $\gamma = 0.02$ in Vasiček and $\gamma\sqrt{0.1} = 0.02$ in CIR). Compare the 10-year ZCB prices.

Solution

Vasiček: 0.3805. CIR with $\gamma = 0.0632$: 0.3799. The models agree closely when calibrated to the same initial volatility; they differ more for long maturities and for distributions of future rates (CIR cannot go negative; its volatility falls as rates fall).

Exercise 8.9

(B 22.5) Show that $f(0, T) = y(0, T) + T \partial_T y(0, T)$, where y is the zero-coupon yield, and that for an increasing yield curve $y_M(0, T) \leq y(0, T) \leq f(0, T)$, y_M being the yield of a coupon bond. Interpret.

Solution

$P(0, T) = e^{-Ty(0, T)}$, so $f = -\partial_T \log P = y + Ty'$. If $y' \geq 0$, then $f \geq y$. A coupon bond's yield is a weighted average of the zero yields at its payment dates, all $\leq T$, so $y_M \leq y(0, T)$. The forward rate is the *marginal* rate at T , the zero yield the *average* over $[0, T]$, and the coupon yield a further average — marginal above average when the curve rises.

Exercise 8.10

(B 22.7) A consol pays 1 at $t = 1, 2, \dots$ forever, with a flat continuously compounded yield y . Compute its price, duration and convexity.

Solution

With $x = e^{-y}$: price $P = \sum_{n \geq 1} x^n = \frac{x}{1-x} = \frac{1}{e^y - 1}$. Duration $D = \frac{1}{P} \sum nx^n = \frac{1}{1-x} = \frac{e^y}{e^y - 1}$ (also $-P'(y)/P$). Convexity $\frac{1}{P} \sum n^2 x^n = \frac{1+x}{(1-x)^2}$. For $y = 5\%$: $P = 19.50$, $D = 20.50$ years, $C = 820$.

Exercise 8.11

(Vasiček bond price, directly) Show that $\int_t^T r_u du$ is Gaussian in the Vasiček model and recover $P_t(T) = e^{A(\tau) - B(\tau)r_t}$ from $\mathbb{E}^Q[e^{-\int_t^T r_u du} | r_t]$.

Solution

It is a linear functional of a Gaussian process, with mean $m = \kappa\tau + (r_t - \kappa)B(\tau)$ and variance $v = \frac{\gamma^2}{\theta^2}(\tau - 2B(\tau) + \frac{1 - e^{-2\theta\tau}}{2\theta})$, $\tau = T - t$. Then $\mathbb{E}[e^{-X}] = e^{-m+v/2}$, which simplifies to $e^{A(\tau) - B(\tau)r_t}$.

8.7. Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 8 ([slides_8](#)). Björk, *Arbitrage Theory in Continuous Time*, Chapters 22–24; Hull, *Options, Futures, and Other Derivatives*, Chapters 7, 29 and 31; Musiela and Rutkowski, *Martingale Methods in Financial Modelling*, Chapters 9–10; further reading: D. Brigo and F. Mercurio, *Interest Rate Models: Theory and Practice*, 2nd ed. Original references: O. Vasiček, J. Financial Econ. **5**, 177 (1977); J. Cox, J. Ingersoll and S. Ross, *Econometrica* **53**, 385 (1985); J. Hull and A. White, *Rev. Financial Stud.* **3**, 573 (1990). The numerical example was computed for these notes.

Part III.

Part III: Practitioner lectures

9. Risk-free rates in practice

Chapter 8 treated the risk-free curve as a given mathematical object. In practice it has to be **built** from market quotes, under precise conventions, and since 2023 from a new family of benchmarks. This chapter, based on a practitioner lecture by Vincenzo Russo (Assicurazioni Generali), covers market conventions, the transition from interbank offered rates (IBORs) to overnight risk-free rates (RFRs), the pricing of overnight indexed swaps, swaptions and caplets on **backward-looking** rates, the bootstrap of an RFR curve, and the use of risk-free curves in the insurance sector under Solvency II.

9.1. The fixed-income market

The fixed-income market trades debt securities that pay predetermined cash flows: government bonds, corporate and covered bonds, municipal bonds, securitised products (MBS, ABS), money-market instruments. Issuers are sovereigns and financial or non-financial corporates; investors (asset managers, insurers, banks, pension funds) seek income, capital preservation, diversification and **liability matching**. Bonds differ by maturity (money market below 1 year, short 1–3, intermediate 3–7, long beyond 7), currency, structure (fixed, floating, inflation-linked, callable or putable, securitised) and credit rating (AAA down to D).

Prices and yields move inversely. The total yield of a risky bond decomposes as

$$\text{yield} = \text{risk-free rate} + \text{credit spread} + \text{liquidity premium},$$

so the risk-free curve is the reference for everything: **discounting** cash flows (time value of money without credit risk), **forwarding** floating rates (forward rates implied by no arbitrage), risk measures (duration, DV01, convexity), the risk-neutral drift in option pricing, and, for insurers, the fair value of both assets and liabilities.

Risk-free proxies. No asset is truly risk-free. Proxies are short-term rates anchored by central banks — government bills of top-rated sovereigns, and the new overnight benchmarks €STR (euro), SOFR (US), SONIA (UK), SARON (Switzerland), TONA (Japan) — and the derivatives written on them: futures, overnight indexed swaps (OIS). Risk-free rates are nominal: real rate plus expected inflation.

9.2. Market conventions

Day counts. The **year fraction** $\tau(t, T)$ between two dates depends on the day-count convention: ACT/360 (actual days divided by 360, standard for money markets), ACT/365, ACT/ACT, 30/360. For ACT/360, $\tau(T_{i-1}, T_i) = (T_i - T_{i-1})/360$ with dates counted in days.

Business-day rules adjust payment dates falling on holidays: *Following* (next business day), *Modified Following* (next business day unless it crosses into the next month, then the previous one), *Unadjusted*.

Compounding. A ZCB price $P(t, T)$ defines a spot rate in each convention:

compounding	definition	rate from price
simple	$P = \frac{1}{1 + \tau R_s}$	$R_s = \frac{1}{\tau} \left(\frac{1}{P} - 1 \right)$

compounding	definition	rate from price
annual	$P = (1 + R_a)^{-\tau}$	$R_a = P^{-1/\tau} - 1$
k times a year	$P = \left(1 + \frac{R_k}{k}\right)^{-k\tau}$	$R_k = k(P^{-1/(k\tau)} - 1)$
continuous	$P = e^{-R_c\tau}$	$R_c = -\frac{\log P}{\tau}$

with $\tau = \tau(t, T)$. The **zero curve** $T \mapsto R_g(t, T)$ depends on the convention g ; the **discount curve** $T \mapsto P(t, T)$ does not, and is decreasing from $P(t, t) = 1$ when rates are positive.

Interpolation. Quotes exist only for some maturities, and are sparse or illiquid at the long end. Missing points are interpolated — linearly, by cubic splines, or with parametric families such as Nelson–Siegel. For par rates $S_{T_{i-1}}, S_{T_i}$ quoted at consecutive maturities, linear interpolation gives, for $T_{i-1} < T_h < T_i$,

$$S_{T_h} = \frac{S_{T_{i-1}}(T_i - T_h) + S_{T_i}(T_h - T_{i-1})}{T_i - T_{i-1}}.$$

Bonds. A fixed-rate bond with coupon rate K and dates $T_1, \dots, T_n$ is worth

$$B_{fx}(t, T_n) = \sum_{i=1}^n P(t, T_i) \tau(T_{i-1}, T_i) K + P(t, T_n).$$

For a **defaultable** issuer, market practice discounts with the risk-free zero rate plus a **zero spread** S : with annual compounding $\bar{P}(t, T) = (1 + R_a(t, T) + S)^{-\tau(t, T)}$, and the bond price uses $\bar{P}$ in place of P .

9.3. From IBORs to RFRs

IBORs (Libor, Euribor) were for decades the benchmark for loans, notes and derivatives, and together with overnight rates (Fed Funds, EONIA) were treated as risk-free. During the 2007–2009 crisis the IBOR–overnight spread widened dramatically (Chapter 1), showing that IBORs contain bank credit and liquidity risk; IBOR quotes were also manipulated. Hence:

- 2013–2014: the Financial Stability Board reviews the major benchmarks and recommends alternatives;
- 2017: the UK FCA announces it will stop supporting Libor panels;
- 2019: ISDA establishes fallbacks to transaction-based overnight RFRs;
- Libor is discontinued (USD Libor on 30 June 2023), replaced by SOFR, SONIA, SARON, TONA; in the euro area €STR replaced EONIA, while **Euribor survives** in reformed, transaction-based form.

The two kinds of rates differ structurally:

- **IBORs are forward-looking term rates:** quoted for tenors from one week to one year as simple rates, fixed at the *start* of the accrual period — a market expectation.
- **RFRs are backward-looking overnight rates:** published daily, with no term structure of their own; the rate for a period is obtained by **compounding the daily fixings in arrears**, so it is known only at the *end* of the period.

Correspondingly, three pricing environments have succeeded one another in a single currency:

1. **Single curve, IBOR** (before 2007): IBOR rates for both discounting and forwarding.
2. **Dual curve** (after 2007): overnight rates (EONIA, Fed Funds, now €STR, SOFR) for discounting, IBOR for forwarding — still the standard for Euribor derivatives.
3. **Single curve, RFR, backward-looking:** the RFR curve for both discounting and forwarding. This is the setting of the rest of the chapter.

9.4. Backward-looking rates

Let $C(t) = \exp(\int_0^t r(u) du)$ be the cash account driven by the instantaneous risk-free rate (Chapter 8), so that $P(t, T) = \mathbb{E}^Q[e^{-\int_t^T r(u) du} | \mathcal{F}_t]$.

Definition 9.1 (Compounded-in-arrears RFR). For an accrual period $[T_{i-1}, T_i)$ with d_i business days, daily fixings $r(i, j)$ and day-count fractions $\delta(i, j)$,

$$R(T_{i-1}, T_i) = \frac{1}{\tau(T_{i-1}, T_i)} \left[\prod_{j=1}^{d_i} (1 + r(i, j)\delta(i, j)) - 1 \right] \approx \frac{1}{\tau(T_{i-1}, T_i)} \left[\frac{C(T_i)}{C(T_{i-1})} - 1 \right],$$

the continuous-time version being the limit of vanishing daily steps.

Proposition 9.1 (Forward value of a backward-looking rate). For $t \leq T_{i-1}$, the expectation of $R(T_{i-1}, T_i)$ under the T_i -forward measure (whose numeraire is $P(\cdot, T_i)$) is

$$R(t, T_{i-1}, T_i) = \frac{1}{\tau(T_{i-1}, T_i)} \left(\frac{P(t, T_{i-1})}{P(t, T_i)} - 1 \right),$$

the same formula as the simple forward rate of the IBOR world. For $T_{i-1} < t < T_i$ part of the rate is already known:

$$R(t, T_{i-1}, T_i) = \frac{1}{\tau(T_{i-1}, T_i)} \left(\frac{e^{\int_{T_{i-1}}^t r(u) du}}{P(t, T_i)} - 1 \right).$$

Proof. Under the T_i -forward measure, $\mathbb{E}^{T_i}[X | \mathcal{F}_t] = \mathbb{E}^Q[e^{-\int_t^{T_i} r(u) du} X | \mathcal{F}_t] / P(t, T_i)$. With $X = C(T_i)/C(T_{i-1})$, $e^{-\int_t^{T_i} r(u) du} X = e^{-\int_t^{T_{i-1}} r(u) du}$, whose Q -expectation is $P(t, T_{i-1})$. $\square$

9.4.1. Overnight indexed swaps

An **OIS** exchanges a fixed rate K against the compounded RFR. A payer OIS with floating dates $T_1, \dots, T_n$, fixed dates $T'_1, \dots, T'_m$ (same start and end), notional N , valued at $t < T_0$:

$$OIS_R(t, T_0, T_n) = N \sum_{i=1}^n P(t, T_i) \tau(T_{i-1}, T_i) R(t, T_{i-1}, T_i) - N \sum_{j=1}^m P(t, T'_j) \tau(T'_{j-1}, T'_j) K.$$

By the proposition, before the start this coincides with the single-curve Libor swap formula (and the floating leg telescopes to $N(P(t, T_0) - P(t, T_n))$). Once an accrual period has started, the known part of the compounded rate must be included: the first floating coupon becomes $N(e^{\int_{T_0}^t r(u) du} - P(t, T_1))$.

9.4.2. Swaptions

Assume the forward swap rate $S_R(t, T_0, T_n)$ is a log-normal martingale under the **swap (annuity) measure**, with numeraire the annuity $A(t) = \sum_j P(t, T'_j) \tau(T'_{j-1}, T'_j)$:

$$\frac{dS_R(t, T_0, T_n)}{S_R(t, T_0, T_n)} = \sigma_S dW_t.$$

Then the payer swaption follows **Black's (1976) formula**

$$PSWT_R(t) = N A(t) [S_R \Phi(d_1) - K \Phi(d_2)], \quad d_{1,2} = \frac{\log(S_R/K) \pm \frac{1}{2} \sigma_S^2 (T_0 - t)}{\sigma_S \sqrt{T_0 - t}}.$$

Swaption valuation is **unaffected** by whether the underlying floating rate is forward- or backward-looking: what matters is the swap rate at T_0 .

9.4.3. Caplets on backward-looking rates

A caplet on $[T_{i-1}, T_i]$ with strike K pays $N\tau(T_{i-1}, T_i)(R(T_{i-1}, T_i) - K)^+$ at T_i . The underlying rate keeps accruing during the period, so its uncertainty is resolved only at T_i . Lyashenko and Mercurio (2019) model the forward rate as a log-normal martingale under the (extended) T_i -forward measure, with a volatility that decays linearly during the accrual period:

$$\frac{dR(t, T_{i-1}, T_i)}{R(t, T_{i-1}, T_i)} = \sigma_R \min\left(\frac{(T_i - t)^+}{T_i - T_{i-1}}, 1\right) dW_t.$$

(The *extended* ZCB price — the bond maturing at T_i , with proceeds rolled over at the RFR after T_i — keeps the numeraire defined for $t > T_i$.) This gives a Black formula

$$CPLT_R(t) = N\tau(T_{i-1}, T_i)P(t, T_i)[R(t, T_{i-1}, T_i)\Phi(d_1) - K\Phi(d_2)], \quad d_{1,2} = \frac{\log(R/K) \pm \frac{1}{2}\Sigma_R^2}{\Sigma_R},$$

with total variance

$$\Sigma_R^2 = \sigma_R^2 \int_t^{T_i} \min\left(\frac{(T_i - s)^+}{T_i - T_{i-1}}, 1\right)^2 ds = \sigma_R^2 \left[\max(T_{i-1} - t, 0) + \frac{(T_i - \max(T_{i-1}, t))^3}{3(T_i - T_{i-1})^2} \right].$$

Since a forward-looking caplet's rate is fixed at T_{i-1} (variance $\sigma^2(T_{i-1} - t)$), $\Sigma_R \geq \Sigma_L$: the backward-looking caplet carries the extra variance of the accrual period, a third of its length.

Example 9.1 (Forward- vs backward-looking caplet)

On the curve bootstrapped below, the at-the-money caplet on $[1, 2]$ (forward rate 3.407%) with $\sigma = 20\%$ is worth 25.5 bp of notional if the rate is forward-looking ($\Sigma_L = 0.2\sqrt{1}$) and 29.4 bp if backward-looking ($\Sigma_R = 0.2\sqrt{1 + 1/3} = 0.231$).

The same structure holds in the one- and two-factor Hull–White models, where the caplet becomes an option on $C(T_i)/C(T_{i-1})$ and Σ_R^2 splits into the variance of the bond's forward price up to T_{i-1} plus the extra variance accumulated over $[T_{i-1}, T_i]$.

9.5. Bootstrapping an RFR curve

RFR curves are stripped from OIS quoted **at par** (zero initial value), typically with annual payments on both legs. With spot start $T_0 = 0$ and par rates $S_i = S_R(0, 0, T_i)$, setting the OIS value to zero for each maturity in turn,

$$P(0, T_1) = \frac{1}{1 + \tau_1 S_1}, \quad P(0, T_i) = \frac{1 - S_i \sum_{j=1}^{i-1} \tau_j P(0, T_j)}{1 + \tau_i S_i}, \quad i = 2, \dots, n,$$

with $\tau_j = \tau(T_{j-1}, T_j)$. (This follows from the par condition $1 - P(0, T_i) = S_i \sum_{j \leq i} \tau_j P(0, T_j)$, the telescoped floating leg equal to the fixed leg.)

Example 9.2 (A five-year OIS bootstrap)

Par rates (annual payments, $\tau = 1$): 1y 3.00%, 2y 3.20%, 3y 3.35%, 5y 3.50%; the 4y rate is linearly interpolated, 3.425%.

T	par rate	$P(0, T)$	zero rate (annual)	zero rate (cont.)
1	3.000%	0.97087	3.000%	2.956%
2	3.200%	0.93889	3.203%	3.153%
3	3.350%	0.90568	3.357%	3.302%
4	3.425%	0.87365	3.435%	3.377%
5	3.500%	0.84143	3.513%	3.453%

With an upward-sloping curve, zero rates lie slightly above par rates, and forward rates above both (the 1-year forward from 4 to 5 years is 3.83%).

In practice (e.g. SOFR on 30 September 2024) the procedure is the same with more instruments, real day counts, a few days' spot lag and an interpolation scheme between pillars.

9.6. Risk-free rates in insurance: Solvency II

Under **Solvency II**, EU insurers compute their solvency ratio from a market-value balance sheet, valuing liabilities (technical provisions) and the Solvency Capital Requirement with a **regulatory risk-free curve** published by EIOPA:

1. collect par rates of IRS (Euribor) or OIS (SARON, SONIA, SOFR, ...) for the liquid maturities (up to about 20 years for the euro);
2. subtract a **credit risk adjustment** (10 bp for Euribor-based curves, zero for RFR-based ones);
3. **bootstrap** zero rates from the adjusted par rates;
4. **extrapolate** beyond the last liquid point up to 150 years with the **Smith–Wilson** method, converging to an ultimate forward rate;
5. optionally add a **volatility adjustment** or **matching adjustment**, which reflect the spread insurers earn on their assets.

Economic scenario generators. Life liabilities contain financial options and guarantees, profit-sharing rules, management actions and policyholder behaviour, so their best estimate (BEL) is computed by Monte Carlo projection of the balance sheet in an asset–liability model, over thousands of **risk-neutral scenarios** that start from the Solvency II curve. Typical rate models are the displaced LIBOR market model with stochastic volatility, the two-factor Hull–White model and the displaced Black–Karasinski model, calibrated to OIS (level) and swaptions (implied volatility). The IBOR→RFR transition did not change these models: OIS and OIS swaptions replace IRS and IRS swaptions, with small adjustments to conventions.

Measuring interest rate risk. Since cash flows of both sides depend on rates, the relevant sensitivity is the **effective duration**, computed by revaluation under parallel shifts $\pm \Delta R$ of the curve:

$$D_A^{ef} = -\frac{1}{A} \frac{\partial A}{\partial R} \approx -\frac{A^+ - A^-}{2A \Delta R}, \quad D_L^{ef} = -\frac{1}{L} \frac{\partial L}{\partial R} \approx -\frac{L^+ - L^-}{2L \Delta R}.$$

Duration matching chooses the asset portfolio so that the value-weighted durations coincide, minimising the **duration gap**

$$D_{Gap} = D_A^{ef} - \frac{L}{A} D_L^{ef},$$

so that a parallel move of rates leaves the net asset value $A - L$ approximately unchanged. Interest rate swaps and swaptions are the main hedging instruments.

9.7. Exercises

Exercise 9.1

A deposit of 100 over 182 days pays 1.60 under ACT/360 simple interest. Find the simple rate and the equivalent continuously compounded rate under ACT/365.

Solution

$\tau_{360} = 182/360 = 0.5056$, so $R_s = 1.60/(100 \times 0.5056) = 3.165\%$. $P = 1/1.016$; with $\tau_{365} = 182/365 = 0.4986$, $R_c = \log(1.016)/0.4986 = 3.184\%$.

Exercise 9.2

Verify the bootstrap formula from the par condition and recompute $P(0, 2)$ in the example.

Solution

At par, the floating leg $1 - P(0, T_i)$ equals the fixed leg $S_i \sum_{j \leq i} \tau_j P(0, T_j)$; isolating $P(0, T_i)$ gives the formula. For T_2 : $P(0, 2) = (1 - 0.032 \times 0.97087)/1.032 = 0.96893/1.032 = 0.93889$.

Exercise 9.3

Compute the ratio Σ_R/Σ_L for a caplet on a 3-month period starting in 1 year, at $t = 0$, and at $t = T_{i-1} + 1.5$ months.

Solution

At $t = 0$: $\Sigma_R^2/\sigma^2 = 1 + 0.25/3 = 1.0833$ against $\Sigma_L^2/\sigma^2 = 1$, ratio 1.041. Midway through the accrual period the forward-looking rate is already fixed ($\Sigma_L = 0$), while $\Sigma_R^2/\sigma^2 = (0.125)^3/(3 \times 0.25^2) = 0.0104$: the backward-looking caplet still has optionality.

Exercise 9.4

(Duration matching; Hull 4.11–4.12) An insurer must pay 100 in 10 years. The curve is flat at 3% (continuous). It can invest in 2-year and 30-year zero-coupon bonds. Find the holdings that match present value and duration, and explain why the hedge is not perfect.

Solution

The liability is worth $L = 100e^{-0.3} = 74.08$ with duration 10. Holding present values x in the 2-year and y in the 30-year bond: $x + y = 74.08$ and $2x + 30y = 10 \times 74.08$, so $y = 21.17$, $x = 52.92$. The portfolio is immunised against small **parallel** shifts, and since its convexity ($\frac{4x+900y}{L} = 260$) exceeds the liability's (100), it gains from larger parallel moves. It remains exposed to **non-parallel** moves (a steepening of the curve between 10 and 30 years), and must be rebalanced as time passes and rates change, since durations drift.

Exercise 9.5

(Hull 4.9) A deposit pays 4% per annum with continuous compounding, but interest is actually paid quarterly. How much interest is paid each quarter on 10 000?

Solution

The equivalent quarterly-compounded rate is $4(e^{0.01} - 1) = 4.0201\%$, so each quarter pays $10\,000 \times 0.040201/4 = 100.50$.

9.8. Sources

V. Russo (Assicurazioni Generali), *Risk-free rates and practical implications*, guest lecture for the Master in Computational Finance, University of Padova, 9 December 2025 ([RFR_Practical](#)). References cited in the lecture: A. Lyashenko and F. Mercurio, *Looking forward to backward-looking rates*, SSRN 3330240 (2019); V. Russo and F. Fabozzi, *J. Fixed Income* **32**, 45 (2023) and **35**, 6 (2025), *J. Derivatives* **31**, 96 (2023); F. Fabozzi, M. Molyboga and V. Russo, *J. Derivatives* **32**, 64 (2024). The numerical examples were computed for these notes.

10. FX options and automated market making

The previous chapters priced derivatives at their arbitrage-free “fair value”. A bank that trades with clients does something different: it **makes a market**, quoting a price at which it buys and a higher one at which it sells, and manages the risk left on its book. This chapter, based on a practitioner lecture by Veronika Lunina, describes how an automated market-making system for FX options works: valuation on the implied volatility surface, spreading, skewing, hedging, dealer intervention and system design. A short review of FX option pricing comes first.

10.1. FX options in brief

An FX option is written on an exchange rate S_t (units of *domestic* currency per unit of *foreign* currency; for EURUSD, USD per EUR). Holding foreign currency earns the foreign interest rate r_f , which acts like a continuous dividend yield. The Black–Scholes argument of Chapter 5 then gives the **Garman–Kohlhagen** formula:

$$C = S_t e^{-r_f(T-t)} \Phi(d_1) - K e^{-r_d(T-t)} \Phi(d_2), \quad d_{1,2} = \frac{\log(S_t/K) + (r_d - r_f \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}},$$

with forward rate $F_t = S_t e^{(r_d - r_f)(T-t)}$ (covered interest parity). The spot Delta is $e^{-r_f(T-t)} \Phi(d_1)$ and the vega $S_t e^{-r_f(T-t)} \varphi(d_1) \sqrt{T-t}$.

Prices are quoted in volatility. In the vanilla market, the price of an option is a **vol**: the number to plug into the Black–Scholes (Garman–Kohlhagen) formula, together with spot, rates and maturity, to obtain the premium in currency. The FX implied volatility surface shows a **smile**: at-the-money options have lower implied volatility than in- and out-of-the-money ones. By convention FX smiles are quoted in terms of Delta rather than strike — at-the-money, 25-delta and 10-delta **risk reversals** (call vol minus put vol, measuring skew) and **butterflies** (average wing vol minus ATM vol, measuring convexity).

10.2. What is market making?

Market making means taking the opposite side of clients’ trades. A market maker does not take directional bets: it earns the **bid–offer spread**, selling at the offer and buying at the bid. Some asset classes are fully electronic (FX spot, cash equities); FX options are less liquid, with much volume still traded OTC and by “voice”. Because holding times are long in illiquid products, risk management is central.

Why automate? Quoting is time-consuming, and a human’s ability to quote while managing risk is limited. Quoting a £100m request takes as much effort as a £1m one, so a voice desk naturally neglects small trades. Automation lets a bank serve more clients and increase market share, providing liquidity on a scalable, 24/7 basis. It **complements** the trading desk; it does not replace it.

10.3. Anatomy of a market-making system

Clients reach the bank through multi-dealer platforms (Bloomberg, Tradeweb, Digital Vega) or the bank’s own single-dealer platform, via two channels: **request for quote** (RFQ) and continuous **price streaming**.

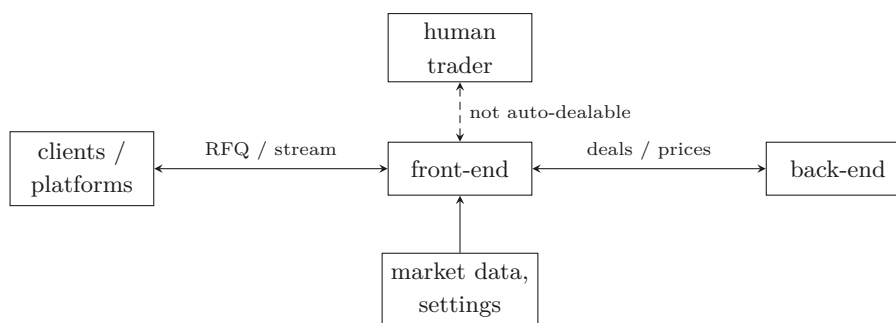


Figure: the front-end handles connectivity, market data and routing; the back-end prices; requests outside automatic limits go to a trader.

- The **front-end** handles connectivity to external platforms, receives price requests, sources market information and passes it, with the desk's settings, to the back-end; it displays results and books trades.
- The **back-end** contains the pricing logic for a collection of deals (a *structure*) and returns the results. It is usually written in C++ or Java.

The back-end performs four steps:

1. **Valuation**: the mid price of the structure and its Greeks (Delta, vega, rho, ...);
2. **Spreading**: bid and offer obtained by subtracting and adding a symmetric spread to the mid;
3. **Skewing**: shifting the quote towards the side that reduces the book's risk;
4. **Hedging**: an FX option is always booked **Delta-hedged** — the Delta is either exchanged with the client or executed in the spot market.

10.4. Where is the mid?

The mid price is an abstraction: participants buy at offers and sell at bids, and the mid must be **estimated**. For vanillas it is read off the implied volatility surface: observed prices of the liquid points (ATM, risk reversals, butterflies at standard tenors) plus a model to interpolate between them give a mid vol for any strike and expiry. Determining the levels of the liquid points is called **marking the surface**; it is done by traders or automatically, for instance with a non-linear Kalman filter that updates the surface as new quotes and trades arrive. Exotic structures are valued with smile-consistent models (local-stochastic volatility, Chapter 7).

10.5. Spreading

Spreading decides how much to charge on top of the mid: too much and nobody trades; too little and the margin does not cover hedging costs and adverse moves. In practice:

- the desk maintains a **base ATM volatility spread** term structure for each currency pair (a value per tenor);
- **multipliers** widen or tighten it depending on market conditions (volatile periods, events), the instrument (strike, Delta, exotic features) and the **counterparty** (client tiering);
- the spread includes the **cost of hedging** the trade (e.g. the bid-offer of the spot hedge);
- for a structure of several deals, spreading accounts for the **netting** of risk between legs.

Since prices are quoted in vols, a spread of $\delta\sigma$ vol points translates into premium through the vega: premium spread $\approx \mathcal{V} \delta\sigma$.

Example 10.1 (*Quoting a EURUSD option*)

Spot $S = 1.10$, $r_{USD} = 4\%$, $r_{EUR} = 2.5\%$, 6-month at-the-money-forward call ($K = F = 1.1083$), mid vol 8%. Garman–Kohlhagen gives a mid premium of 0.02451 USD per EUR, spot Delta 0.505 and vega 0.00306 USD per EUR per vol point. On a 10m EUR notional: mid premium 245 127 USD, vega 30 633 USD per vol point. Quoting 7.85% / 8.15% (a spread of 0.15 vol on each side) gives bid and offer premiums of 240 532 and 249 722 USD: about 4 600 USD of margin per side. If the client buys, the bank books the option together with a Delta hedge: buying about 5.05m EUR spot.

10.6. Skewing

A market maker does not choose its trades: if client flow is one-sided, risk accumulates. Ideally the book stays “flat”. Risk can be reduced **actively** by executing hedges in the market — expensive, especially for vega in illiquid options — or **passively** by **skewing**: quoting tighter (closer to mid) on the side that would reduce the current risk, and wider on the side that would increase it. This raises the probability of favourable trades without paying to hedge.

Skewing needs a **single quantitative measure** of the book’s risk. For a Delta-hedged option book the main risks are **vega** (sensitivity to implied volatility, across tenors and strikes) and sensitivity to scheduled **market events** (central bank rate decisions, non-farm payrolls, ...). With such a measure one computes the risk of the current portfolio, of the portfolio plus the trade and of the portfolio minus the trade, and makes the skew a function of these numbers: if buying the option from the client reduces risk, the bid moves up towards mid.

10.7. Dealability and dealer intervention

The system has **auto-dealable limits** on trade size, depending on the currency pair, the option’s Delta, the time of day, and whether an event is under way. Requests outside the limits are **re-routed to a human trader**, who sees the prices suggested by the system and accepts or modifies them. Requests are also re-routed when anomalies are detected in the input market data or in the system’s outputs.

10.8. System design

The system must be **object-oriented and scalable**: it supports different product types with possibly different spreading logic and is easy to extend to new products. Traders configure parameters that flow in as real-time inputs. Independent pieces of functionality are encapsulated in separate classes. Every price request, together with its market data and settings, can be **serialised** (saved as text) for testing and for investigating issues.

10.9. Glossary

term	meaning
two-way price	a bid and an offer
RFQ	request for quote
widen / tighten	show worse / better prices to a counterparty
spreading	the charge added on top of fair value

term	meaning
skewing	decreasing the spread on one side
margin P&L	profit at inception from the spread charged
position P&L	profit or loss after inception from market moves
smile hedge	Delta computed with a smile model
Fenics hedge	Black–Scholes Delta at the strike's quoted vol
client tiering	client groups with different spreads
client P&L attribution	tracing how much was made or lost per client, to adjust pricing

 Tip

The difference between a **smile hedge** and a **Fenics (Black–Scholes) hedge** is a direct application of Chapter 7: when the smile moves with spot, the true Delta of an option includes the effect of the vol change, $\Delta_{\text{smile}} = \Delta_{BS} + \mathcal{V} \partial \sigma_{\text{imp}} / \partial S$.

10.10. Exercises

Exercise 10.1

Show that the Garman–Kohlhagen formula follows from Black–Scholes by treating the foreign currency as an asset paying a continuous dividend yield r_f .

Solution

One unit of foreign currency deposited abroad grows to $e^{r_f T}$ units, worth $S_T e^{r_f T}$ in domestic currency. The traded asset $X_t = S_t e^{r_f t}$ pays no dividends and, under the domestic risk-neutral measure, grows at r_d ; so S has risk-neutral drift $r_d - r_f$. Repeating the Black–Scholes computation with $\mathbb{E}^Q[S_T] = S_t e^{(r_d - r_f)(T-t)}$ replaces S_t by $S_t e^{-r_f(T-t)}$ in the formula.

Exercise 10.2

A desk is short 2m USD of vega (per vol point) in 1-year EURUSD. A client asks for a two-way price on a 1-year option with 20k USD vega per vol point. With a base spread of 0.2 vol each side, how should the quote be skewed?

Solution

Buying the option from the client (the client sells) would *reduce* the short vega position: the bank should improve its bid, e.g. quoting the bid only 0.1 vol below mid and the offer 0.3 vol above. The margin per side is roughly vega $\times$ spread: 2k USD on the bid side, 6k USD on the offer side.

Exercise 10.3

(Hull 19.10) A bank has sold 1000 seven-month European calls on the yen (each on one yen). Spot 0.80 US cents per yen, strike 0.81, $r_{USD} = 8\%$, $r_{JPY} = 5\%$, volatility 15%. Compute the Delta, Gamma, vega, Theta and rho of the position and interpret them.

Solution

With Garman–Kohlhagen, $d_1 = 0.1016$, and per option (prices in cents): $\Delta = e^{-r_f T} \Phi(d_1) = 0.525$, $\Gamma = 4.206$, $\mathcal{V} = 0.2355$ (per unit of volatility), $\Theta = -0.0399$ per year, $\rho = 0.223$ (per unit of r_{USD}). The short position has: **Delta** -525 (the position loses 5.25 cents if the yen rises by 0.01 cent; hedge by buying 525 yen); **Gamma** -4206 (Delta falls by 4.2 for each 0.001-cent rise); **vega** -235.5 , i.e. -2.36 cents per volatility point; **Theta** $+39.9$ cents per year, i.e. about 0.11 cent per day earned by time decay; **rho** -223 , i.e. -2.23 cents per percentage point of the US rate. Short options are short Gamma and vega and long Theta.

10.11. Sources

V. Lunina, *FX Options Automated Market Making*, guest lecture for the MSc in Computational Finance, University of Padova, 12 November 2025. Hull, *Options, Futures, and Other Derivatives*, Chapter 17 (currency options); for the FX smile conventions, I. Clark, *Foreign Exchange Option Pricing*, Wiley (2011). Original reference: M. Garman and S. Kohlhagen, *J. Int. Money Finance* **2**, 231 (1983). The quoting example was computed for these notes.

